MIDDLESEX SCHOOL
MATHEMATICS DEPARTMENT

Full Curriculum Report

Math 12 through Math 49

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TABLE OF CONTENTS

Select any entry to jump directly to that section

Math 12

Intermediate Algebra

Math 12 Course at a Glance

Intermediate Algebra

TEXTBOOK | OpenStax Intermediate Algebra 2e

Review Content

REVIEW SECTIONS 1.1, 1.2, 1.3, 1.4, 1.5

LEARNING OBJECTIVES

M12-REV-001	I can evaluate numerical expressions using the order of operations.
M12-REV-002	I can perform operations with signed integers and fractions.
M12-REV-003	I can simplify an algebraic expression by combining like terms.
M12-REV-004	I can use the distributive property to rewrite an expression.
M12-REV-005	I can determine the prime factorization of a whole number.

Linear Equations and Applications

REQUIRED SECTIONS 1.1, 2.1, 2.2, 2.3, 2.4

LEARNING OBJECTIVES

M12-EQN-001	I can solve a linear equation and present equivalent steps clearly.
M12-EQN-002	I can solve a linear equation containing fractions.
M12-EQN-003	I can rearrange a formula to isolate a specified variable.
M12-EQN-004	I can translate a verbal relationship into a linear equation.
M12-EQN-005	I can solve and interpret a number, mixture, or motion problem using a linear equation.

Inequalities and Absolute-Value Equations

REQUIRED SECTIONS 2.5, 2.6, 2.7

LEARNING OBJECTIVES

M12-INE-001	I can solve and graph a one-variable linear inequality.
M12-INE-002	I can solve a compound inequality involving AND.
M12-INE-003	I can solve a compound inequality involving OR.
M12-INE-004	I can express an inequality solution using interval notation.
M12-INE-005	I can solve an absolute-value equation and check its solutions.

Lines and Introductory Functions

REQUIRED SECTIONS 3.1, 3.2, 3.3, 3.5, 3.6

LEARNING OBJECTIVES

M12-LIN-001	I can determine and interpret the slope of a line.
M12-LIN-002	I can graph a line from an equation in a common form.
M12-LIN-003	I can write an equation of a line from given information.
M12-LIN-004	I can determine whether two lines are parallel, perpendicular, or neither.
M12-LIN-005	I can evaluate a function using function notation.
M12-LIN-006	I can determine whether a relation is a function using the vertical line test or input-output data.
M12-LIN-008	I can determine domain and range from a graph or table.
M12-LIN-009	I can interpret domain and range in context.

Math 12 Course at a Glance

Intermediate Algebra

TEXTBOOK | OpenStax Intermediate Algebra 2e

Systems of Linear Equations

REQUIRED **SECTIONS** 4.1, 4.2, 4.3

LEARNING OBJECTIVES

M12-SYS-001	I can interpret the solution of a linear system as an intersection point.
M12-SYS-002	I can solve a two-variable linear system by graphing.
M12-SYS-003	I can solve a two-variable linear system by substitution.
M12-SYS-004	I can solve a two-variable linear system by elimination.
M12-SYS-005	I can create and solve a two-variable linear system for an application.

Polynomial Operations and Basic Factoring

REQUIRED **SECTIONS** 5.1, 5.2, 5.3, 5.4, 6.1

LEARNING OBJECTIVES

M12-POL-001	I can simplify expressions using integer exponent rules.
M12-POL-002	I can add and subtract polynomials.
M12-POL-003	I can multiply polynomials, including binomials.
M12-POL-004	I can divide a polynomial by a monomial.
M12-POL-005	I can factor a greatest common factor from a polynomial.

Extension Content

EXTENSIO N **SECTIONS** 2.7, 3.4, 3.6

LEARNING OBJECTIVES

M12-EXT-001	I can solve and graph an absolute-value inequality.
M12-EXT-002	I can graph a system of linear inequalities and identify its solution region.
M12-EXT-003	I can factor a quadratic trinomial.
M12-EXT-004	I can identify a relationship as discrete or continuous.

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Review Content

REVIEW

UNIT TEXTBOOK COVERAGE

1.1 Use the Language of Algebra; 1.2 Integers; 1.3 Fractions; 1.4 Decimals; 1.5 Properties of Real Numbers

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M12-REV-001 I can evaluate numerical expressions using the order of operations.

TEXTBOOK SECTIONS 1.1 Use the Language of Algebra

SUPPORTING SKILLS

R	Apply the order of operations to a numerical expression. (inherited from middle school) SK-NUM-ORDER-OPERATIONS · Reinforce

M12-REV-002 I can perform operations with signed integers and fractions.

TEXTBOOK SECTIONS 1.2 Integers; 1.3 Fractions; 1.4 Decimals

SUPPORTING SKILLS

R	Add, subtract, multiply, and divide fractions. (inherited from middle school) SK-NUM-FRACTION-OPERATE · Reinforce
R	Add, subtract, multiply, and divide signed integers. (inherited from middle school) SK-NUM-INTEGER-OPERATE · Reinforce

M12-REV-003 I can simplify an algebraic expression by combining like terms.

TEXTBOOK SECTIONS 1.5 Properties of Real Numbers

SUPPORTING SKILLS

R	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Reinforce
R	Identify and combine like terms. (inherited from middle school) SK-ALG-LIKE-TERMS · Reinforce

M12-REV-004 I can use the distributive property to rewrite an expression.

TEXTBOOK SECTIONS 1.5 Properties of Real Numbers

SUPPORTING SKILLS

R	Apply the distributive property in either direction. (inherited from middle school) SK-ALG-DISTRIBUTE · Reinforce

M12-REV-005 I can determine the prime factorization of a whole number.

TEXTBOOK SECTIONS 1.1 Use the Language of Algebra

SUPPORTING SKILLS

R	Express a whole number as a product of prime factors. (inherited from middle school) SK-NUM-PRIME-FACTOR · Reinforce

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Linear Equations and Applications

REQUIRED

UNIT TEXTBOOK COVERAGE

1.1 Use the Language of Algebra; 2.1 Use a General Strategy to Solve Linear Equations; 2.2 Use a Problem Solving Strategy; 2.3 Solve a Formula for a Specific Variable; 2.4 Solve Mixture and Uniform Motion Applications

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M12-EQN-001 I can solve a linear equation and present equivalent steps clearly.

TEXTBOOK SECTIONS 2.1 Use a General Strategy to Solve Linear Equations

SUPPORTING SKILLS

I	Check a proposed solution in the original equation. SK-ALG-CHECK-SOLUTION · Introduce
A	Apply the distributive property in either direction. (inherited from middle school) SK-ALG-DISTRIBUTE · Apply
I	Use inverse operations to maintain an equivalent equation. SK-ALG-INVERSE-OPERATIONS · Introduce
A	Identify and combine like terms. (inherited from middle school) SK-ALG-LIKE-TERMS · Apply

M12-EQN-002 I can solve a linear equation containing fractions.

TEXTBOOK SECTIONS 2.1 Use a General Strategy to Solve Linear Equations

SUPPORTING SKILLS

A	Check a proposed solution in the original equation. (first introduced in Math 12) SK-ALG-CHECK-SOLUTION · Apply
I	Clear fractions from an equation using a common denominator. SK-ALG-CLEAR-FRACTIONS · Introduce
A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
A	Add, subtract, multiply, and divide fractions. (inherited from middle school) SK-NUM-FRACTION-OPERATE · Apply

M12-EQN-003 I can rearrange a formula to isolate a specified variable.

TEXTBOOK SECTIONS 2.3 Solve a Formula for a Specific Variable

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Isolate a specified variable in an equation or formula. SK-ALG-VARIABLE-ISOLATE · Introduce

M12-EQN-004 I can translate a verbal relationship into a linear equation.

TEXTBOOK SECTIONS 1.1 Use the Language of Algebra; 2.2 Use a Problem Solving Strategy

SUPPORTING SKILLS

A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Translate quantities and relationships into an equation. SK-ALG-TRANSLATE-EQUATION · Introduce

M12-EQN-005 I can solve and interpret a number, mixture, or motion problem using a linear equation.

TEXTBOOK SECTIONS 2.2 Use a Problem Solving Strategy; 2.4 Solve Mixture and Uniform Motion Applications

SUPPORTING SKILLS

A	Check a proposed solution in the original equation. (first introduced in Math 12) SK-ALG-CHECK-SOLUTION · Apply
A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
D	Translate quantities and relationships into an equation. (first introduced in Math 12) SK-ALG-TRANSLATE-EQUATION · Deepen

SKILL PROGRESSION

I	Introduce
R	Reinforce
D	Deepen
A	Apply

Inequalities and Absolute-Value Equations

REQUIRED

UNIT TEXTBOOK COVERAGE

2.5 Solve Linear Inequalities; 2.6 Solve Compound Inequalities; 2.7 Solve Absolute Value Inequalities

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M12-INE-001 I can solve and graph a one-variable linear inequality.

TEXTBOOK SECTIONS 2.5 Solve Linear Inequalities

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Represent an inequality solution on a number line. SK-INE-NUMBER-LINE · Introduce
I	Reverse an inequality when multiplying or dividing by a negative value. SK-INE-REVERSE-SIGN · Introduce

M12-INE-002 I can solve a compound inequality involving AND.

TEXTBOOK SECTIONS 2.6 Solve Compound Inequalities

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Interpret AND as the intersection of two inequality solution sets. SK-INE-COMPOUND-AND · Introduce
A	Represent an inequality solution on a number line. (first introduced in Math 12) SK-INE-NUMBER-LINE · Apply
A	Reverse an inequality when multiplying or dividing by a negative value. (first introduced in Math 12) SK-INE-REVERSE-SIGN · Apply

M12-INE-003 I can solve a compound inequality involving OR.

TEXTBOOK SECTIONS 2.6 Solve Compound Inequalities

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Interpret OR as the union of two inequality solution sets. SK-INE-COMPOUND-OR · Introduce
A	Represent an inequality solution on a number line. (first introduced in Math 12) SK-INE-NUMBER-LINE · Apply
A	Reverse an inequality when multiplying or dividing by a negative value. (first introduced in Math 12) SK-INE-REVERSE-SIGN · Apply

M12-INE-004 I can express an inequality solution using interval notation.

TEXTBOOK SECTIONS 2.5 Solve Linear Inequalities; 2.6 Solve Compound Inequalities

SUPPORTING SKILLS

A	Represent an inequality solution on a number line. (first introduced in Math 12) SK-INE-NUMBER-LINE · Apply
I	Translate between inequality and interval notation. SK-NOT-INTERVAL · Introduce

M12-INE-005 I can solve an absolute-value equation and check its solutions.

TEXTBOOK SECTIONS 2.7 Solve Absolute Value Inequalities

SUPPORTING SKILLS

I	Rewrite an absolute-value equation as two cases. SK-ABS-SPLIT-EQUATION · Introduce
A	Check a proposed solution in the original equation. (first introduced in Math 12) SK-ALG-CHECK-SOLUTION · Apply
A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Lines and Introductory Functions

REQUIRED

UNIT TEXTBOOK COVERAGE

3.1 Graph Linear Equations in Two Variables; 3.2 Slope of a Line; 3.3 Find the Equation of a Line; 3.5 Relations and Functions; 3.6 Graphs of Functions

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M12-LIN-01 I can determine and interpret the slope of a line.

TEXTBOOK SECTIONS 3.2 Slope of a Line

SUPPORTING SKILLS	
I	Determine slope from a graph. SK-LIN-SLOPE-GRAPH · Introduce
I	Calculate slope from two points. SK-LIN-SLOPE-POINTS · Introduce
I	Attach and interpret units for rates, inputs, and outputs. SK-REP-UNITS · Introduce

M12-LIN-02 I can graph a line from an equation in a common form.

TEXTBOOK SECTIONS 3.1 Graph Linear Equations in Two Variables

SUPPORTING SKILLS	
I	Convert among common forms of a linear equation. SK-LIN-FORM-CONVERT · Introduce
I	Use slope to generate additional points on a line. SK-LIN-GRAPH-SLOPE · Introduce
I	Plot an intercept accurately on a coordinate plane. SK-LIN-PLOT-INTERCEPT · Introduce

M12-LIN-03 I can write an equation of a line from given information.

TEXTBOOK SECTIONS 3.3 Find the Equation of a Line

SUPPORTING SKILLS	
A	Isolate a specified variable in an equation or formula. (first introduced in Math 12) SK-ALG-VARIABLE-ISOLATE · Apply
D	Convert among common forms of a linear equation. (first introduced in Math 12) SK-LIN-FORM-CONVERT · Deepen
A	Calculate slope from two points. (first introduced in Math 12) SK-LIN-SLOPE-POINTS · Apply

M12-LIN-04 I can determine whether two lines are parallel, perpendicular, or neither.

TEXTBOOK SECTIONS 3.2 Slope of a Line; 3.3 Find the Equation of a Line

SUPPORTING SKILLS	
I	Recognize that distinct parallel lines have equal slopes. SK-LIN-PARALLEL-SLOPE · Introduce
I	Determine the negative reciprocal of a nonzero slope. SK-LIN-PERPENDICULAR-SLOPE · Introduce

M12-LIN-05 I can evaluate a function using function notation.

TEXTBOOK SECTIONS 3.5 Relations and Functions; 3.6 Graphs of Functions

SUPPORTING SKILLS	
A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Identify input and output values in a formula, graph, table, or context. SK-FUNC-INPUT-OUTPUT · Introduce
I	Identify the input and output represented by function notation. SK-FUNC-NOTATION-READ · Introduce

Math 12 In-Depth Curriculum

Intermediate Algebra

TEXTBOOK | OpenStax Intermediate Algebra 2e

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M12-LIN-006 — I can determine whether a relation is a function using the vertical line test or input-output data.

TEXTBOOK SECTIONS 3.5 Relations and Functions

SUPPORTING SKILLS

D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
I	Apply the vertical line test to a graph. SK-FUNC-VERTICAL-LINE · Introduce

M12-LIN-008 — I can determine domain and range from a graph or table.

TEXTBOOK SECTIONS 3.5 Relations and Functions; 3.6 Graphs of Functions

SUPPORTING SKILLS

I	Read domain and range from a graph. SK-FUNC-DOMAIN-RANGE-GRAPH · Introduce
I	Read domain and range from a table. SK-FUNC-DOMAIN-RANGE-TABLE · Introduce

M12-LIN-009 — I can interpret domain and range in context.

TEXTBOOK SECTIONS 3.6 Graphs of Functions

SUPPORTING SKILLS

I	Restrict domain to values meaningful in a context. SK-FUNC-CONTEXT-DOMAIN · Introduce
I	Interpret the range of a function in a context. SK-FUNC-CONTEXT-RANGE · Introduce
I	Restrict a model to meaningful contextual inputs. SK-MODEL-VALID-DOMAIN · Introduce

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Systems of Linear Equations

REQUIRED

UNIT TEXTBOOK COVERAGE

4.1 Solve Systems of Linear Equations with Two Variables; 4.2 Solve Applications with Systems of Equations; 4.3 Solve Mixture Applications with Systems of Equations

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M12-SYS-001 I can interpret the solution of a linear system as an intersection point.

TEXTBOOK SECTIONS 4.1 Solve Systems of Linear Equations with Two Variables

SUPPORTING SKILLS

I	Locate and interpret the intersection of two lines. SK-SYS-GRAPH-INTERSECTION · Introduce

M12-SYS-002 I can solve a two-variable linear system by graphing.

TEXTBOOK SECTIONS 4.1 Solve Systems of Linear Equations with Two Variables

SUPPORTING SKILLS

A	Use slope to generate additional points on a line. (first introduced in Math 12) SK-LIN-GRAPH-SLOPE · Apply
A	Plot an intercept accurately on a coordinate plane. (first introduced in Math 12) SK-LIN-PLOT-INTERCEPT · Apply
I	Check an ordered pair in both equations of a system. SK-SYS-CHECK · Introduce
D	Locate and interpret the intersection of two lines. (first introduced in Math 12) SK-SYS-GRAPH-INTERSECTION · Deepen

M12-SYS-003 I can solve a two-variable linear system by substitution.

TEXTBOOK SECTIONS 4.1 Solve Systems of Linear Equations with Two Variables

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
A	Check an ordered pair in both equations of a system. (first introduced in Math 12) SK-SYS-CHECK · Apply
I	Substitute an equivalent expression for a variable. SK-SYS-SUBSTITUTE · Introduce

M12-SYS-004 I can solve a two-variable linear system by elimination.

TEXTBOOK SECTIONS 4.1 Solve Systems of Linear Equations with Two Variables

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
A	Check an ordered pair in both equations of a system. (first introduced in Math 12) SK-SYS-CHECK · Apply
I	Add equations to eliminate a variable. SK-SYS-ELIMINATE · Introduce

M12-SYS-005 I can create and solve a two-variable linear system for an application.

TEXTBOOK SECTIONS 4.2 Solve Applications with Systems of Equations; 4.3 Solve Mixture Applications with Systems of Equations

SUPPORTING SKILLS

A	Check an ordered pair in both equations of a system. (first introduced in Math 12) SK-SYS-CHECK · Apply
A	Add equations to eliminate a variable. (first introduced in Math 12) SK-SYS-ELIMINATE · Apply
I	Define two variables and write two equations for a context. SK-SYS-MODEL · Introduce
A	Substitute an equivalent expression for a variable. (first introduced in Math 12) SK-SYS-SUBSTITUTE · Apply

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Polynomial Operations and Basic Factoring

REQUIRED

UNIT TEXTBOOK COVERAGE

5.1 Add and Subtract Polynomials; 5.2 Properties of Exponents and Scientific Notation; 5.3 Multiply Polynomials; 5.4 Dividing Polynomials; 6.1 Greatest Common Factor and Factor by Grouping

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M12-POL-001 I can simplify expressions using integer exponent rules.

TEXTBOOK SECTIONS 5.2 Properties of Exponents and Scientific Notation

SUPPORTING SKILLS	
I	Apply the power rule for exponents. SK-EXP-POWER · Introduce
I	Apply the product rule for exponents. SK-EXP-PRODUCT · Introduce
I	Apply the quotient rule for exponents. SK-EXP-QUOTIENT · Introduce

M12-POL-002 I can add and subtract polynomials.

TEXTBOOK SECTIONS 5.1 Add and Subtract Polynomials

SUPPORTING SKILLS	
A	Identify and combine like terms. (inherited from middle school) SK-ALG-LIKE-TERMS · Apply
I	Identify like polynomial terms by variable part and exponent. SK-POLY-LIKE-TERMS · Introduce
I	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. SK-POLY-VOCABULARY · Introduce

M12-POL-003 I can multiply polynomials, including binomials.

TEXTBOOK SECTIONS 5.3 Multiply Polynomials

SUPPORTING SKILLS	
A	Apply the distributive property in either direction. (inherited from middle school) SK-ALG-DISTRIBUTE · Apply
I	Multiply two binomials using distribution. SK-POLY-BINOMIAL-MULTIPLY · Introduce
I	Multiply a monomial by a polynomial. SK-POLY-DISTRIBUTE · Introduce
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

M12-POL-004 I can divide a polynomial by a monomial.

TEXTBOOK SECTIONS 5.4 Dividing Polynomials

SUPPORTING SKILLS	
A	Apply the quotient rule for exponents. (first introduced in Math 12) SK-EXP-QUOTIENT · Apply
I	Divide each term of a polynomial by a monomial. SK-POLY-DIVIDE-MONOMIAL · Introduce
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

M12-POL-005 I can factor a greatest common factor from a polynomial.

TEXTBOOK SECTIONS 6.1 Greatest Common Factor and Factor by Grouping

SUPPORTING SKILLS	
A	Apply the distributive property in either direction. (inherited from middle school) SK-ALG-DISTRIBUTE · Apply
I	Factor the greatest common monomial factor from a polynomial. SK-FACTOR-GCF · Introduce
I	Determine the numerical and variable greatest common factor of polynomial terms. SK-POLY-GCF · Introduce
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

SKILL PROGRESSION

I Introduce R Reinforce D Deepen A Apply

Extension Content

EXTENSION

UNIT TEXTBOOK COVERAGE
2.7 Solve Absolute Value Inequalities; 3.4 Graph Linear Inequalities in Two Variables; 3.6 Graphs of Functions

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M12-EXT-01 I can solve and graph an absolute-value inequality.

TEXTBOOK SECTIONS 2.7 Solve Absolute Value Inequalities

SUPPORTING SKILLS

I	Rewrite an absolute-value inequality as a compound inequality. SK-ABS-SPLIT-INEQUALITY · Introduce
A	Interpret AND as the intersection of two inequality solution sets. (first introduced in Math 12) SK-INE-COMPOUND-AND · Apply
A	Interpret OR as the union of two inequality solution sets. (first introduced in Math 12) SK-INE-COMPOUND-OR · Apply
A	Represent an inequality solution on a number line. (first introduced in Math 12) SK-INE-NUMBER-LINE · Apply

M12-EXT-02 I can graph a system of linear inequalities and identify its solution region.

TEXTBOOK SECTIONS 3.4 Graph Linear Inequalities in Two Variables

SUPPORTING SKILLS

I	Graph the boundary line of a linear inequality with the correct line type. SK-INE-GRAPH-BOUNDARY · Introduce
I	Identify the overlapping solution region of multiple linear inequalities. SK-INE-SYSTEM-REGION · Introduce
A	Use slope to generate additional points on a line. (first introduced in Math 12) SK-LIN-GRAPH-SLOPE · Apply

M12-EXT-03 I can factor a quadratic trinomial.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

I	Verify a factorization by multiplication. SK-FACTOR-CHECK · Introduce
I	Factor a quadratic trinomial with a nonunit leading coefficient. SK-FACTOR-TRINOMIAL-GENERAL · Introduce
I	Factor a monic quadratic trinomial. SK-FACTOR-TRINOMIAL-MONIC · Introduce

M12-EXT-04 I can identify a relationship as discrete or continuous.

TEXTBOOK SECTIONS 3.6 Graphs of Functions

SUPPORTING SKILLS

I	Distinguish discrete inputs from values varying continuously over an interval. SK-FUNC-DISCRETE-CONTINUOUS · Introduce

Math 21

Algebra and its Functions

Math 21 Course at a Glance

Algebra and its Functions

TEXTBOOK | OpenStax Intermediate Algebra 2e (Marecek and Mathis, 2020)

Review Content

REVIEW SECTIONS 3.1, 3.2, 3.3, 5.2

LEARNING OBJECTIVES

M21-REV-001	I can determine and interpret the slope of a line.
M21-REV-002	I can graph a line from an equation and write an equation from given information.
M21-REV-003	I can apply a linear function to a context.
M21-REV-004	I can identify the solution of a linear system as an intersection point.
M21-REV-005	I can solve a simple two-variable linear system using an appropriate method.
M21-REV-006	I can simplify an expression using basic positive-integer exponent rules.

Functions and Function Interpretation

REQUIRED SECTIONS 3.5, 3.6

LEARNING OBJECTIVES

M21-FUN-001	I can evaluate a function from a formula, graph, or table.
M21-FUN-002	I can determine whether a relation is a function.
M21-FUN-003	I can determine domain and range from a graph or table.
M21-FUN-004	I can solve for an input that produces a specified output.
M21-FUN-005	I can interpret inputs, outputs, domain, and range in context.
M21-FUN-006	I can calculate and interpret average rate of change.
M21-FUN-007	I can compare functions represented by formulas, graphs, or tables.

Polynomials and Factoring

REQUIRED SECTIONS 5.1, 5.3, 5.4, 6.1, 6.2, 6.3

LEARNING OBJECTIVES

M21-POL-001	I can add and subtract polynomials.
M21-POL-002	I can multiply polynomials.
M21-POL-003	I can factor out a greatest common factor.
M21-POL-004	I can factor a quadratic trinomial.
M21-POL-005	I can factor a difference of squares and other required special cases.
M21-POL-006	I can use factoring to determine the zeros of a polynomial.

Rational Expressions and Equations

REQUIRED SECTIONS 7.1, 7.2, 7.3, 7.4, 7.5

LEARNING OBJECTIVES

M21-RAT-001	I can state excluded values for a rational expression.
M21-RAT-002	I can simplify a rational expression while preserving its restrictions.
M21-RAT-003	I can multiply and divide rational expressions.
M21-RAT-004	I can add and subtract rational expressions using a common denominator.
M21-RAT-005	I can solve a rational equation and reject extraneous solutions.

Math 21 Course at a Glance

Algebra and its Functions

TEXTBOOK | OpenStax Intermediate Algebra 2e (Marecek and Mathis, 2020)

Exponents and Radicals

REQUIRED **SECTIONS** 8.1, 8.2, 8.3, 8.4

LEARNING OBJECTIVES

M21-EXP-001	I can simplify expressions containing negative exponents.
M21-EXP-002	I can rewrite expressions between radical and rational-exponent forms.
M21-EXP-003	I can simplify radical expressions.
M21-EXP-004	I can apply exponent rules to expressions with integer and rational exponents.
M21-EXP-005	I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.

Quadratic Functions and Equations

REQUIRED **SECTIONS** 9.1, 9.2, 9.3, 9.4, 9.5, 9.6, 9.7

LEARNING OBJECTIVES

M21-QUAD-001	I can identify key features of a quadratic function from its graph or equation.
M21-QUAD-002	I can connect standard, factored, and vertex forms to features of a parabola.
M21-QUAD-003	I can solve a quadratic equation by factoring.
M21-QUAD-004	I can solve a quadratic equation using the square-root property.
M21-QUAD-005	I can solve a quadratic equation by completing the square.
M21-QUAD-006	I can solve a quadratic equation using the quadratic formula.
M21-QUAD-007	I can choose an efficient method for solving a quadratic equation.
M21-QUAD-008	I can graph a quadratic function using its vertex, intercepts, and symmetry.
M21-QUAD-009	I can create and interpret a quadratic model for a projectile, area, or optimization problem.

Extension Content

EXTENSION **SECTIONS** 9.3

LEARNING OBJECTIVES

M21-EXT-001	I can perform basic arithmetic with complex numbers.
M21-EXT-002	I can express the nonreal solutions of a quadratic equation using complex numbers.

Review Content

REVIEW

UNIT TEXTBOOK COVERAGE

3.1 Graph Linear Equations in Two Variables; 3.2 Slope of a Line; 3.3 Find the Equation of a Line; 5.2 Properties of Exponents and Scientific Notation

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M21-REV-001 I can determine and interpret the slope of a line.

TEXTBOOK SECTIONS 3.2 Slope of a Line

SUPPORTING SKILLS

R	Determine slope from a graph. (first introduced in Math 12) SK-LIN-SLOPE-GRAPH · Reinforce
R	Calculate slope from two points. (first introduced in Math 12) SK-LIN-SLOPE-POINTS · Reinforce
R	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Reinforce

M21-REV-002 I can graph a line from an equation and write an equation from given information.

TEXTBOOK SECTIONS 3.1 Graph Linear Equations in Two Variables; 3.3 Find the Equation of a Line

SUPPORTING SKILLS

R	Convert among common forms of a linear equation. (first introduced in Math 12) SK-LIN-FORM-CONVERT · Reinforce
R	Use slope to generate additional points on a line. (first introduced in Math 12) SK-LIN-GRAPH-SLOPE · Reinforce
R	Plot an intercept accurately on a coordinate plane. (first introduced in Math 12) SK-LIN-PLOT-INTERCEPT · Reinforce
A	Calculate slope from two points. (first introduced in Math 12) SK-LIN-SLOPE-POINTS · Apply

M21-REV-003 I can apply a linear function to a context.

TEXTBOOK SECTIONS 3.3 Find the Equation of a Line

SUPPORTING SKILLS

I	Explain a model parameter in the language of its context. SK-MODEL-PARAMETER-INTERPRET · Introduce
R	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Reinforce
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

M21-REV-004 I can identify the solution of a linear system as an intersection point.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

R	Locate and interpret the intersection of two lines. (first introduced in Math 12) SK-SYS-GRAPH-INTERSECTION · Reinforce

M21-REV-005 I can solve a simple two-variable linear system using an appropriate method.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

R	Check an ordered pair in both equations of a system. (first introduced in Math 12) SK-SYS-CHECK · Reinforce
R	Add equations to eliminate a variable. (first introduced in Math 12) SK-SYS-ELIMINATE · Reinforce
R	Locate and interpret the intersection of two lines. (first introduced in Math 12) SK-SYS-GRAPH-INTERSECTION · Reinforce
R	Substitute an equivalent expression for a variable. (first introduced in Math 12) SK-SYS-SUBSTITUTE · Reinforce

M21-REV-006 I can simplify an expression using basic positive-integer exponent rules.

TEXTBOOK SECTIONS 5.2 Properties of Exponents and Scientific Notation

SUPPORTING SKILLS	
R	Apply the power rule for exponents. (first introduced in Math 12) SK-EXP-POWER · Reinforce
R	Apply the product rule for exponents. (first introduced in Math 12) SK-EXP-PRODUCT · Reinforce
R	Apply the quotient rule for exponents. (first introduced in Math 12) SK-EXP-QUOTIENT · Reinforce

Functions and Function Interpretation

REQUIRED

UNIT TEXTBOOK COVERAGE
3.5 Relations and Functions; 3.6 Graphs of Functions

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M21-FUN-001 — I can evaluate a function from a formula, graph, or table.

TEXTBOOK SECTIONS 3.5 Relations and Functions; 3.6 Graphs of Functions

	SUPPORTING SKILLS
A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
A	Identify the input and output represented by function notation. (first introduced in Math 12) SK-FUNC-NOTATION-READ · Apply

M21-FUN-002 — I can determine whether a relation is a function.

TEXTBOOK SECTIONS 3.5 Relations and Functions; 3.6 Graphs of Functions

	SUPPORTING SKILLS
D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
D	Apply the vertical line test to a graph. (first introduced in Math 12) SK-FUNC-VERTICAL-LINE · Deepen

M21-FUN-003 — I can determine domain and range from a graph or table.

TEXTBOOK SECTIONS 3.5 Relations and Functions; 3.6 Graphs of Functions

	SUPPORTING SKILLS
D	Read domain and range from a graph. (first introduced in Math 12) SK-FUNC-DOMAIN-RANGE-GRAPH · Deepen
D	Read domain and range from a table. (first introduced in Math 12) SK-FUNC-DOMAIN-RANGE-TABLE · Deepen

M21-FUN-004 — I can solve for an input that produces a specified output.

TEXTBOOK SECTIONS 3.5 Relations and Functions; 3.6 Graphs of Functions

	SUPPORTING SKILLS
A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Solve for an input associated with a specified output. SK-FUNC-SOLVE-INPUT · Introduce

M21-FUN-005 — I can interpret inputs, outputs, domain, and range in context.

TEXTBOOK SECTIONS 3.5 Relations and Functions; 3.6 Graphs of Functions

	SUPPORTING SKILLS
D	Restrict domain to values meaningful in a context. (first introduced in Math 12) SK-FUNC-CONTEXT-DOMAIN · Deepen
D	Interpret the range of a function in a context. (first introduced in Math 12) SK-FUNC-CONTEXT-RANGE · Deepen
D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
D	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Deepen

M21-FUN-006 — I can calculate and interpret average rate of change.

TEXTBOOK SECTIONS No mapped textbook section

	SUPPORTING SKILLS
I	Calculate average rate of change over an interval. SK-FUNC-AROC · Introduce
I	Interpret average rate of change with contextual units. SK-FUNC-AROC-INTERPRET · Introduce
D	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Deepen

M21-FUN-007 I can compare functions represented by formulas, graphs, or tables.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

I	Compare functions shown in different representations. SK-FUNC-COMPARE-REP · Introduce
I	Connect parameters in a formula to features of its graph. SK-REP-FORMULA-GRAPH · Introduce
I	Use tabular patterns to predict graph shape and behavior. SK-REP-TABLE-GRAPH · Introduce

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Polynomials and Factoring

REQUIRED

UNIT TEXTBOOK COVERAGE

5.1 Add and Subtract Polynomials; 5.3 Multiply Polynomials; 5.4 Dividing Polynomials; 6.1 Greatest Common Factor and Factor by Grouping; 6.2 Factor Trinomials; 6.3 Factor Special Products

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M21-POL-001 I can add and subtract polynomials.

TEXTBOOK SECTIONS 5.1 Add and Subtract Polynomials; 5.4 Dividing Polynomials

	SUPPORTING SKILLS
A	Identify and combine like terms. (inherited from middle school) SK-ALG-LIKE-TERMS · Apply
D	Identify like polynomial terms by variable part and exponent. (first introduced in Math 12) SK-POLY-LIKE-TERMS · Deepen
R	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Reinforce

M21-POL-002 I can multiply polynomials.

TEXTBOOK SECTIONS 5.3 Multiply Polynomials; 5.4 Dividing Polynomials

	SUPPORTING SKILLS
A	Apply the distributive property in either direction. (inherited from middle school) SK-ALG-DISTRIBUTE · Apply
D	Multiply two binomials using distribution. (first introduced in Math 12) SK-POLY-BINOMIAL-MULTIPLY · Deepen
D	Multiply a monomial by a polynomial. (first introduced in Math 12) SK-POLY-DISTRIBUTE · Deepen
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

M21-POL-003 I can factor out a greatest common factor.

TEXTBOOK SECTIONS 6.1 Greatest Common Factor and Factor by Grouping

	SUPPORTING SKILLS
R	Verify a factorization by multiplication. (first introduced in Math 12) SK-FACTOR-CHECK · Reinforce
D	Factor the greatest common monomial factor from a polynomial. (first introduced in Math 12) SK-FACTOR-GCF · Deepen
A	Determine the numerical and variable greatest common factor of polynomial terms. (first introduced in Math 12) SK-POLY-GCF · Apply
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

M21-POL-004 I can factor a quadratic trinomial.

TEXTBOOK SECTIONS 6.1 Greatest Common Factor and Factor by Grouping; 6.2 Factor Trinomials

	SUPPORTING SKILLS
A	Verify a factorization by multiplication. (first introduced in Math 12) SK-FACTOR-CHECK · Apply
R	Factor a quadratic trinomial with a nonunit leading coefficient. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-GENERAL · Reinforce
R	Factor a monic quadratic trinomial. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-MONIC · Reinforce
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

M21-POL-005 I can factor a difference of squares and other required special cases.

TEXTBOOK SECTIONS 6.3 Factor Special Products

SUPPORTING SKILLS

A	Verify a factorization by multiplication. (first introduced in Math 12) SK-FACTOR-CHECK · Apply
I	Factor a difference of two squares. SK-FACTOR-DIFF-SQUARES · Introduce
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

M21-POL-006 I can use factoring to determine the zeros of a polynomial.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

A	Factor a difference of two squares. (first introduced in Math 21) SK-FACTOR-DIFF-SQUARES · Apply
A	Factor a quadratic trinomial with a nonunit leading coefficient. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-GENERAL · Apply
A	Factor a monic quadratic trinomial. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-MONIC · Apply
I	Apply the zero-product property. SK-FACTOR-ZERO-PRODUCT · Introduce
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

SKILL PROGRESSION

I Introduce | R Reinforce | D Deepen | A Apply

Rational Expressions and Equations

REQUIRED

UNIT TEXTBOOK COVERAGE

7.1 Simplify Rational Expressions; 7.2 Multiply and Divide Rational Expressions; 7.3 Add and Subtract Rational Expressions with a Common Denominator; 7.4 Add and Subtract Rational Expressions with Unlike Denominators; 7.5 Solve Rational Equations

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M21-RAT-001 I can state excluded values for a rational expression.

TEXTBOOK SECTIONS 7.1 Simplify Rational Expressions; 7.2 Multiply and Divide Rational Expressions; 7.3 Add and Subtract Rational Expressions with a Common Denominator; 7.4 Add and Subtract Rational Expressions with Unlike Denominators

SUPPORTING SKILLS	
I	Determine excluded values from an original denominator. SK-RAT-RESTRICTIONS · Introduce

M21-RAT-002 I can simplify a rational expression while preserving its restrictions.

TEXTBOOK SECTIONS 7.1 Simplify Rational Expressions; 7.2 Multiply and Divide Rational Expressions

SUPPORTING SKILLS	
I	Cancel common factors rather than individual terms. SK-RAT-CANCEL-FACTORS · Introduce
I	Factor numerators and denominators completely. SK-RAT-FACTOR · Introduce
A	Determine excluded values from an original denominator. (first introduced in Math 21) SK-RAT-RESTRICTIONS · Apply

M21-RAT-003 I can multiply and divide rational expressions.

TEXTBOOK SECTIONS 7.2 Multiply and Divide Rational Expressions

SUPPORTING SKILLS	
A	Add, subtract, multiply, and divide fractions. (inherited from middle school) SK-NUM-FRACTION-OPERATE · Apply
A	Cancel common factors rather than individual terms. (first introduced in Math 21) SK-RAT-CANCEL-FACTORS · Apply
A	Factor numerators and denominators completely. (first introduced in Math 21) SK-RAT-FACTOR · Apply
A	Determine excluded values from an original denominator. (first introduced in Math 21) SK-RAT-RESTRICTIONS · Apply

M21-RAT-004 I can add and subtract rational expressions using a common denominator.

TEXTBOOK SECTIONS 7.3 Add and Subtract Rational Expressions with a Common Denominator; 7.4 Add and Subtract Rational Expressions with Unlike Denominators

SUPPORTING SKILLS	
A	Add, subtract, multiply, and divide fractions. (inherited from middle school) SK-NUM-FRACTION-OPERATE · Apply
I	Combine rational expressions over a common denominator. SK-RAT-COMBINE · Introduce
I	Determine the least common denominator of rational expressions. SK-RAT-LCD · Introduce

M21-RAT-005 I can solve a rational equation and reject extraneous solutions.

TEXTBOOK SECTIONS 7.5 Solve Rational Equations

SUPPORTING SKILLS	
A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Clear denominators from a rational equation. SK-RAT-CLEAR-DENOMINATORS · Introduce
I	Check candidate solutions against denominator restrictions. SK-RAT-EXTRANEOUS · Introduce
A	Determine excluded values from an original denominator. (first introduced in Math 21) SK-RAT-RESTRICTIONS · Apply

SKILL PROGRESSION

I Introduce | R Reinforce | D Deepen | A Apply

Exponents and Radicals

REQUIRED

UNIT TEXTBOOK COVERAGE

8.1 Simplify Expressions with Roots; 8.2 Simplify Radical Expressions; 8.3 Simplify Rational Exponents; 8.4 Add, Subtract, and Multiply Radical Expressions

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M21-EXP-001 — **I can simplify expressions containing negative exponents.**

TEXTBOOK SECTIONS 8.3 Simplify Rational Exponents

SUPPORTING SKILLS

I	Rewrite a negative exponent using a reciprocal. SK-EXP-NEGATIVE · Introduce
A	Apply the product rule for exponents. (first introduced in Math 12) SK-EXP-PRODUCT · Apply
A	Apply the quotient rule for exponents. (first introduced in Math 12) SK-EXP-QUOTIENT · Apply

M21-EXP-002 — **I can rewrite expressions between radical and rational-exponent forms.**

TEXTBOOK SECTIONS 8.3 Simplify Rational Exponents

SUPPORTING SKILLS

I	Interpret numerator and denominator roles in a rational exponent. SK-EXP-RATIONAL · Introduce
I	Convert between radical and rational-exponent notation. SK-RAD-CONVERT · Introduce

M21-EXP-003 — **I can simplify radical expressions.**

TEXTBOOK SECTIONS 8.1 Simplify Expressions with Roots; 8.2 Simplify Radical Expressions; 8.4 Add, Subtract, and Multiply Radical Expressions

SUPPORTING SKILLS

A	Express a whole number as a product of prime factors. (inherited from middle school) SK-NUM-PRIME-FACTOR · Apply
I	Extract perfect-power factors from a radical. SK-RAD-FACTOR-PERFECT · Introduce

M21-EXP-004 — **I can apply exponent rules to expressions with integer and rational exponents.**

TEXTBOOK SECTIONS 8.3 Simplify Rational Exponents; 8.4 Add, Subtract, and Multiply Radical Expressions

SUPPORTING SKILLS

A	Rewrite a negative exponent using a reciprocal. (first introduced in Math 21) SK-EXP-NEGATIVE · Apply
D	Apply the power rule for exponents. (first introduced in Math 12) SK-EXP-POWER · Deepen
D	Apply the product rule for exponents. (first introduced in Math 12) SK-EXP-PRODUCT · Deepen
D	Apply the quotient rule for exponents. (first introduced in Math 12) SK-EXP-QUOTIENT · Deepen
A	Interpret numerator and denominator roles in a rational exponent. (first introduced in Math 21) SK-EXP-RATIONAL · Apply
I	Apply the zero-exponent rule. SK-EXP-ZERO · Introduce

M21-EXP-005 — **I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.**

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Require an even-root radicand to be nonnegative when determining domain. SK-FUNC-DOMAIN-EVEN-ROOT · Introduce
I	Check a candidate solution for extraneous results. SK-RAD-CHECK · Introduce
I	Isolate a radical expression before solving. SK-RAD-ISOLATE · Introduce

SKILL PROGRESSION

I Introduce | R Reinforce | D Deepen | A Apply

Quadratic Functions and Equations

REQUIRED

UNIT TEXTBOOK COVERAGE

9.1 Solve Quadratic Equations Using the Square Root Property; 9.2 Solve Quadratic Equations by Completing the Square; 9.3 Solve Quadratic Equations Using the Quadratic Formula; 9.4 Solve Equations in Quadratic Form; 9.5 Solve Applications of Quadratic Equations; 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M21-QUAD-001 — I can identify key features of a quadratic function from its graph or equation.

TEXTBOOK SECTIONS 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations

SUPPORTING SKILLS

I	Identify standard, factored, and vertex forms. SK-QUAD-FORM-IDENTIFY · Introduce
I	Determine x- and y-intercepts of a quadratic function. SK-QUAD-INTERCEPTS · Introduce
I	Determine the vertex and axis of symmetry. SK-QUAD-VERTEX · Introduce

M21-QUAD-002 — I can connect standard, factored, and vertex forms to features of a parabola.

TEXTBOOK SECTIONS 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations

SUPPORTING SKILLS

I	Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method. SK-QUAD-FORM-CONVERT · Introduce
D	Identify standard, factored, and vertex forms. (first introduced in Math 21) SK-QUAD-FORM-IDENTIFY · Deepen
A	Determine x- and y-intercepts of a quadratic function. (first introduced in Math 21) SK-QUAD-INTERCEPTS · Apply
A	Determine the vertex and axis of symmetry. (first introduced in Math 21) SK-QUAD-VERTEX · Apply
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

M21-QUAD-003 — I can solve a quadratic equation by factoring.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

A	Factor a quadratic trinomial with a nonunit leading coefficient. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-GENERAL · Apply
A	Factor a monic quadratic trinomial. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-MONIC · Apply
D	Apply the zero-product property. (first introduced in Math 21) SK-FACTOR-ZERO-PRODUCT · Deepen
I	Determine real and complex zeros using an appropriate method. SK-POLY-ZEROS · Introduce

M21-QUAD-004 — I can solve a quadratic equation using the square-root property.

TEXTBOOK SECTIONS 9.1 Solve Quadratic Equations Using the Square Root Property

SUPPORTING SKILLS

I	Apply the square-root property with both signs. SK-QUAD-SQUARE-ROOT · Introduce
A	Extract perfect-power factors from a radical. (first introduced in Math 21) SK-RAD-FACTOR-PERFECT · Apply

M21-QUAD-005 — I can solve a quadratic equation by completing the square.

TEXTBOOK SECTIONS 9.2 Solve Quadratic Equations by Completing the Square

SUPPORTING SKILLS

A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Create a perfect-square trinomial by completing the square. SK-QUAD-COMPLETE-SQUARE · Introduce

Math 21 In-Depth Curriculum

Algebra and its Functions

TEXTBOOK | OpenStax Intermediate Algebra 2e (Marecek and Mathis, 2020)

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M21-QUAD-006 — I can solve a quadratic equation using the quadratic formula.

TEXTBOOK SECTIONS 9.3 Solve Quadratic Equations Using the Quadratic Formula

SUPPORTING SKILLS

I	Use the discriminant to predict the number and type of solutions. SK-QUAD-DISCRIMINANT · Introduce
I	Substitute coefficients accurately into the quadratic formula. SK-QUAD-FORMULA · Introduce
A	Extract perfect-power factors from a radical. (first introduced in Math 21) SK-RAD-FACTOR-PERFECT · Apply

M21-QUAD-007 — I can choose an efficient method for solving a quadratic equation.

TEXTBOOK SECTIONS 9.1 Solve Quadratic Equations Using the Square Root Property; 9.2 Solve Quadratic Equations by Completing the Square; 9.3 Solve Quadratic Equations Using the Quadratic Formula; 9.4 Solve Equations in Quadratic Form

SUPPORTING SKILLS

A	Apply the zero-product property. (first introduced in Math 21) SK-FACTOR-ZERO-PRODUCT · Apply
A	Create a perfect-square trinomial by completing the square. (first introduced in Math 21) SK-QUAD-COMPLETE-SQUARE · Apply
A	Substitute coefficients accurately into the quadratic formula. (first introduced in Math 21) SK-QUAD-FORMULA · Apply
I	Select an efficient quadratic-solving method from equation structure. SK-QUAD-METHOD-SELECT · Introduce
A	Apply the square-root property with both signs. (first introduced in Math 21) SK-QUAD-SQUARE-ROOT · Apply

M21-QUAD-008 — I can graph a quadratic function using its vertex, intercepts, and symmetry.

TEXTBOOK SECTIONS 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations

SUPPORTING SKILLS

A	Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method. (first introduced in Math 21) SK-QUAD-FORM-CONVERT · Apply
D	Determine x- and y-intercepts of a quadratic function. (first introduced in Math 21) SK-QUAD-INTERCEPTS · Deepen
D	Determine the vertex and axis of symmetry. (first introduced in Math 21) SK-QUAD-VERTEX · Deepen
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

M21-QUAD-009 — I can create and interpret a quadratic model for a projectile, area, or optimization problem.

TEXTBOOK SECTIONS 9.5 Solve Applications of Quadratic Equations

SUPPORTING SKILLS

D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Apply
I	Interpret vertex and zeros in an application. SK-QUAD-MODEL-FEATURES · Introduce
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

SKILL PROGRESSION

I Introduce R Reinforce D Deepen A Apply

Extension Content

EXTENSION

UNIT TEXTBOOK COVERAGE

9.3 Solve Quadratic Equations Using the Quadratic Formula

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M21-EXT-001 **I can perform basic arithmetic with complex numbers.**

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS	
I	Simplify integer powers of the imaginary unit. SK-CPLX-I-POWERS · Introduce
I	Add, subtract, and multiply complex numbers. SK-CPLX-OPERATE · Introduce

M21-EXT-002 **I can express the nonreal solutions of a quadratic equation using complex numbers.**

TEXTBOOK SECTIONS 9.3 Solve Quadratic Equations Using the Quadratic Formula

SUPPORTING SKILLS	
A	Simplify integer powers of the imaginary unit. (first introduced in Math 21) SK-CPLX-I-POWERS · Apply
I	Express a quadratic solution with a negative discriminant in complex form. SK-CPLX-QUAD-SOLUTION · Introduce
A	Use the discriminant to predict the number and type of solutions. (first introduced in Math 21) SK-QUAD-DISCRIMINANT · Apply

Math 22

Geometry

Math 22 Course at a Glance

Geometry

Review Content

REVIEW

LEARNING OBJECTIVES

M22-REV-001	I can solve a linear equation arising from a geometric relationship.
M22-REV-002	I can solve a proportion and interpret the resulting scale factor.
M22-REV-003	I can simplify an exact square root and approximate it when needed.
M22-REV-004	I can calculate distance, midpoint, and slope on the coordinate plane.
M22-REV-005	I can substitute measurements into a formula with consistent units.

Geometry Basics

REQUIRED

LEARNING OBJECTIVES

M22-BAS-001	I can name and represent points, lines, planes, segments, and rays using correct notation.
M22-BAS-002	I can identify collinear, coplanar, intersecting, and concurrent figures.
M22-BAS-003	I can calculate segment lengths using betweenness and the segment-addition postulate.
M22-BAS-004	I can determine a midpoint or use a segment bisector relationship.
M22-BAS-005	I can classify and measure angles.
M22-BAS-006	I can calculate angle measures using angle addition and angle bisectors.
M22-BAS-007	I can identify and use complementary, supplementary, vertical, and linear-pair relationships.
M22-BAS-008	I can distinguish a definition, postulate, theorem, converse, and counterexample.
M22-BAS-009	I can determine angle relationships formed by parallel lines and a transversal.
M22-BAS-010	I can use angle relationships to determine whether lines are parallel.
M22-BAS-011	I can solve problems involving parallel, perpendicular, vertical, or horizontal lines.
M22-BAS-012	I can construct a logical geometric argument from stated facts and accepted results.

Triangle Congruence and Proof

REQUIRED

LEARNING OBJECTIVES

M22-CON-001	I can classify triangles by sides and angles.
M22-CON-002	I can use triangle angle-sum and exterior-angle relationships.
M22-CON-003	I can write a congruence statement with corresponding vertices in the correct order.
M22-CON-004	I can determine whether triangles are congruent by SAS.
M22-CON-005	I can determine whether triangles are congruent by SSS.
M22-CON-006	I can determine whether right triangles are congruent by HL.
M22-CON-007	I can determine whether triangles are congruent by ASA or AAS.
M22-CON-008	I can apply isosceles and equilateral triangle theorems and their converses.
M22-CON-009	I can use corresponding parts of congruent triangles after proving congruence.
M22-CON-010	I can write a structured proof involving triangle congruence.

Quadrilaterals, Perimeter, and Area

REQUIRED

LEARNING OBJECTIVES

M22-QUAD-001	I can calculate interior or exterior angle measures of a polygon.
M22-QUAD-002	I can calculate perimeter and area of common polygons.
M22-QUAD-003	I can apply properties of parallelograms to determine unknown measures.
M22-QUAD-004	I can determine whether a quadrilateral is a parallelogram.
M22-QUAD-005	I can apply properties of rectangles, rhombi, and squares.
M22-QUAD-006	I can classify a quadrilateral from its stated or marked properties.
M22-QUAD-007	I can apply trapezoid and trapezoid-midsegment relationships.
M22-QUAD-008	I can apply properties of kites.
M22-QUAD-009	I can calculate the perimeter or area of a composite polygon.
M22-QUAD-010	I can justify a quadrilateral conclusion using definitions and theorems.

Math 22 Course at a Glance

Geometry

Similarity and Proportional Reasoning

REQUIRED

LEARNING OBJECTIVES

M22-SIM-001	I can determine image lengths and coordinates under a dilation.
M22-SIM-002	I can identify the scale factor of a dilation or similarity relationship.
M22-SIM-003	I can write a similarity statement with corresponding vertices in the correct order.
M22-SIM-004	I can use proportional corresponding sides to solve similar-polygon problems.
M22-SIM-005	I can relate similarity scale factor to perimeter and area scale factors.
M22-SIM-006	I can prove triangles similar by AA, SSS, or SAS.
M22-SIM-007	I can apply triangle proportionality theorems and their converses.
M22-SIM-008	I can solve and justify an indirect-measurement or other similarity application.

Right Triangles

REQUIRED

LEARNING OBJECTIVES

M22-RGT-001	I can apply the Pythagorean theorem to determine a missing side.
M22-RGT-002	I can use the converse of the Pythagorean theorem to classify a triangle.
M22-RGT-003	I can apply triangle-inequality relationships.
M22-RGT-004	I can determine exact side lengths in 45-45-90 and 30-60-90 triangles.
M22-RGT-005	I can apply right-triangle similarity relationships.
M22-RGT-006	I can use geometric-mean relationships created by an altitude to the hypotenuse.
M22-RGT-007	I can select sine, cosine, or tangent for a right-triangle problem.
M22-RGT-008	I can determine a missing side or angle of a right triangle.
M22-RGT-009	I can solve a contextual right-triangle problem.
M22-RGT-010	I can justify a right-triangle solution using an appropriate theorem or ratio.

Circles

REQUIRED

LEARNING OBJECTIVES

M22-CIR-001	I can identify and name radii, diameters, chords, secants, tangents, arcs, sectors, and segments.
M22-CIR-002	I can calculate circumference and arc length.
M22-CIR-003	I can calculate the area of a circle, sector, or segment.
M22-CIR-004	I can determine angle and arc measures involving central and inscribed angles.
M22-CIR-005	I can apply relationships among chords, arcs, and distances from the center.
M22-CIR-006	I can apply tangent-radius and tangent-segment relationships.
M22-CIR-007	I can determine angle measures formed by chords, secants, or tangents.
M22-CIR-008	I can determine segment lengths formed by intersecting chords.
M22-CIR-009	I can determine segment lengths formed by secants or tangents.
M22-CIR-010	I can solve a multi-step circle problem and justify each relationship used.

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Review Content

REVIEW

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M22-REV-001 I can solve a linear equation arising from a geometric relationship.

SUPPORTING SKILLS

A	Check a proposed solution in the original equation. (first introduced in Math 12) SK-ALG-CHECK-SOLUTION · Apply
R	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Reinforce
A	Translate quantities and relationships into an equation. (first introduced in Math 12) SK-ALG-TRANSLATE-EQUATION · Apply

M22-REV-002 I can solve a proportion and interpret the resulting scale factor.

SUPPORTING SKILLS

R	Use cross multiplication when two ratios form a proportion. (inherited from middle school) SK-ALG-CROSS-MULTIPLY · Reinforce
I	Calculate and interpret a similarity or dilation scale factor. SK-SIM-SCALE-FACTOR · Introduce

M22-REV-003 I can simplify an exact square root and approximate it when needed.

SUPPORTING SKILLS

I	Distinguish exact values from measured or rounded approximations. SK-GEO-EXACT-APPROX · Introduce
R	Extract perfect-power factors from a radical. (first introduced in Math 21) SK-RAD-FACTOR-PERFECT · Reinforce

M22-REV-004 I can calculate distance, midpoint, and slope on the coordinate plane.

SUPPORTING SKILLS

R	Calculate horizontal or vertical distance from the difference of coordinates. (inherited horizontal/vertical coordinate distance) SK-COORD-DISTANCE-AXIS · Reinforce
I	Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula. SK-COORD-DISTANCE-PYTHAGOREAN · Introduce
R	Calculate the midpoint of a segment from endpoint coordinates. (inherited from earlier mathematics) SK-COORD-MIDPOINT · Reinforce
R	Calculate slope from two points. (first introduced in Math 12) SK-LIN-SLOPE-POINTS · Reinforce

M22-REV-005 I can substitute measurements into a formula with consistent units.

SUPPORTING SKILLS

I	Select and report appropriate linear, square, or angular units. SK-GEO-UNITS · Introduce
A	Apply the order of operations to a numerical expression. (inherited from middle school) SK-NUM-ORDER-OPERATIONS · Apply

Math 22 In-Depth Curriculum

Geometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Geometry Basics

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M22-BAS-001 — I can name and represent points, lines, planes, segments, and rays using correct notation.

SUPPORTING SKILLS

I	Add congruence, parallel, perpendicular, and angle markings to a diagram. SK-GEO-DIAGRAM-MARK · Introduce
I	Read and write geometric notation correctly. SK-GEO-NOTATION · Introduce

M22-BAS-002 — I can identify collinear, coplanar, intersecting, and concurrent figures.

SUPPORTING SKILLS

I	Distinguish stated or marked facts from visual appearance. SK-GEO-DIAGRAM-INFER · Introduce
A	Read and write geometric notation correctly. (first introduced in Math 22) SK-GEO-NOTATION · Apply

M22-BAS-003 — I can calculate segment lengths using betweenness and the segment-addition postulate.

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
A	Translate quantities and relationships into an equation. (first introduced in Math 12) SK-ALG-TRANSLATE-EQUATION · Apply
I	Translate segment addition into an equation. SK-GEO-SEGMENT-ADDITION · Introduce

M22-BAS-004 — I can determine a midpoint or use a segment bisector relationship.

SUPPORTING SKILLS

I	Use a bisector to create equal measures. SK-GEO-BISECTOR · Introduce
I	Use a midpoint to create equal segment lengths. SK-GEO-MIDPOINT · Introduce
A	Translate segment addition into an equation. (first introduced in Math 22) SK-GEO-SEGMENT-ADDITION · Apply

M22-BAS-005 — I can classify and measure angles.

SUPPORTING SKILLS

I	Identify complementary, supplementary, vertical, and linear-pair relationships. SK-ANG-PAIR-IDENTIFY · Introduce
I	Measure or calculate an angle accurately. SK-GEO-ANGLE-MEASURE · Introduce

M22-BAS-006 — I can calculate angle measures using angle addition and angle bisectors.

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Translate angle addition into an equation. SK-GEO-ANGLE-ADDITION · Introduce
A	Use a bisector to create equal measures. (first introduced in Math 22) SK-GEO-BISECTOR · Apply

Math 22 In-Depth Curriculum

Geometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M22-BAS-007 — I can identify and use complementary, supplementary, vertical, and linear-pair relationships.

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
A	Translate quantities and relationships into an equation. (first introduced in Math 12) SK-ALG-TRANSLATE-EQUATION · Apply
D	Identify complementary, supplementary, vertical, and linear-pair relationships. (first introduced in Math 22) SK-ANG-PAIR-IDENTIFY · Deepen

M22-BAS-008 — I can distinguish a definition, postulate, theorem, converse, and counterexample.

SUPPORTING SKILLS

I	Recognize and apply the converse of a conditional statement. SK-PROOF-CONVERSE · Introduce
I	Disprove a universal statement with a counterexample. SK-PROOF-COUNTEREXAMPLE · Introduce
I	Apply a geometric definition as justification. SK-PROOF-DEFINITION · Introduce
I	Select a theorem that connects known facts to a target statement. SK-PROOF-THEOREM-SELECT · Introduce

M22-BAS-009 — I can determine angle relationships formed by parallel lines and a transversal.

SUPPORTING SKILLS

I	Use parallel-line angle relationships. SK-ANG-PARALLEL-APPLY · Introduce
I	Identify corresponding, alternate, and same-side angle pairs. SK-ANG-TRANSVERSAL-IDENTIFY · Introduce

M22-BAS-010 — I can use angle relationships to determine whether lines are parallel.

SUPPORTING SKILLS

I	Use angle relationships to prove lines parallel. SK-ANG-PARALLEL-CONVERSE · Introduce
A	Identify corresponding, alternate, and same-side angle pairs. (first introduced in Math 22) SK-ANG-TRANSVERSAL-IDENTIFY · Apply

M22-BAS-011 — I can solve problems involving parallel, perpendicular, vertical, or horizontal lines.

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
A	Use parallel-line angle relationships. (first introduced in Math 22) SK-ANG-PARALLEL-APPLY · Apply
I	Use right-angle relationships created by perpendicular lines. SK-LIN-PERPENDICULAR-ANGLE · Introduce

M22-BAS-012 — I can construct a logical geometric argument from stated facts and accepted results.

SUPPORTING SKILLS

I	Translate givens into diagram markings and statements. SK-PROOF-GIVEN · Introduce
I	Organize statements and reasons into a valid proof sequence. SK-PROOF-STRUCTURE · Introduce
A	Select a theorem that connects known facts to a target statement. (first introduced in Math 22) SK-PROOF-THEOREM-SELECT · Apply

Math 22 In-Depth Curriculum

Geometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Triangle Congruence and Proof

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M22-CON-001 — I can classify triangles by sides and angles.

SUPPORTING SKILLS

A	Add congruence, parallel, perpendicular, and angle markings to a diagram. (first introduced in Math 22) SK-GEO-DIAGRAM-MARK · Apply
I	Classify a triangle by side lengths and angle measures. SK-TRI-CLASSIFY · Introduce

M22-CON-002 — I can use triangle angle-sum and exterior-angle relationships.

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Apply the triangle angle-sum theorem. SK-TRI-ANGLE-SUM · Introduce
I	Apply the triangle exterior-angle theorem. SK-TRI-EXTERIOR-ANGLE · Introduce

M22-CON-003 — I can write a congruence statement with corresponding vertices in the correct order.

SUPPORTING SKILLS

I	Match corresponding vertices, sides, and angles in order. SK-GEO-CORRESPONDENCE · Introduce
I	Write congruence statements in corresponding order. SK-PROOF-CONGRUENCE-ORDER · Introduce

M22-CON-004 — I can determine whether triangles are congruent by SAS.

SUPPORTING SKILLS

A	Match corresponding vertices, sides, and angles in order. (first introduced in Math 22) SK-GEO-CORRESPONDENCE · Apply
D	Select a theorem that connects known facts to a target statement. (first introduced in Math 22) SK-PROOF-THEOREM-SELECT · Deepen
I	Select SAS, SSS, HL, ASA, or AAS from marked information. SK-TRI-CONGRUENCE-SELECT · Introduce

M22-CON-005 — I can determine whether triangles are congruent by SSS.

SUPPORTING SKILLS

A	Match corresponding vertices, sides, and angles in order. (first introduced in Math 22) SK-GEO-CORRESPONDENCE · Apply
D	Select SAS, SSS, HL, ASA, or AAS from marked information. (first introduced in Math 22) SK-TRI-CONGRUENCE-SELECT · Deepen

M22-CON-006 — I can determine whether right triangles are congruent by HL.

SUPPORTING SKILLS

A	Use right-angle relationships created by perpendicular lines. (first introduced in Math 22) SK-LIN-PERPENDICULAR-ANGLE · Apply
D	Select SAS, SSS, HL, ASA, or AAS from marked information. (first introduced in Math 22) SK-TRI-CONGRUENCE-SELECT · Deepen

M22-CON-007 — I can determine whether triangles are congruent by ASA or AAS.

SUPPORTING SKILLS

A	Apply the triangle angle-sum theorem. (first introduced in Math 22) SK-TRI-ANGLE-SUM · Apply
D	Select SAS, SSS, HL, ASA, or AAS from marked information. (first introduced in Math 22) SK-TRI-CONGRUENCE-SELECT · Deepen

Math 22 In-Depth Curriculum

Geometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M22-CON-008 — I can apply isosceles and equilateral triangle theorems and their converses.

SUPPORTING SKILLS

D	Recognize and apply the converse of a conditional statement. (first introduced in Math 22) SK-PROOF-CONVERSE · Deepen
I	Apply isosceles-triangle theorems and converses. SK-TRI-ISOSCELES · Introduce

M22-CON-009 — I can use corresponding parts of congruent triangles after proving congruence.

SUPPORTING SKILLS

A	Write congruence statements in corresponding order. (first introduced in Math 22) SK-PROOF-CONGRUENCE-ORDER · Apply
I	Use corresponding parts only after triangle congruence is established. SK-PROOF-CPCTC · Introduce

M22-CON-010 — I can write a structured proof involving triangle congruence.

SUPPORTING SKILLS

A	Use corresponding parts only after triangle congruence is established. (first introduced in Math 22) SK-PROOF-CPCTC · Apply
A	Translate givens into diagram markings and statements. (first introduced in Math 22) SK-PROOF-GIVEN · Apply
D	Organize statements and reasons into a valid proof sequence. (first introduced in Math 22) SK-PROOF-STRUCTURE · Deepen
A	Select SAS, SSS, HL, ASA, or AAS from marked information. (first introduced in Math 22) SK-TRI-CONGRUENCE-SELECT · Apply

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Quadrilaterals, Perimeter, and Area

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M22-QUA D-001 — I can calculate interior or exterior angle measures of a polygon.

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Calculate polygon interior or exterior angle sums. SK-POLY-ANGLE-SUM · Introduce

M22-QUA D-002 — I can calculate perimeter and area of common polygons.

SUPPORTING SKILLS

I	Add boundary lengths without omitting or duplicating sides. SK-GEO-PERIMETER · Introduce
D	Select and report appropriate linear, square, or angular units. (first introduced in Math 22) SK-GEO-UNITS · Deepen
A	Apply the order of operations to a numerical expression. (inherited from middle school) SK-NUM-ORDER-OPERATIONS · Apply

M22-QUA D-003 — I can apply properties of parallelograms to determine unknown measures.

SUPPORTING SKILLS

A	Translate quantities and relationships into an equation. (first introduced in Math 12) SK-ALG-TRANSLATE-EQUATION · Apply
I	Apply parallelogram properties and tests. SK-QUAD-PARALLELOGRAM · Introduce

M22-QUA D-004 — I can determine whether a quadrilateral is a parallelogram.

SUPPORTING SKILLS

A	Recognize and apply the converse of a conditional statement. (first introduced in Math 22) SK-PROOF-CONVERSE · Apply
D	Apply parallelogram properties and tests. (first introduced in Math 22) SK-QUAD-PARALLELOGRAM · Deepen

M22-QUA D-005 — I can apply properties of rectangles, rhombi, and squares.

SUPPORTING SKILLS

I	Use the inclusive hierarchy of quadrilateral classifications. SK-QUAD-HIERARCHY · Introduce
I	Distinguish and apply rectangle, rhombus, and square properties. SK-QUAD-RECT-RHOM-SQUARE · Introduce

M22-QUA D-006 — I can classify a quadrilateral from its stated or marked properties.

SUPPORTING SKILLS

A	Distinguish stated or marked facts from visual appearance. (first introduced in Math 22) SK-GEO-DIAGRAM-INFER · Apply
D	Use the inclusive hierarchy of quadrilateral classifications. (first introduced in Math 22) SK-QUAD-HIERARCHY · Deepen

M22-QUA D-007 — I can apply trapezoid and trapezoid-midsegment relationships.

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Apply trapezoid and midsegment relationships. SK-QUAD-TRAPEZOID · Introduce

Math 22 In-Depth Curriculum

Geometry

SKILL PROGRESSION

I	Introduce
R	Reinforce
D	Deepen
A	Apply

M22-QUAD-008 — I can apply properties of kites.

SUPPORTING SKILLS

I	Apply kite properties. SK-QUAD-KITE · Introduce

M22-QUAD-009 — I can calculate the perimeter or area of a composite polygon.

SUPPORTING SKILLS

I	Decompose a composite region into familiar shapes. SK-GEO-AREA-DECOMPOSE · Introduce
D	Add boundary lengths without omitting or duplicating sides. (first introduced in Math 22) SK-GEO-PERIMETER · Deepen
A	Select and report appropriate linear, square, or angular units. (first introduced in Math 22) SK-GEO-UNITS · Apply

M22-QUAD-010 — I can justify a quadrilateral conclusion using definitions and theorems.

SUPPORTING SKILLS

D	Apply a geometric definition as justification. (first introduced in Math 22) SK-PROOF-DEFINITION · Deepen
A	Organize statements and reasons into a valid proof sequence. (first introduced in Math 22) SK-PROOF-STRUCTURE · Apply
D	Select a theorem that connects known facts to a target statement. (first introduced in Math 22) SK-PROOF-THEOREM-SELECT · Deepen

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Similarity and Proportional Reasoning

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M22-SIM-01 — I can determine image lengths and coordinates under a dilation.

SUPPORTING SKILLS

A	Match corresponding vertices, sides, and angles in order. (first introduced in Math 22) SK-GEO-CORRESPONDENCE · Apply
D	Calculate and interpret a similarity or dilation scale factor. (first introduced in Math 22) SK-SIM-SCALE-FACTOR · Deepen

M22-SIM-02 — I can identify the scale factor of a dilation or similarity relationship.

SUPPORTING SKILLS

A	Use cross multiplication when two ratios form a proportion. (inherited from middle school) SK-ALG-CROSS-MULTIPLY · Apply
D	Calculate and interpret a similarity or dilation scale factor. (first introduced in Math 22) SK-SIM-SCALE-FACTOR · Deepen

M22-SIM-03 — I can write a similarity statement with corresponding vertices in the correct order.

SUPPORTING SKILLS

D	Match corresponding vertices, sides, and angles in order. (first introduced in Math 22) SK-GEO-CORRESPONDENCE · Deepen
I	Write similarity statements in corresponding order. SK-PROOF-SIMILARITY-ORDER · Introduce

M22-SIM-04 — I can use proportional corresponding sides to solve similar-polygon problems.

SUPPORTING SKILLS

A	Use cross multiplication when two ratios form a proportion. (inherited from middle school) SK-ALG-CROSS-MULTIPLY · Apply
A	Match corresponding vertices, sides, and angles in order. (first introduced in Math 22) SK-GEO-CORRESPONDENCE · Apply
I	Set up proportions using corresponding sides. SK-SIM-PROPORTION · Introduce

M22-SIM-05 — I can relate similarity scale factor to perimeter and area scale factors.

SUPPORTING SKILLS

I	Relate length scale factor to perimeter and area factors. SK-SIM-PERIMETER-AREA · Introduce
A	Calculate and interpret a similarity or dilation scale factor. (first introduced in Math 22) SK-SIM-SCALE-FACTOR · Apply

M22-SIM-06 — I can prove triangles similar by AA, SSS, or SAS.

SUPPORTING SKILLS

A	Write similarity statements in corresponding order. (first introduced in Math 22) SK-PROOF-SIMILARITY-ORDER · Apply
D	Organize statements and reasons into a valid proof sequence. (first introduced in Math 22) SK-PROOF-STRUCTURE · Deepen
I	Recognize AA triangle similarity. SK-SIM-AA · Introduce
I	Test SSS or SAS triangle similarity. SK-SIM-SSS-SAS · Introduce

Math 22 In-Depth Curriculum

Geometry

SKILL PROGRESSION

I Introduce | R Reinforce | D Deepen | A Apply

M22-SIM-007 I can apply triangle proportionality theorems and their converses.

SUPPORTING SKILLS

A	Recognize and apply the converse of a conditional statement. (first introduced in Math 22) SK-PROOF-CONVERSE · Apply
I	Apply triangle proportionality relationships. SK-SIM-TRI-PROPORTIONALITY · Introduce

M22-SIM-008 I can solve and justify an indirect-measurement or other similarity application.

SUPPORTING SKILLS

A	Select a theorem that connects known facts to a target statement. (first introduced in Math 22) SK-PROOF-THEOREM-SELECT · Apply
D	Set up proportions using corresponding sides. (first introduced in Math 22) SK-SIM-PROPORTION · Deepen
A	Calculate and interpret a similarity or dilation scale factor. (first introduced in Math 22) SK-SIM-SCALE-FACTOR · Apply

Math 22 In-Depth Curriculum

Geometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Right Triangles

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M22-RGT-001 — I can apply the Pythagorean theorem to determine a missing side.

SUPPORTING SKILLS

A	Extract perfect-power factors from a radical. (first introduced in Math 21) SK-RAD-FACTOR-PERFECT · Apply
I	Apply the Pythagorean theorem or its converse. SK-TRI-PYTHAGOREAN · Introduce

M22-RGT-002 — I can use the converse of the Pythagorean theorem to classify a triangle.

SUPPORTING SKILLS

A	Recognize and apply the converse of a conditional statement. (first introduced in Math 22) SK-PROOF-CONVERSE · Apply
D	Apply the Pythagorean theorem or its converse. (first introduced in Math 22) SK-TRI-PYTHAGOREAN · Deepen

M22-RGT-003 — I can apply triangle-inequality relationships.

SUPPORTING SKILLS

A	Select a theorem that connects known facts to a target statement. (first introduced in Math 22) SK-PROOF-THEOREM-SELECT · Apply
A	Classify a triangle by side lengths and angle measures. (first introduced in Math 22) SK-TRI-CLASSIFY · Apply

M22-RGT-004 — I can determine exact side lengths in 45-45-90 and 30-60-90 triangles.

SUPPORTING SKILLS

A	Distinguish exact values from measured or rounded approximations. (first introduced in Math 22) SK-GEO-EXACT-APPROX · Apply
R	Extract perfect-power factors from a radical. (first introduced in Math 21) SK-RAD-FACTOR-PERFECT · Reinforce
I	Recall special-right-triangle ratios. SK-TRI-SPECIAL-RATIOS · Introduce

M22-RGT-005 — I can apply right-triangle similarity relationships.

SUPPORTING SKILLS

A	Recognize AA triangle similarity. (first introduced in Math 22) SK-SIM-AA · Apply
D	Set up proportions using corresponding sides. (first introduced in Math 22) SK-SIM-PROPORTION · Deepen

M22-RGT-006 — I can use geometric-mean relationships created by an altitude to the hypotenuse.

SUPPORTING SKILLS

A	Set up proportions using corresponding sides. (first introduced in Math 22) SK-SIM-PROPORTION · Apply
I	Apply geometric-mean relationships in a right triangle split by an altitude. SK-TRI-GEOMETRIC-MEAN · Introduce

Math 22 In-Depth Curriculum

Geometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M22-RGT-007 — I can select sine, cosine, or tangent for a right-triangle problem.

SUPPORTING SKILLS

A	Add congruence, parallel, perpendicular, and angle markings to a diagram. (first introduced in Math 22) SK-GEO-DIAGRAM-MARK · Apply
I	Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. SK-TRI-TRIG-RATIO · Introduce
I	Label sides relative to a selected angle. SK-TRIG-TRIANGLE-LABEL · Introduce

M22-RGT-008 — I can determine a missing side or angle of a right triangle.

SUPPORTING SKILLS

A	Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. (first introduced in Math 22) SK-TRI-TRIG-RATIO · Apply
I	Use an inverse trigonometric function to determine a missing angle. SK-TRI-TRIG-SOLVE-ANGLE · Introduce
I	Use a trigonometric ratio to determine a missing side. SK-TRI-TRIG-SOLVE-SIDE · Introduce

M22-RGT-009 — I can solve a contextual right-triangle problem.

SUPPORTING SKILLS

A	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Apply
D	Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. (first introduced in Math 22) SK-TRI-TRIG-RATIO · Deepen
A	Use an inverse trigonometric function to determine a missing angle. (first introduced in Math 22) SK-TRI-TRIG-SOLVE-ANGLE · Apply
A	Use a trigonometric ratio to determine a missing side. (first introduced in Math 22) SK-TRI-TRIG-SOLVE-SIDE · Apply

M22-RGT-010 — I can justify a right-triangle solution using an appropriate theorem or ratio.

SUPPORTING SKILLS

D	Select a theorem that connects known facts to a target statement. (first introduced in Math 22) SK-PROOF-THEOREM-SELECT · Deepen
A	Apply the Pythagorean theorem or its converse. (first introduced in Math 22) SK-TRI-PYTHAGOREAN · Apply
A	Recall special-right-triangle ratios. (first introduced in Math 22) SK-TRI-SPECIAL-RATIOS · Apply
A	Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. (first introduced in Math 22) SK-TRI-TRIG-RATIO · Apply

Math 22 In-Depth Curriculum

Geometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Circles

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M22-CIR-001 — I can identify and name radii, diameters, chords, secants, tangents, arcs, sectors, and segments.

SUPPORTING SKILLS	
I	Identify circle parts and relationships from a diagram. SK-CIR-VOCAB · Introduce
A	Read and write geometric notation correctly. (first introduced in Math 22) SK-GEO-NOTATION · Apply

M22-CIR-002 — I can calculate circumference and arc length.

SUPPORTING SKILLS	
I	Relate a central angle to a fraction of a circle. SK-CIR-ARC-FRACTION · Introduce
I	Calculate arc length from radius and central angle. SK-CIR-ARC-LENGTH · Introduce
I	Calculate circumference from radius or diameter. SK-CIR-CIRCUMFERENCE · Introduce
A	Select and report appropriate linear, square, or angular units. (first introduced in Math 22) SK-GEO-UNITS · Apply

M22-CIR-003 — I can calculate the area of a circle, sector, or segment.

SUPPORTING SKILLS	
I	Calculate the area of a circle. SK-CIR-AREA · Introduce
I	Calculate sector area from radius and central angle. SK-CIR-SECTOR-AREA · Introduce
I	Calculate segment area by subtracting a triangle from a sector. SK-CIR-SEGMENT-AREA · Introduce
A	Select and report appropriate linear, square, or angular units. (first introduced in Math 22) SK-GEO-UNITS · Apply

M22-CIR-004 — I can determine angle and arc measures involving central and inscribed angles.

SUPPORTING SKILLS	
A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
D	Relate a central angle to a fraction of a circle. (first introduced in Math 22) SK-CIR-ARC-FRACTION · Deepen
I	Relate an inscribed angle to its intercepted arc. SK-CIR-INSCRIBED-ANGLE · Introduce

M22-CIR-005 — I can apply relationships among chords, arcs, and distances from the center.

SUPPORTING SKILLS	
I	Apply equal-chord, equal-arc, and center-distance relationships. SK-CIR-CHORD-RELATION · Introduce
A	Identify circle parts and relationships from a diagram. (first introduced in Math 22) SK-CIR-VOCAB · Apply

M22-CIR-006 — I can apply tangent-radius and tangent-segment relationships.

SUPPORTING SKILLS	
I	Select and apply an intersecting-segment product relationship. SK-CIR-SEGMENT-PRODUCT · Introduce
I	Use the perpendicular relationship between a radius and tangent. SK-CIR-TANGENT-RADIUS · Introduce

M22-CIR-007 — I can determine angle measures formed by chords, secants, or tangents.

SUPPORTING SKILLS	
I	Select the correct angle formula for chords, secants, or tangents. SK-CIR-ANGLE-RELATION · Introduce
A	Identify circle parts and relationships from a diagram. (first introduced in Math 22) SK-CIR-VOCAB · Apply

Math 22 In-Depth Curriculum

Geometry

SKILL PROGRESSION

I Introduce | R Reinforce | D Deepen | A Apply

M22-CIR-008 — I can determine segment lengths formed by intersecting chords.

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
I	Select and apply an intersecting-segment product relationship. SK-CIR-SEGMENT-PRODUCT · Introduce

M22-CIR-009 — I can determine segment lengths formed by secants or tangents.

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
D	Select and apply an intersecting-segment product relationship. (first introduced in Math 22) SK-CIR-SEGMENT-PRODUCT · Deepen

M22-CIR-010 — I can solve a multi-step circle problem and justify each relationship used.

SUPPORTING SKILLS

A	Select the correct angle formula for chords, secants, or tangents. (first introduced in Math 22) SK-CIR-ANGLE-RELATION · Apply
A	Select and apply an intersecting-segment product relationship. (first introduced in Math 22) SK-CIR-SEGMENT-PRODUCT · Apply
A	Identify circle parts and relationships from a diagram. (first introduced in Math 22) SK-CIR-VOCAB · Apply
D	Select a theorem that connects known facts to a target statement. (first introduced in Math 22) SK-PROOF-THEOREM-SELECT · Deepen

Math 31

Advanced Algebra

Math 31 Course at a Glance

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

Review Content

REVIEW — SECTIONS No mapped textbook section

LEARNING OBJECTIVES

M31-REV-001	I can calculate a percentage, percent increase, or percent decrease.
M31-REV-002	I can simplify an expression using exponent properties.
M31-REV-003	I can use polynomial vocabulary to distinguish terms, factors, coefficients, degree, and polynomial structure.
M31-REV-004	I can expand a product of two binomials.

Linear Functions and Function Notation

REQUIRED — SECTIONS 3.1, 3.2, 3.3, 4.1, 4.2

LEARNING OBJECTIVES

M31-LIN-001	I can interpret function notation as a statement about input and output.
M31-LIN-002	I can determine whether a graph or table represents a function.
M31-LIN-003	I can evaluate a function or solve for an input using a graph, table, or formula.
M31-LIN-004	I can identify independent and dependent variables in a relationship.
M31-LIN-005	I can calculate average rate of change from two points.
M31-LIN-006	I can interpret interval notation and identify increasing or decreasing intervals.
M31-LIN-007	I can determine total change from an average rate of change over an interval.
M31-LIN-008	I can determine and interpret the units of an average rate of change.
M31-LIN-009	I can identify a linear function from a table, graph, or formula.
M31-LIN-010	I can interpret the slope and intercept of a linear model.
M31-LIN-011	I can determine the intersection of two linear relationships graphically or algebraically.
M31-LIN-012	I can write equations of horizontal, vertical, parallel, and perpendicular lines through a specified point.

Function Behavior and Structure

REQUIRED — SECTIONS 3.1, 3.2, 3.3, 3.4, 3.5, 3.7

LEARNING OBJECTIVES

M31-FUN-001	I can determine whether a relation is a function.
M31-FUN-002	I can evaluate a function from a formula, graph, or table.
M31-FUN-003	I can determine domain and range from a formula, graph, or table.
M31-FUN-004	I can determine algebraic domain restrictions caused by denominators and even roots.
M31-FUN-005	I can graph a piecewise function from its formula.
M31-FUN-006	I can write a piecewise formula that represents a graph.
M31-FUN-007	I can graph a transformed function using shifts, stretches, compressions, and reflections.
M31-FUN-008	I can describe how a transformation affects domain, range, or asymptotes.
M31-FUN-009	I can evaluate a composition from formulas, graphs, or tables.
M31-FUN-010	I can write a formula for a composition of two functions.
M31-FUN-011	I can find an inverse function algebraically.
M31-FUN-012	I can evaluate and interpret an inverse from a graph, formula, or table.
M31-FUN-013	I can determine and interpret the units of an inverse function.
M31-FUN-014	I can explain how the domain and range of a function relate to those of its inverse.
M31-FUN-015	I can determine the range of a function by analyzing the domain of its inverse.
M31-FUN-016	I can verify algebraically that two functions are inverses.
M31-FUN-017	I can interpret the meaning and units of a composition.
M31-FUN-018	I can classify a graph or table as concave up or concave down using changes in average rate of change.
M31-FUN-019	I can determine whether a function is invertible using the horizontal line test.

Function Behavior and Structure

M31-FUN-020	I can graph an inverse function as a reflection across the line `y = x`.

Math 31 Course at a Glance

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

Quadratic Functions and Equations

REQUIRED SECTIONS 2.5, 5.1

LEARNING OBJECTIVES

M31-QUAD-001	I can identify the vertex, intercepts, axis of symmetry, and concavity of a quadratic function.
M31-QUAD-002	I can connect standard, factored, and vertex forms to features of a quadratic graph.
M31-QUAD-003	I can rewrite a quadratic function in a form that reveals a requested feature.
M31-QUAD-004	I can solve a quadratic equation by factoring or using the quadratic formula.
M31-QUAD-005	I can graph a quadratic function using its algebraic structure and key features.
M31-QUAD-006	I can construct a quadratic function from specified zeros, a vertex, or graphical information.
M31-QUAD-007	I can create and interpret a quadratic trajectory model.

Exponential Functions and Models

REQUIRED SECTIONS 6.1, 6.2, 6.7

LEARNING OBJECTIVES

M31-EXP-001	I can distinguish linear change from exponential change using differences, ratios, or percent change.
M31-EXP-002	I can identify initial value and growth or decay factor in an exponential model.
M31-EXP-003	I can write an exponential function from a table, graph, or context.
M31-EXP-004	I can evaluate and interpret an exponential model.
M31-EXP-005	I can calculate growth under periodic compounding.
M31-EXP-006	I can calculate growth under continuous compounding.
M31-EXP-007	I can determine an exponential formula from two points or other sufficient data.
M31-EXP-008	I can determine domain, range, intercepts, and asymptotes of an exponential function.
M31-EXP-009	I can compare the growth rates of two exponential functions.
M31-EXP-010	I can determine the percent growth or decay represented by a continuous exponential model.

Logarithmic Functions and Models

REQUIRED SECTIONS 6.3, 6.4, 6.5, 6.6, 6.7

LEARNING OBJECTIVES

M31-LOG-001	I can convert between logarithmic and exponential forms.
M31-LOG-002	I can evaluate a logarithm using its exponential meaning.
M31-LOG-003	I can expand or condense an expression using product, quotient, and power properties.
M31-LOG-004	I can solve an exponential equation using logarithms.
M31-LOG-005	I can solve a logarithmic equation using exponential form and check domain restrictions.
M31-LOG-006	I can calculate and interpret doubling time or half-life.
M31-LOG-007	I can determine domain, range, intercepts, and asymptotes of a logarithmic function.
M31-LOG-008	I can explain how exponential and logarithmic functions undo one another.
M31-LOG-009	I can solve a multi-step exponential or logarithmic equation.
M31-LOG-010	I can determine an intersection of exponential functions graphically or algebraically.
M31-LOG-011	I can rewrite a discrete exponential model as an equivalent continuous exponential model.
M31-LOG-012	I can solve and interpret an exponential or logarithmic modeling problem.

Extension Content

EXTENSION SECTIONS 3.6

LEARNING OBJECTIVES

M31-EXT-001	I can analyze an absolute-value function using transformations.
M31-EXT-002	I can determine even or odd symmetry from a graph or formula.
M31-EXT-003	I can interpret a contextual logarithmic scale.

SKILL PROGRESSION

I	Introduce
R	Reinforce
D	Deepen
A	Apply

Review Content

REVIEW

UNIT TEXTBOOK COVERAGE

No mapped textbook section

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M31-REV-001 I can calculate a percentage, percent increase, or percent decrease.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS	
R	Calculate a percentage of a quantity or determine what percent one quantity is of another. (inherited from middle school) SK-NUM-PERCENT-CALCULATE · Reinforce
R	Calculate and interpret percent increase or decrease. (inherited from middle school) SK-NUM-PERCENT-CHANGE · Reinforce

M31-REV-002 I can simplify an expression using exponent properties.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS	
R	Rewrite a negative exponent using a reciprocal. (first introduced in Math 21) SK-EXP-NEGATIVE · Reinforce
R	Apply the power rule for exponents. (first introduced in Math 12) SK-EXP-POWER · Reinforce
R	Apply the product rule for exponents. (first introduced in Math 12) SK-EXP-PRODUCT · Reinforce
R	Apply the quotient rule for exponents. (first introduced in Math 12) SK-EXP-QUOTIENT · Reinforce
R	Apply the zero-exponent rule. (first introduced in Math 21) SK-EXP-ZERO · Reinforce

M31-REV-003 I can use polynomial vocabulary to distinguish terms, factors, coefficients, degree, and polynomial structure.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS	
R	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Reinforce

M31-REV-004 I can expand a product of two binomials.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS	
A	Apply the distributive property in either direction. (inherited from middle school) SK-ALG-DISTRIBUTE · Apply
R	Multiply two binomials using distribution. (first introduced in Math 12) SK-POLY-BINOMIAL-MULTIPLY · Reinforce

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Linear Functions and Function Notation

REQUIRED

UNIT TEXTBOOK COVERAGE

3.1 Functions and Function Notation; 3.2 Domain and Range; 3.3 Rates of Change and Behavior of Graphs; 4.1 Linear Functions; 4.2 Modeling with Linear Functions

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M31-LIN-01 I can interpret function notation as a statement about input and output.

TEXTBOOK SECTIONS 3.1 Functions and Function Notation

SUPPORTING SKILLS

D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
A	Identify the input and output represented by function notation. (first introduced in Math 12) SK-FUNC-NOTATION-READ · Apply
I	Use function notation to express and solve an input-output question. SK-FUNC-NOTATION-SOLVE · Introduce

M31-LIN-02 I can determine whether a graph or table represents a function.

TEXTBOOK SECTIONS 3.1 Functions and Function Notation

SUPPORTING SKILLS

D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
D	Apply the vertical line test to a graph. (first introduced in Math 12) SK-FUNC-VERTICAL-LINE · Deepen

M31-LIN-03 I can evaluate a function or solve for an input using a graph, table, or formula.

TEXTBOOK SECTIONS 3.1 Functions and Function Notation

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
A	Use function notation to express and solve an input-output question. (first introduced in Math 31) SK-FUNC-NOTATION-SOLVE · Apply
D	Solve for an input associated with a specified output. (first introduced in Math 21) SK-FUNC-SOLVE-INPUT · Deepen

M31-LIN-04 I can identify independent and dependent variables in a relationship.

TEXTBOOK SECTIONS 3.1 Functions and Function Notation

SUPPORTING SKILLS

I	Identify independent and dependent variables. SK-FUNC-INDEPENDENT-DEPENDENT · Introduce
A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply

M31-LIN-05 I can calculate average rate of change from two points.

TEXTBOOK SECTIONS 3.3 Rates of Change and Behavior of Graphs; 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

D	Calculate average rate of change over an interval. (first introduced in Math 21) SK-FUNC-AROC · Deepen
A	Calculate slope from two points. (first introduced in Math 12) SK-LIN-SLOPE-POINTS · Apply

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M31-LIN-006 — I can interpret interval notation and identify increasing or decreasing intervals.

TEXTBOOK SECTIONS 3.2 Domain and Range; 3.3 Rates of Change and Behavior of Graphs

SUPPORTING SKILLS

I	Describe increasing or decreasing behavior over intervals. SK-FUNC-INTERVAL-BEHAVIOR · Introduce
D	Translate between inequality and interval notation. (first introduced in Math 12) SK-NOT-INTERVAL · Deepen

M31-LIN-007 — I can determine total change from an average rate of change over an interval.

TEXTBOOK SECTIONS 3.3 Rates of Change and Behavior of Graphs; 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

A	Calculate average rate of change over an interval. (first introduced in Math 21) SK-FUNC-AROC · Apply
I	Use average rate of change and interval length to calculate total change. SK-FUNC-AROC-TOTAL-CHANGE · Introduce
A	Add, subtract, multiply, and divide signed integers. (inherited from middle school) SK-NUM-INTEGER-OPERATE · Apply

M31-LIN-008 — I can determine and interpret the units of an average rate of change.

TEXTBOOK SECTIONS 3.3 Rates of Change and Behavior of Graphs; 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

I	Determine rate units from input- and output-axis units. SK-FUNC-AROC-UNITS · Introduce
D	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Deepen

M31-LIN-009 — I can identify a linear function from a table, graph, or formula.

TEXTBOOK SECTIONS 4.1 Linear Functions; 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

I	Use constant differences or ratios to distinguish linear and exponential patterns. SK-EXP-DIFFERENCE-RATIO · Introduce
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen
D	Use tabular patterns to predict graph shape and behavior. (first introduced in Math 21) SK-REP-TABLE-GRAPH · Deepen

M31-LIN-010 — I can interpret the slope and intercept of a linear model.

TEXTBOOK SECTIONS 4.1 Linear Functions; 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

A	Plot an intercept accurately on a coordinate plane. (first introduced in Math 12) SK-LIN-PLOT-INTERCEPT · Apply
A	Determine slope from a graph. (first introduced in Math 12) SK-LIN-SLOPE-GRAPH · Apply
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

M31-LIN-011 — I can determine the intersection of two linear relationships graphically or algebraically.

TEXTBOOK SECTIONS 4.2 Modeling with Linear Functions

SUPPORTING SKILLS

A	Check an ordered pair in both equations of a system. (first introduced in Math 12) SK-SYS-CHECK · Apply
A	Add equations to eliminate a variable. (first introduced in Math 12) SK-SYS-ELIMINATE · Apply
A	Locate and interpret the intersection of two lines. (first introduced in Math 12) SK-SYS-GRAPH-INTERSECTION · Apply
A	Substitute an equivalent expression for a variable. (first introduced in Math 12) SK-SYS-SUBSTITUTE · Apply

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M31-LIN-012

I can write equations of horizontal, vertical, parallel, and perpendicular lines through a specified point.

TEXTBOOK SECTIONS 4.1 Linear Functions

SUPPORTING SKILLS

A	Translate quantities and relationships into an equation. (first introduced in Math 12) SK-ALG-TRANSLATE-EQUATION · Apply
A	Convert among common forms of a linear equation. (first introduced in Math 12) SK-LIN-FORM-CONVERT · Apply
D	Recognize that distinct parallel lines have equal slopes. (first introduced in Math 12) SK-LIN-PARALLEL-SLOPE · Deepen
D	Determine the negative reciprocal of a nonzero slope. (first introduced in Math 12) SK-LIN-PERPENDICULAR-SLOPE · Deepen

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Function Behavior and Structure

REQUIRED

UNIT TEXTBOOK COVERAGE

3.1 Functions and Function Notation; 3.2 Domain and Range; 3.3 Rates of Change and Behavior of Graphs; 3.4 Composition of Functions; 3.5 Transformation of Functions; 3.7 Inverse Functions

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M31-FUN-001 — I can determine whether a relation is a function.

TEXTBOOK SECTIONS 3.1 Functions and Function Notation

SUPPORTING SKILLS	
D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
A	Apply the vertical line test to a graph. (first introduced in Math 12) SK-FUNC-VERTICAL-LINE · Apply

M31-FUN-002 — I can evaluate a function from a formula, graph, or table.

TEXTBOOK SECTIONS 3.1 Functions and Function Notation

SUPPORTING SKILLS	
A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
D	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Deepen
A	Solve for an input associated with a specified output. (first introduced in Math 21) SK-FUNC-SOLVE-INPUT · Apply

M31-FUN-003 — I can determine domain and range from a formula, graph, or table.

TEXTBOOK SECTIONS 3.2 Domain and Range

SUPPORTING SKILLS	
I	Determine domain restrictions from denominators, even roots, and logarithms. SK-FUNC-DOMAIN-FORMULA · Introduce
D	Read domain and range from a graph. (first introduced in Math 12) SK-FUNC-DOMAIN-RANGE-GRAPH · Deepen
D	Read domain and range from a table. (first introduced in Math 12) SK-FUNC-DOMAIN-RANGE-TABLE · Deepen

M31-FUN-004 — I can determine algebraic domain restrictions caused by denominators and even roots.

TEXTBOOK SECTIONS 3.2 Domain and Range

SUPPORTING SKILLS	
A	Factor a quadratic trinomial with a nonunit leading coefficient. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-GENERAL · Apply
A	Factor a monic quadratic trinomial. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-MONIC · Apply
D	Require an even-root radicand to be nonnegative when determining domain. (first introduced in Math 21) SK-FUNC-DOMAIN-EVEN-ROOT · Deepen
D	Determine domain restrictions from denominators, even roots, and logarithms. (first introduced in Math 31) SK-FUNC-DOMAIN-FORMULA · Deepen
A	Determine excluded values from an original denominator. (first introduced in Math 21) SK-RAT-RESTRICTIONS · Apply

M31-FUN-005 — I can graph a piecewise function from its formula.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS	
A	Describe increasing or decreasing behavior over intervals. (first introduced in Math 31) SK-FUNC-INTERVAL-BEHAVIOR · Apply
I	Select the correct rule of a piecewise function for a given input. SK-FUNC-PIECEWISE-EVALUATE · Introduce
I	Graph each rule on its stated domain with correct endpoints. SK-FUNC-PIECEWISE-GRAPH · Introduce

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION: I Introduce · R Reinforce · D Deepen · A Apply

M31-FUN-006 — I can write a piecewise formula that represents a graph.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

A	Describe increasing or decreasing behavior over intervals. (first introduced in Math 31) SK-FUNC-INTERVAL-BEHAVIOR · Apply
I	Translate a segmented graph into a piecewise formula. SK-FUNC-PIECEWISE-WRITE · Introduce
A	Convert among common forms of a linear equation. (first introduced in Math 12) SK-LIN-FORM-CONVERT · Apply

M31-FUN-007 — I can graph a transformed function using shifts, stretches, compressions, and reflections.

TEXTBOOK SECTIONS 3.5 Transformation of Functions

SUPPORTING SKILLS

I	Interpret transformations applied to function inputs. SK-FUNC-TRANSFORM-HORIZONTAL · Introduce
I	Distinguish stretches from compressions and horizontal effects from vertical effects. SK-FUNC-TRANSFORM-SCALE · Introduce
I	Interpret transformations applied to function outputs. SK-FUNC-TRANSFORM-VERTICAL · Introduce
A	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Apply

M31-FUN-008 — I can describe how a transformation affects domain, range, or asymptotes.

TEXTBOOK SECTIONS 3.5 Transformation of Functions

SUPPORTING SKILLS

I	Exchange domain and range between a function and its inverse. SK-FUNC-INVERSE-DOMAIN-RANGE · Introduce
D	Interpret transformations applied to function inputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-HORIZONTAL · Deepen
D	Distinguish stretches from compressions and horizontal effects from vertical effects. (first introduced in Math 31) SK-FUNC-TRANSFORM-SCALE · Deepen
D	Interpret transformations applied to function outputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-VERTICAL · Deepen

M31-FUN-009 — I can evaluate a composition from formulas, graphs, or tables.

TEXTBOOK SECTIONS 3.4 Composition of Functions

SUPPORTING SKILLS

I	Substitute one function formula into another. SK-FUNC-COMPOSE-FORMULA · Introduce
I	Follow intermediate values to compose functions from tables. SK-FUNC-COMPOSE-TABLE · Introduce
A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply

M31-FUN-010 — I can write a formula for a composition of two functions.

TEXTBOOK SECTIONS 3.4 Composition of Functions

SUPPORTING SKILLS

A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
D	Substitute one function formula into another. (first introduced in Math 31) SK-FUNC-COMPOSE-FORMULA · Deepen

M31-FUN-011 — I can find an inverse function algebraically.

TEXTBOOK SECTIONS 3.7 Inverse Functions

SUPPORTING SKILLS

A	Isolate a specified variable in an equation or formula. (first introduced in Math 12) SK-ALG-VARIABLE-ISOLATE · Apply
I	Exchange variables and solve to obtain an inverse formula. SK-FUNC-INVERSE-ALGEBRA · Introduce
I	Exchange input and output values when interpreting an inverse. SK-FUNC-INVERSE-SWAP · Introduce

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION: I Introduce · R Reinforce · D Deepen · A Apply

M31-FUN-012 I can evaluate and interpret an inverse from a graph, formula, or table.

TEXTBOOK SECTIONS 3.7 Inverse Functions

SUPPORTING SKILLS

A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply
A	Exchange variables and solve to obtain an inverse formula. (first introduced in Math 31) SK-FUNC-INVERSE-ALGEBRA · Apply
D	Exchange input and output values when interpreting an inverse. (first introduced in Math 31) SK-FUNC-INVERSE-SWAP · Deepen

M31-FUN-013 I can determine and interpret the units of an inverse function.

TEXTBOOK SECTIONS 3.7 Inverse Functions

SUPPORTING SKILLS

I	Reverse input and output units when interpreting an inverse. SK-FUNC-INVERSE-UNITS · Introduce
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

M31-FUN-014 I can explain how the domain and range of a function relate to those of its inverse.

TEXTBOOK SECTIONS 3.7 Inverse Functions

SUPPORTING SKILLS

I	Exchange domain and range between a function and its inverse. SK-FUNC-INVERSE-DOMAIN-RANGE · Introduce
A	Exchange input and output values when interpreting an inverse. (first introduced in Math 31) SK-FUNC-INVERSE-SWAP · Apply

M31-FUN-015 I can determine the range of a function by analyzing the domain of its inverse.

TEXTBOOK SECTIONS 3.7 Inverse Functions

SUPPORTING SKILLS

A	Determine domain restrictions from denominators, even roots, and logarithms. (first introduced in Math 31) SK-FUNC-DOMAIN-FORMULA · Apply
D	Exchange domain and range between a function and its inverse. (first introduced in Math 31) SK-FUNC-INVERSE-DOMAIN-RANGE · Deepen

M31-FUN-016 I can verify algebraically that two functions are inverses.

TEXTBOOK SECTIONS 3.7 Inverse Functions

SUPPORTING SKILLS

A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
A	Substitute one function formula into another. (first introduced in Math 31) SK-FUNC-COMPOSE-FORMULA · Apply
I	Verify inverse functions using composition. SK-FUNC-INVERSE-VERIFY · Introduce

M31-FUN-017 I can interpret the meaning and units of a composition.

TEXTBOOK SECTIONS 3.4 Composition of Functions

SUPPORTING SKILLS

D	Substitute one function formula into another. (first introduced in Math 31) SK-FUNC-COMPOSE-FORMULA · Deepen
D	Follow intermediate values to compose functions from tables. (first introduced in Math 31) SK-FUNC-COMPOSE-TABLE · Deepen
A	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Apply
D	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Deepen

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M31-FUN-018

I can classify a graph or table as concave up or concave down using changes in average rate of change.

TEXTBOOK SECTIONS 3.3 Rates of Change and Behavior of Graphs

SUPPORTING SKILLS

A	Calculate average rate of change over an interval. (first introduced in Math 21) SK-FUNC-AROC · Apply
I	Use successive average rates of change to classify concavity. SK-FUNC-CONCAVITY-AROC · Introduce
D	Use tabular patterns to predict graph shape and behavior. (first introduced in Math 21) SK-REP-TABLE-GRAPH · Deepen

M31-FUN-019

I can determine whether a function is invertible using the horizontal line test.

TEXTBOOK SECTIONS 3.7 Inverse Functions

SUPPORTING SKILLS

I	Apply the horizontal line test for invertibility. SK-FUNC-INVERSE-HLT · Introduce
I	Test whether a function is one-to-one on a domain. SK-FUNC-ONE-TO-ONE · Introduce

M31-FUN-020

I can graph an inverse function as a reflection across the line `y = x`.

TEXTBOOK SECTIONS 3.7 Inverse Functions

SUPPORTING SKILLS

A	Exchange domain and range between a function and its inverse. (first introduced in Math 31) SK-FUNC-INVERSE-DOMAIN-RANGE · Apply
D	Exchange input and output values when interpreting an inverse. (first introduced in Math 31) SK-FUNC-INVERSE-SWAP · Deepen
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Quadratic Functions and Equations

REQUIRED

UNIT TEXTBOOK COVERAGE

2.5 Quadratic Equations; 5.1 Quadratic Functions

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M31-QUA D-001 — I can identify the vertex, intercepts, axis of symmetry, and concavity of a quadratic function.

TEXTBOOK SECTIONS 5.1 Quadratic Functions

SUPPORTING SKILLS

A	Use successive average rates of change to classify concavity. (first introduced in Math 31) SK-FUNC-CONCAVITY-AROC · Apply
A	Identify standard, factored, and vertex forms. (first introduced in Math 21) SK-QUAD-FORM-IDENTIFY · Apply
D	Determine x- and y-intercepts of a quadratic function. (first introduced in Math 21) SK-QUAD-INTERCEPTS · Deepen
D	Determine the vertex and axis of symmetry. (first introduced in Math 21) SK-QUAD-VERTEX · Deepen

M31-QUA D-002 — I can connect standard, factored, and vertex forms to features of a quadratic graph.

TEXTBOOK SECTIONS 5.1 Quadratic Functions

SUPPORTING SKILLS

D	Identify standard, factored, and vertex forms. (first introduced in Math 21) SK-QUAD-FORM-IDENTIFY · Deepen
A	Determine x- and y-intercepts of a quadratic function. (first introduced in Math 21) SK-QUAD-INTERCEPTS · Apply
A	Determine the vertex and axis of symmetry. (first introduced in Math 21) SK-QUAD-VERTEX · Apply
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

M31-QUA D-003 — I can rewrite a quadratic function in a form that reveals a requested feature.

TEXTBOOK SECTIONS 5.1 Quadratic Functions

SUPPORTING SKILLS

A	Factor a quadratic trinomial with a nonunit leading coefficient. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-GENERAL · Apply
A	Factor a monic quadratic trinomial. (first introduced in Math 12) SK-FACTOR-TRINOMIAL-MONIC · Apply
D	Create a perfect-square trinomial by completing the square. (first introduced in Math 21) SK-QUAD-COMPLETE-SQUARE · Deepen
D	Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method. (first introduced in Math 21) SK-QUAD-FORM-CONVERT · Deepen

M31-QUA D-004 — I can solve a quadratic equation by factoring or using the quadratic formula.

TEXTBOOK SECTIONS 2.5 Quadratic Equations; 5.1 Quadratic Functions

SUPPORTING SKILLS

A	Apply the zero-product property. (first introduced in Math 21) SK-FACTOR-ZERO-PRODUCT · Apply
A	Use the discriminant to predict the number and type of solutions. (first introduced in Math 21) SK-QUAD-DISCRIMINANT · Apply
A	Substitute coefficients accurately into the quadratic formula. (first introduced in Math 21) SK-QUAD-FORMULA · Apply
D	Select an efficient quadratic-solving method from equation structure. (first introduced in Math 21) SK-QUAD-METHOD-SELECT · Deepen

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M31-QUAD-005

I can graph a quadratic function using its algebraic structure and key features.

TEXTBOOK SECTIONS 5.1 Quadratic Functions

SUPPORTING SKILLS

A	Identify standard, factored, and vertex forms. (first introduced in Math 21) SK-QUAD-FORM-IDENTIFY · Apply
D	Determine x- and y-intercepts of a quadratic function. (first introduced in Math 21) SK-QUAD-INTERCEPTS · Deepen
D	Determine the vertex and axis of symmetry. (first introduced in Math 21) SK-QUAD-VERTEX · Deepen
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

M31-QUAD-006

I can construct a quadratic function from specified zeros, a vertex, or graphical information.

TEXTBOOK SECTIONS 5.1 Quadratic Functions

SUPPORTING SKILLS

A	Translate quantities and relationships into an equation. (first introduced in Math 12) SK-ALG-TRANSLATE-EQUATION · Apply
I	Build a polynomial from zeros, multiplicities, and a point. SK-POLY-CONSTRUCT · Introduce
A	Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method. (first introduced in Math 21) SK-QUAD-FORM-CONVERT · Apply

M31-QUAD-007

I can create and interpret a quadratic trajectory model.

TEXTBOOK SECTIONS 5.1 Quadratic Functions

SUPPORTING SKILLS

D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Apply
D	Interpret vertex and zeros in an application. (first introduced in Math 21) SK-QUAD-MODEL-FEATURES · Deepen
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Exponential Functions and Models

REQUIRED

UNIT TEXTBOOK COVERAGE

6.1 Exponential Functions; 6.2 Graphs of Exponential Functions; 6.7 Exponential and Logarithmic Models

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M31-EXP-001 I can distinguish linear change from exponential change using differences, ratios, or percent change.

TEXTBOOK SECTIONS 6.1 Exponential Functions

SUPPORTING SKILLS

D	Use constant differences or ratios to distinguish linear and exponential patterns. (first introduced in Math 31) SK-EXP-DIFFERENCE-RATIO · Deepen
D	Compare functions shown in different representations. (first introduced in Math 21) SK-FUNC-COMPARE-REP · Deepen
A	Calculate and interpret percent increase or decrease. (inherited from middle school) SK-NUM-PERCENT-CHANGE · Apply

M31-EXP-002 I can identify initial value and growth or decay factor in an exponential model.

TEXTBOOK SECTIONS 6.1 Exponential Functions; 6.2 Graphs of Exponential Functions

SUPPORTING SKILLS

I	Identify initial amount, rate, and time interval in context. SK-EXP-MODEL-CONTEXT · Introduce
I	Convert a percent increase or decrease to a growth or decay factor. SK-EXP-PERCENT-FACTOR · Introduce
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen

M31-EXP-003 I can write an exponential function from a table, graph, or context.

TEXTBOOK SECTIONS 6.1 Exponential Functions

SUPPORTING SKILLS

D	Identify initial amount, rate, and time interval in context. (first introduced in Math 31) SK-EXP-MODEL-CONTEXT · Deepen
I	Determine an exponential model from tabular values. SK-EXP-MODEL-TABLE · Introduce
I	Solve for an exponential model using two points. SK-EXP-MODEL-TWO-POINTS · Introduce
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

M31-EXP-004 I can evaluate and interpret an exponential model.

TEXTBOOK SECTIONS 6.1 Exponential Functions; 6.2 Graphs of Exponential Functions; 6.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

D	Identify initial amount, rate, and time interval in context. (first introduced in Math 31) SK-EXP-MODEL-CONTEXT · Deepen
A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

M31-EXP-005 I can calculate growth under periodic compounding.

TEXTBOOK SECTIONS 6.1 Exponential Functions; 6.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

I	Substitute correctly into a periodic-compounding model. SK-EXP-COMPOUND-PERIODIC · Introduce
A	Convert a percent increase or decrease to a growth or decay factor. (first introduced in Math 31) SK-EXP-PERCENT-FACTOR · Apply
A	Calculate a percentage of a quantity or determine what percent one quantity is of another. (inherited from middle school) SK-NUM-PERCENT-CALCULATE · Apply

Math 31 In-Depth Curriculum

Advanced Algebra

TEXTBOOK | OpenStax College Algebra 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M31-EXP-006 — I can calculate growth under continuous compounding.

TEXTBOOK SECTIONS 6.1 Exponential Functions; 6.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

I	Substitute correctly into a continuous-compounding model. SK-EXP-COMPOUND-CONTINUOUS · Introduce
A	Identify initial amount, rate, and time interval in context. (first introduced in Math 31) SK-EXP-MODEL-CONTEXT · Apply

M31-EXP-007 — I can determine an exponential formula from two points or other sufficient data.

TEXTBOOK SECTIONS 6.1 Exponential Functions

SUPPORTING SKILLS

A	Isolate a specified variable in an equation or formula. (first introduced in Math 12) SK-ALG-VARIABLE-ISOLATE · Apply
D	Solve for an exponential model using two points. (first introduced in Math 31) SK-EXP-MODEL-TWO-POINTS · Deepen
A	Interpret numerator and denominator roles in a rational exponent. (first introduced in Math 21) SK-EXP-RATIONAL · Apply

M31-EXP-008 — I can determine domain, range, intercepts, and asymptotes of an exponential function.

TEXTBOOK SECTIONS 6.2 Graphs of Exponential Functions

SUPPORTING SKILLS

I	Identify horizontal asymptotic behavior of an exponential graph. SK-EXP-ASYMPTOTE · Introduce
D	Determine domain restrictions from denominators, even roots, and logarithms. (first introduced in Math 31) SK-FUNC-DOMAIN-FORMULA · Deepen
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

M31-EXP-009 — I can compare the growth rates of two exponential functions.

TEXTBOOK SECTIONS 6.2 Graphs of Exponential Functions

SUPPORTING SKILLS

I	Compare exponential growth rates expressed in different forms. SK-EXP-COMPARE-GROWTH · Introduce
D	Compare functions shown in different representations. (first introduced in Math 21) SK-FUNC-COMPARE-REP · Deepen

M31-EXP-010 — I can determine the percent growth or decay represented by a continuous exponential model.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

I	Convert between equivalent discrete- and continuous-growth forms. SK-EXP-DISCRETE-CONTINUOUS · Introduce
D	Convert a percent increase or decrease to a growth or decay factor. (first introduced in Math 31) SK-EXP-PERCENT-FACTOR · Deepen
A	Calculate and interpret percent increase or decrease. (inherited from middle school) SK-NUM-PERCENT-CHANGE · Apply

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Logarithmic Functions and Models

REQUIRED

UNIT TEXTBOOK COVERAGE

6.3 Logarithmic Functions; 6.4 Graphs of Logarithmic Functions; 6.5 Logarithmic Properties; 6.6 Exponential and Logarithmic Equations; 6.7 Exponential and Logarithmic Models

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M31-LOG-001 I can convert between logarithmic and exponential forms.

TEXTBOOK SECTIONS 6.3 Logarithmic Functions

SUPPORTING SKILLS

I	Convert between logarithmic and exponential forms. SK-LOG-CONVERT · Introduce
I	Interpret a logarithm as an exponent. SK-LOG-DEFINITION · Introduce

M31-LOG-002 I can evaluate a logarithm using its exponential meaning.

TEXTBOOK SECTIONS 6.3 Logarithmic Functions

SUPPORTING SKILLS

I	Rewrite exponential expressions using a common base when possible. SK-EXP-REWRITE-BASE · Introduce
D	Interpret a logarithm as an exponent. (first introduced in Math 31) SK-LOG-DEFINITION · Deepen

M31-LOG-003 I can expand or condense an expression using product, quotient, and power properties.

TEXTBOOK SECTIONS 6.5 Logarithmic Properties

SUPPORTING SKILLS

A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Apply the logarithm power property in either direction. SK-LOG-POWER · Introduce
I	Apply the logarithm product property in either direction. SK-LOG-PRODUCT · Introduce
I	Apply the logarithm quotient property in either direction. SK-LOG-QUOTIENT · Introduce

M31-LOG-004 I can solve an exponential equation using logarithms.

TEXTBOOK SECTIONS 6.6 Exponential and Logarithmic Equations

SUPPORTING SKILLS

A	Isolate a specified variable in an equation or formula. (first introduced in Math 12) SK-ALG-VARIABLE-ISOLATE · Apply
A	Apply the logarithm power property in either direction. (first introduced in Math 31) SK-LOG-POWER · Apply
I	Isolate an exponential expression and apply a logarithm. SK-LOG-SOLVE-EXP · Introduce

M31-LOG-005 I can solve a logarithmic equation using exponential form and check domain restrictions.

TEXTBOOK SECTIONS 6.6 Exponential and Logarithmic Equations

SUPPORTING SKILLS

A	Convert between logarithmic and exponential forms. (first introduced in Math 31) SK-LOG-CONVERT · Apply
I	Reject solutions that make a logarithm argument nonpositive. SK-LOG-DOMAIN-CHECK · Introduce
I	Combine or isolate logarithms before converting to exponential form. SK-LOG-SOLVE-LOG · Introduce

M31-LOG-006 I can calculate and interpret doubling time or half-life.

TEXTBOOK SECTIONS 6.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

I	Translate a doubling-time or half-life question into an exponential equation. SK-LOG-DOUBLING-HALF · Introduce
A	Isolate an exponential expression and apply a logarithm. (first introduced in Math 31) SK-LOG-SOLVE-EXP · Apply
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

Math 31 In-Depth Curriculum

Advanced Algebra

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SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M31-LOG-007 I can determine domain, range, intercepts, and asymptotes of a logarithmic function.

TEXTBOOK SECTIONS 6.4 Graphs of Logarithmic Functions

SUPPORTING SKILLS

D	Determine domain restrictions from denominators, even roots, and logarithms. (first introduced in Math 31) SK-FUNC-DOMAIN-FORMULA · Deepen
I	Identify vertical asymptotic behavior of a logarithmic graph. SK-LOG-ASYMPTOTE · Introduce
A	Reject solutions that make a logarithm argument nonpositive. (first introduced in Math 31) SK-LOG-DOMAIN-CHECK · Apply
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

M31-LOG-008 I can explain how exponential and logarithmic functions undo one another.

TEXTBOOK SECTIONS 6.3 Logarithmic Functions; 6.4 Graphs of Logarithmic Functions

SUPPORTING SKILLS

A	Exchange input and output values when interpreting an inverse. (first introduced in Math 31) SK-FUNC-INVERSE-SWAP · Apply
A	Verify inverse functions using composition. (first introduced in Math 31) SK-FUNC-INVERSE-VERIFY · Apply
D	Interpret a logarithm as an exponent. (first introduced in Math 31) SK-LOG-DEFINITION · Deepen

M31-LOG-009 I can solve a multi-step exponential or logarithmic equation.

TEXTBOOK SECTIONS 6.6 Exponential and Logarithmic Equations

SUPPORTING SKILLS

A	Reject solutions that make a logarithm argument nonpositive. (first introduced in Math 31) SK-LOG-DOMAIN-CHECK · Apply
A	Apply the logarithm power property in either direction. (first introduced in Math 31) SK-LOG-POWER · Apply
A	Apply the logarithm product property in either direction. (first introduced in Math 31) SK-LOG-PRODUCT · Apply
A	Apply the logarithm quotient property in either direction. (first introduced in Math 31) SK-LOG-QUOTIENT · Apply
D	Isolate an exponential expression and apply a logarithm. (first introduced in Math 31) SK-LOG-SOLVE-EXP · Deepen
D	Combine or isolate logarithms before converting to exponential form. (first introduced in Math 31) SK-LOG-SOLVE-LOG · Deepen

M31-LOG-010 I can determine an intersection of exponential functions graphically or algebraically.

TEXTBOOK SECTIONS 6.6 Exponential and Logarithmic Equations

SUPPORTING SKILLS

A	Compare exponential growth rates expressed in different forms. (first introduced in Math 31) SK-EXP-COMPARE-GROWTH · Apply
D	Isolate an exponential expression and apply a logarithm. (first introduced in Math 31) SK-LOG-SOLVE-EXP · Deepen
A	Check an ordered pair in both equations of a system. (first introduced in Math 12) SK-SYS-CHECK · Apply
A	Locate and interpret the intersection of two lines. (first introduced in Math 12) SK-SYS-GRAPH-INTERSECTION · Apply

Math 31 In-Depth Curriculum

Advanced Algebra

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SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M31-LOG-011

I can rewrite a discrete exponential model as an equivalent continuous exponential model.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

D	Convert between equivalent discrete- and continuous-growth forms. (first introduced in Math 31) SK-EXP-DISCRETE-CONTINUOUS · Deepen
A	Convert a percent increase or decrease to a growth or decay factor. (first introduced in Math 31) SK-EXP-PERCENT-FACTOR · Apply
A	Isolate an exponential expression and apply a logarithm. (first introduced in Math 31) SK-LOG-SOLVE-EXP · Apply

M31-LOG-012

I can solve and interpret an exponential or logarithmic modeling problem.

TEXTBOOK SECTIONS 6.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

D	Identify initial amount, rate, and time interval in context. (first introduced in Math 31) SK-EXP-MODEL-CONTEXT · Deepen
A	Isolate an exponential expression and apply a logarithm. (first introduced in Math 31) SK-LOG-SOLVE-EXP · Apply
A	Combine or isolate logarithms before converting to exponential form. (first introduced in Math 31) SK-LOG-SOLVE-LOG · Apply
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Apply
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Extension Content

EXTENSION

UNIT TEXTBOOK COVERAGE
3.6 Absolute Value Functions

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M31-EXT-01 I can analyze an absolute-value function using transformations.

TEXTBOOK SECTIONS 3.6 Absolute Value Functions

SUPPORTING SKILLS

A	Interpret transformations applied to function inputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-HORIZONTAL · Apply
A	Distinguish stretches from compressions and horizontal effects from vertical effects. (first introduced in Math 31) SK-FUNC-TRANSFORM-SCALE · Apply
A	Interpret transformations applied to function outputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-VERTICAL · Apply
A	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Apply

M31-EXT-02 I can determine even or odd symmetry from a graph or formula.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Test a formula or graph for even or odd symmetry. SK-FUNC-SYMMETRY · Introduce

M31-EXT-03 I can interpret a contextual logarithmic scale.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

A	Interpret a logarithm as an exponent. (first introduced in Math 31) SK-LOG-DEFINITION · Apply
I	Interpret equal intervals on a logarithmic scale as equal multiplicative changes. SK-LOG-SCALE-INTERPRET · Introduce
A	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Apply

Math 32

Precalculus: Trigonometry

Math 32 Course at a Glance

Precalculus: Trigonometry

Trigonometric Foundations

REQUIRED

LEARNING OBJECTIVES

M32-FND-001	I can use the Pythagorean theorem to determine an unknown side of a right triangle.
M32-FND-002	I can determine exact side lengths in 30-60-90 and 45-45-90 triangles.
M32-FND-003	I can convert angle measures between degrees and radians.
M32-FND-004	I can determine coterminal and reference angles for a given angle.
M32-FND-005	I can calculate arc length from a central angle and radius.
M32-FND-006	I can determine exact coordinates on the unit circle.
M32-FND-007	I can evaluate all six trigonometric functions exactly for common angles.
M32-FND-008	I can solve right triangles using trigonometric ratios.
M32-FND-009	I can apply right-triangle trigonometry to contextual problems.
M32-FND-010	I can evaluate trigonometric functions from a point on a circle or terminal side.
M32-FND-011	I can solve a basic trigonometric equation for all angles in a specified interval.

Periodic Functions and Modeling

REQUIRED

LEARNING OBJECTIVES

M32-GRF-001	I can graph basic sine and cosine functions using key features.
M32-GRF-002	I can identify amplitude, midline, period, and phase shift from an equation or graph.
M32-GRF-003	I can graph transformed sine and cosine functions.
M32-GRF-004	I can write sine and cosine equations that represent a periodic graph.
M32-GRF-005	I can construct a sinusoidal model from contextual information or data.
M32-GRF-006	I can interpret the parameters and outputs of a sinusoidal model in context.
M32-GRF-007	I can graph tangent, secant, and cosecant functions with their asymptotes.

Inverse Functions and Trigonometric Equations

REQUIRED

LEARNING OBJECTIVES

M32-INV-001	I can evaluate inverse sine, inverse cosine, and inverse tangent expressions.
M32-INV-002	I can explain how restricted domains make inverse trigonometric functions possible.
M32-INV-003	I can solve trigonometric equations exactly or approximately on a specified interval.
M32-INV-004	I can express all solutions of a trigonometric equation in general form.
M32-INV-005	I can evaluate compositions of trigonometric and inverse trigonometric functions.

Non-Right Triangles

REQUIRED

LEARNING OBJECTIVES

M32-TRI-002	I can solve a non-right triangle using the Law of Sines.
M32-TRI-003	I can determine the number of possible triangles in the ambiguous SSA case.
M32-TRI-004	I can solve a non-right triangle using the Law of Cosines.
M32-TRI-005	I can select and apply an appropriate method to solve a triangle in context.

Math 32 Course at a Glance

Precalculus: Trigonometry

Trigonometric Identities

REQUIRED

LEARNING OBJECTIVES

M32-IDN-001	I can rewrite trigonometric expressions using reciprocal and quotient identities, including `tan(theta) = sin(theta)/cos(theta)`.
M32-IDN-002	I can rewrite trigonometric expressions using the Pythagorean identities.
M32-IDN-003	I can use even/odd and cofunction identities to rewrite trigonometric expressions.
M32-IDN-004	I can verify a trigonometric identity using the required identity families.

Additional Trigonometric Identities

EXTENSIO
N

LEARNING OBJECTIVES

M32-IDX-001	I can apply double-angle identities.
M32-IDX-002	I can apply half-angle identities.
M32-IDX-003	I can apply angle sum and difference identities.
M32-IDX-004	I can determine exact trigonometric values using extension identities.

Triangle Area from SAS

EXTENSIO
N

LEARNING OBJECTIVES

M32-TRX-001	I can calculate the area of a triangle from two sides and the included angle.

Polar Coordinates

REQUIRED

LEARNING OBJECTIVES

M32-POL-001	I can plot and identify points in polar coordinates.
M32-POL-002	I can represent a polar point using equivalent coordinate pairs.
M32-POL-003	I can convert points between polar and rectangular coordinates.

Math 32 Course at a Glance

Precalculus: Trigonometry

Polar Functions

EXTENSIO
N

LEARNING OBJECTIVES

M32-PLX-001	I can graph and interpret basic polar functions.
M32-PLX-002	I can convert equations between polar and rectangular forms.

Vectors

EXTENSIO
N

LEARNING OBJECTIVES

M32-VEC-001	I can represent a vector geometrically and in component form.
M32-VEC-002	I can add, subtract, and multiply vectors by scalars.
M32-VEC-003	I can determine vector magnitude, direction, and unit vector.
M32-VEC-004	I can resolve a vector into horizontal and vertical components.
M32-VEC-005	I can perform basic operations with vectors in three dimensions.

Parametric Equations

EXTENSIO
N

LEARNING OBJECTIVES

M32-PAR-00 1	I can graph a plane curve defined by parametric equations with its orientation.
M32-PAR-00 2	I can eliminate a parameter to obtain a rectangular equation.
M32-PAR-00 3	I can create or interpret parametric equations for motion in the plane.

Math 32 In-Depth Curriculum

Precalculus: Trigonometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Trigonometric Foundations

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-FND-001 — I can use the Pythagorean theorem to determine an unknown side of a right triangle.

SUPPORTING SKILLS

A	Extract perfect-power factors from a radical. (first introduced in Math 21) SK-RAD-FACTOR-PERFECT · Apply
R	Apply the Pythagorean theorem or its converse. (first introduced in Math 22) SK-TRI-PYTHAGOREAN · Reinforce

M32-FND-002 — I can determine exact side lengths in 30-60-90 and 45-45-90 triangles.

SUPPORTING SKILLS

A	Distinguish exact values from measured or rounded approximations. (first introduced in Math 22) SK-GEO-EXACT-APPROX · Apply
A	Extract perfect-power factors from a radical. (first introduced in Math 21) SK-RAD-FACTOR-PERFECT · Apply
R	Recall special-right-triangle ratios. (first introduced in Math 22) SK-TRI-SPECIAL-RATIOS · Reinforce

M32-FND-003 — I can convert angle measures between degrees and radians.

SUPPORTING SKILLS

A	Add, subtract, multiply, and divide fractions. (inherited from middle school) SK-NUM-FRACTION-OPERATE · Apply
I	Convert between degree and radian measure. SK-TRIG-ANGLE-CONVERT · Introduce

M32-FND-004 — I can determine coterminal and reference angles for a given angle.

SUPPORTING SKILLS

I	Generate and recognize coterminal angles. SK-TRIG-ANGLE-COTERMINAL · Introduce
I	Determine a reference angle and its quadrant. SK-TRIG-ANGLE-REFERENCE · Introduce
I	Determine the sign of a trigonometric function by quadrant. SK-TRIG-SIGN-QUADRANT · Introduce

M32-FND-005 — I can calculate arc length from a central angle and radius.

SUPPORTING SKILLS

D	Calculate arc length from radius and central angle. (first introduced in Math 22) SK-CIR-ARC-LENGTH · Deepen
A	Select and report appropriate linear, square, or angular units. (first introduced in Math 22) SK-GEO-UNITS · Apply
A	Convert between degree and radian measure. (first introduced in Math 32) SK-TRIG-ANGLE-CONVERT · Apply

M32-FND-006 — I can determine exact coordinates on the unit circle.

SUPPORTING SKILLS

A	Recall special-right-triangle ratios. (first introduced in Math 22) SK-TRI-SPECIAL-RATIOS · Apply
A	Determine a reference angle and its quadrant. (first introduced in Math 32) SK-TRIG-ANGLE-REFERENCE · Apply
I	Determine unit-circle coordinates for common angles. SK-TRIG-UNIT-CIRCLE-COORD · Introduce

Math 32 In-Depth Curriculum

Precalculus: Trigonometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M32-FND-007 — I can evaluate all six trigonometric functions exactly for common angles.

SUPPORTING SKILLS

I	Express common trigonometric values exactly. SK-TRIG-EXACT-VALUE · Introduce
I	Relate sine/cosecant, cosine/secant, and tangent/cotangent. SK-TRIG-RECIPROCAL · Introduce
A	Determine the sign of a trigonometric function by quadrant. (first introduced in Math 32) SK-TRIG-SIGN-QUADRANT · Apply
A	Determine unit-circle coordinates for common angles. (first introduced in Math 32) SK-TRIG-UNIT-CIRCLE-COORD · Apply

M32-FND-008 — I can solve right triangles using trigonometric ratios.

SUPPORTING SKILLS

D	Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. (first introduced in Math 22) SK-TRI-TRIG-RATIO · Deepen
D	Use an inverse trigonometric function to determine a missing angle. (first introduced in Math 22) SK-TRI-TRIG-SOLVE-ANGLE · Deepen
D	Use a trigonometric ratio to determine a missing side. (first introduced in Math 22) SK-TRI-TRIG-SOLVE-SIDE · Deepen
R	Label sides relative to a selected angle. (first introduced in Math 22) SK-TRIG-TRIANGLE-LABEL · Reinforce

M32-FND-009 — I can apply right-triangle trigonometry to contextual problems.

SUPPORTING SKILLS

A	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Apply
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply
D	Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. (first introduced in Math 22) SK-TRI-TRIG-RATIO · Deepen
A	Use an inverse trigonometric function to determine a missing angle. (first introduced in Math 22) SK-TRI-TRIG-SOLVE-ANGLE · Apply
A	Use a trigonometric ratio to determine a missing side. (first introduced in Math 22) SK-TRI-TRIG-SOLVE-SIDE · Apply

M32-FND-010 — I can evaluate trigonometric functions from a point on a circle or terminal side.

SUPPORTING SKILLS

A	Apply the Pythagorean theorem or its converse. (first introduced in Math 22) SK-TRI-PYTHAGOREAN · Apply
I	Use coordinates and radius to evaluate trigonometric functions. SK-TRIG-CIRCLE-POINT · Introduce
A	Relate sine/cosecant, cosine/secant, and tangent/cotangent. (first introduced in Math 32) SK-TRIG-RECIPROCAL · Apply
A	Determine the sign of a trigonometric function by quadrant. (first introduced in Math 32) SK-TRIG-SIGN-QUADRANT · Apply

Math 32 In-Depth Curriculum

Precalculus: Trigonometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M32-FND-011	**I can solve a basic trigonometric equation for all angles in a specified interval.**
SUPPORTING SKILLS	
A	Translate between inequality and interval notation. (first introduced in Math 12) SK-NOT-INTERVAL · Apply
A	Determine a reference angle and its quadrant. (first introduced in Math 32) SK-TRIG-ANGLE-REFERENCE · Apply
I	Use reference angles and periodicity to find every solution in an interval. SK-TRIG-EQUATION-INTERVAL · Introduce
I	Isolate a trigonometric expression before solving. SK-TRIG-EQUATION-ISOLATE · Introduce
A	Determine the sign of a trigonometric function by quadrant. (first introduced in Math 32) SK-TRIG-SIGN-QUADRANT · Apply

Math 32 In-Depth Curriculum

Precalculus: Trigonometry

SKILL PROGRESSION

I Introduce R Reinforce D Deepen A Apply

Periodic Functions and Modeling

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-GRF-001 I can graph basic sine and cosine functions using key features.

SUPPORTING SKILLS

D	Use tabular patterns to predict graph shape and behavior. (first introduced in Math 21) SK-REP-TABLE-GRAPH · Deepen
I	Locate quarter-period key points of a periodic function. SK-TRIG-GRAPH-KEYPOINTS · Introduce
A	Determine unit-circle coordinates for common angles. (first introduced in Math 32) SK-TRIG-UNIT-CIRCLE-COORD · Apply

M32-GRF-002 I can identify amplitude, midline, period, and phase shift from an equation or graph.

SUPPORTING SKILLS

I	Determine amplitude and vertical reflection. SK-TRIG-GRAPH-AMPLITUDE · Introduce
I	Determine a sinusoid's midline and vertical shift. SK-TRIG-GRAPH-MIDLINE · Introduce
I	Determine period from a graph or equation. SK-TRIG-GRAPH-PERIOD · Introduce
I	Determine horizontal shift from a graph or equation. SK-TRIG-GRAPH-PHASE · Introduce

M32-GRF-003 I can graph transformed sine and cosine functions.

SUPPORTING SKILLS

A	Interpret transformations applied to function inputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-HORIZONTAL · Apply
D	Distinguish stretches from compressions and horizontal effects from vertical effects. (first introduced in Math 31) SK-FUNC-TRANSFORM-SCALE · Deepen
A	Interpret transformations applied to function outputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-VERTICAL · Apply
D	Locate quarter-period key points of a periodic function. (first introduced in Math 32) SK-TRIG-GRAPH-KEYPOINTS · Deepen

M32-GRF-004 I can write sine and cosine equations that represent a periodic graph.

SUPPORTING SKILLS

D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen
D	Determine amplitude and vertical reflection. (first introduced in Math 32) SK-TRIG-GRAPH-AMPLITUDE · Deepen
D	Determine a sinusoid's midline and vertical shift. (first introduced in Math 32) SK-TRIG-GRAPH-MIDLINE · Deepen
D	Determine period from a graph or equation. (first introduced in Math 32) SK-TRIG-GRAPH-PERIOD · Deepen
D	Determine horizontal shift from a graph or equation. (first introduced in Math 32) SK-TRIG-GRAPH-PHASE · Deepen

Math 32 In-Depth Curriculum

Precalculus: Trigonometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M32-GRF-005 — I can construct a sinusoidal model from contextual information or data.

SUPPORTING SKILLS

A	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Apply
A	Determine amplitude and vertical reflection. (first introduced in Math 32) SK-TRIG-GRAPH-AMPLITUDE · Apply
A	Determine a sinusoid's midline and vertical shift. (first introduced in Math 32) SK-TRIG-GRAPH-MIDLINE · Apply
A	Determine period from a graph or equation. (first introduced in Math 32) SK-TRIG-GRAPH-PERIOD · Apply
I	Translate contextual maximum, minimum, cycle, and starting position into model parameters. SK-TRIG-MODEL-PARAMETERS · Introduce

M32-GRF-006 — I can interpret the parameters and outputs of a sinusoidal model in context.

SUPPORTING SKILLS

A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply
D	Translate contextual maximum, minimum, cycle, and starting position into model parameters. (first introduced in Math 32) SK-TRIG-MODEL-PARAMETERS · Deepen

M32-GRF-007 — I can graph tangent, secant, and cosecant functions with their asymptotes.

SUPPORTING SKILLS

A	Interpret transformations applied to function inputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-HORIZONTAL · Apply
I	Locate vertical asymptotes of tangent, secant, and cosecant. SK-TRIG-GRAPH-ASYMPTOTE · Introduce
A	Locate quarter-period key points of a periodic function. (first introduced in Math 32) SK-TRIG-GRAPH-KEYPOINTS · Apply
D	Relate sine/cosecant, cosine/secant, and tangent/cotangent. (first introduced in Math 32) SK-TRIG-RECIPROCAL · Deepen

SKILL PROGRESSION

I	Introduce
R	Reinforce
D	Deepen
A	Apply

Inverse Functions and Trigonometric Equations

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-INV-001 I can evaluate inverse sine, inverse cosine, and inverse tangent expressions.

SUPPORTING SKILLS

A	Express common trigonometric values exactly. (first introduced in Math 32) SK-TRIG-EXACT-VALUE · Apply
I	Evaluate an inverse trigonometric expression exactly or approximately. SK-TRIG-INVERSE-EVALUATE · Introduce
I	Recall the principal-value ranges of inverse trigonometric functions. SK-TRIG-INVERSE-RANGE · Introduce

M32-INV-002 I can explain how restricted domains make inverse trigonometric functions possible.

SUPPORTING SKILLS

A	Exchange domain and range between a function and its inverse. (first introduced in Math 31) SK-FUNC-INVERSE-DOMAIN-RANGE · Apply
A	Apply the horizontal line test for invertibility. (first introduced in Math 31) SK-FUNC-INVERSE-HLT · Apply
A	Test whether a function is one-to-one on a domain. (first introduced in Math 31) SK-FUNC-ONE-TO-ONE · Apply
D	Recall the principal-value ranges of inverse trigonometric functions. (first introduced in Math 32) SK-TRIG-INVERSE-RANGE · Deepen

M32-INV-003 I can solve trigonometric equations exactly or approximately on a specified interval.

SUPPORTING SKILLS

A	Check a proposed solution in the original equation. (first introduced in Math 12) SK-ALG-CHECK-SOLUTION · Apply
A	Determine a reference angle and its quadrant. (first introduced in Math 32) SK-TRIG-ANGLE-REFERENCE · Apply
D	Use reference angles and periodicity to find every solution in an interval. (first introduced in Math 32) SK-TRIG-EQUATION-INTERVAL · Deepen
D	Isolate a trigonometric expression before solving. (first introduced in Math 32) SK-TRIG-EQUATION-ISOLATE · Deepen
A	Evaluate an inverse trigonometric expression exactly or approximately. (first introduced in Math 32) SK-TRIG-INVERSE-EVALUATE · Apply

M32-INV-004 I can express all solutions of a trigonometric equation in general form.

SUPPORTING SKILLS

D	Generate and recognize coterminal angles. (first introduced in Math 32) SK-TRIG-ANGLE-COTERMINAL · Deepen
I	Express a periodic solution family in general form. SK-TRIG-EQUATION-GENERAL · Introduce
A	Use reference angles and periodicity to find every solution in an interval. (first introduced in Math 32) SK-TRIG-EQUATION-INTERVAL · Apply

Math 32 In-Depth Curriculum

Precalculus: Trigonometry

SKILL PROGRESSION

I Introduce | R Reinforce | D Deepen | A Apply

M32-INV-005	I can evaluate compositions of trigonometric and inverse trigonometric functions.
SUPPORTING SKILLS	
A	Substitute one function formula into another. (first introduced in Math 31) SK-FUNC-COMPOSE-FORMULA · Apply
A	Verify inverse functions using composition. (first introduced in Math 31) SK-FUNC-INVERSE-VERIFY · Apply
D	Evaluate an inverse trigonometric expression exactly or approximately. (first introduced in Math 32) SK-TRIG-INVERSE-EVALUATE · Deepen
A	Recall the principal-value ranges of inverse trigonometric functions. (first introduced in Math 32) SK-TRIG-INVERSE-RANGE · Apply

Math 32 In-Depth Curriculum

Precalculus: Trigonometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Non-Right Triangles

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-TRI-002 — I can solve a non-right triangle using the Law of Sines.

SUPPORTING SKILLS

A	Evaluate an inverse trigonometric expression exactly or approximately. (first introduced in Math 32) SK-TRIG-INVERSE-EVALUATE · Apply
I	Apply the Law of Sines to determine missing sides or angles. SK-TRIG-LAW-SINES · Introduce
D	Classify triangle data as SSS, SAS, ASA, AAS, or SSA. (first introduced in Math 32) SK-TRIG-TRIANGLE-DATA · Deepen

M32-TRI-003 — I can determine the number of possible triangles in the ambiguous SSA case.

SUPPORTING SKILLS

I	Test and interpret the possible outcomes of SSA data. SK-TRIG-AMBIGUOUS-CASE · Introduce
D	Apply the Law of Sines to determine missing sides or angles. (first introduced in Math 32) SK-TRIG-LAW-SINES · Deepen
A	Classify triangle data as SSS, SAS, ASA, AAS, or SSA. (first introduced in Math 32) SK-TRIG-TRIANGLE-DATA · Apply

M32-TRI-004 — I can solve a non-right triangle using the Law of Cosines.

SUPPORTING SKILLS

A	Evaluate an inverse trigonometric expression exactly or approximately. (first introduced in Math 32) SK-TRIG-INVERSE-EVALUATE · Apply
I	Apply the Law of Cosines to determine missing sides or angles. SK-TRIG-LAW-COSINES · Introduce
D	Classify triangle data as SSS, SAS, ASA, AAS, or SSA. (first introduced in Math 32) SK-TRIG-TRIANGLE-DATA · Deepen

M32-TRI-005 — I can select and apply an appropriate method to solve a triangle in context.

SUPPORTING SKILLS

A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply
A	Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. (first introduced in Math 22) SK-TRI-TRIG-RATIO · Apply
A	Apply the Law of Cosines to determine missing sides or angles. (first introduced in Math 32) SK-TRIG-LAW-COSINES · Apply
A	Apply the Law of Sines to determine missing sides or angles. (first introduced in Math 32) SK-TRIG-LAW-SINES · Apply
D	Classify triangle data as SSS, SAS, ASA, AAS, or SSA. (first introduced in Math 32) SK-TRIG-TRIANGLE-DATA · Deepen
I	Select right-triangle ratios, Law of Sines, or Law of Cosines from the given data. SK-TRIG-TRIANGLE-METHOD · Introduce

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Trigonometric Identities

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-IDN-001 — I can rewrite trigonometric expressions using reciprocal and quotient identities, including `tan(theta) = sin(theta)/cos(theta)`.

SUPPORTING SKILLS

I	Apply reciprocal and quotient identities. SK-TRIG-IDENTITY-RECIPROCAL-QUOTIENT · Introduce
I	Substitute an identity to rewrite an expression. SK-TRIG-IDENTITY-SUBSTITUTE · Introduce

M32-IDN-002 — I can rewrite trigonometric expressions using the Pythagorean identities.

SUPPORTING SKILLS

I	Apply the Pythagorean trigonometric identities. SK-TRIG-IDENTITY-PYTHAGOREAN · Introduce
D	Substitute an identity to rewrite an expression. (first introduced in Math 32) SK-TRIG-IDENTITY-SUBSTITUTE · Deepen

M32-IDN-003 — I can use even/odd and cofunction identities to rewrite trigonometric expressions.

SUPPORTING SKILLS

D	Test a formula or graph for even or odd symmetry. (first introduced in Math 31) SK-FUNC-SYMMETRY · Deepen
I	Apply sine/cosine cofunction identities. SK-TRIG-IDENTITY-COFUNCTION · Introduce
I	Apply even and odd trigonometric identities. SK-TRIG-IDENTITY-EVEN-ODD · Introduce
D	Substitute an identity to rewrite an expression. (first introduced in Math 32) SK-TRIG-IDENTITY-SUBSTITUTE · Deepen

M32-IDN-004 — I can verify a trigonometric identity using the required identity families.

SUPPORTING SKILLS

A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Choose a productive identity or algebraic manipulation. SK-TRIG-IDENTITY-STRATEGY · Introduce
D	Substitute an identity to rewrite an expression. (first introduced in Math 32) SK-TRIG-IDENTITY-SUBSTITUTE · Deepen
I	Maintain equivalence while transforming one side of an identity. SK-TRIG-IDENTITY-VERIFY · Introduce

SKILL PROGRESSION

I	Introduce
R	Reinforce
D	Deepen
A	Apply

Additional Trigonometric Identities

EXTENSION

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-IDX-001 I can apply double-angle identities.

SUPPORTING SKILLS

I	Apply double-angle identities. SK-TRIG-IDENTITY-DOUBLE-ANGLE · Introduce
A	Substitute an identity to rewrite an expression. (first introduced in Math 32) SK-TRIG-IDENTITY-SUBSTITUTE · Apply

M32-IDX-002 I can apply half-angle identities.

SUPPORTING SKILLS

A	Apply double-angle identities. (first introduced in Math 32) SK-TRIG-IDENTITY-DOUBLE-ANGLE · Apply
I	Apply half-angle identities. SK-TRIG-IDENTITY-HALF-ANGLE · Introduce

M32-IDX-003 I can apply angle sum and difference identities.

SUPPORTING SKILLS

A	Substitute an identity to rewrite an expression. (first introduced in Math 32) SK-TRIG-IDENTITY-SUBSTITUTE · Apply
I	Apply angle sum and difference identities. SK-TRIG-IDENTITY-SUM-DIFFERENCE · Introduce

M32-IDX-004 I can determine exact trigonometric values using extension identities.

SUPPORTING SKILLS

D	Express common trigonometric values exactly. (first introduced in Math 32) SK-TRIG-EXACT-VALUE · Deepen
A	Apply double-angle identities. (first introduced in Math 32) SK-TRIG-IDENTITY-DOUBLE-ANGLE · Apply
A	Apply half-angle identities. (first introduced in Math 32) SK-TRIG-IDENTITY-HALF-ANGLE · Apply
A	Apply angle sum and difference identities. (first introduced in Math 32) SK-TRIG-IDENTITY-SUM-DIFFERENCE · Apply

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Triangle Area from SAS

EXTENSION

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-TRX-001	I can calculate the area of a triangle from two sides and the included angle.
SUPPORTING SKILLS	
A	Express common trigonometric values exactly. (first introduced in Math 32) SK-TRIG-EXACT-VALUE · Apply
I	Calculate triangle area from two sides and the included angle. SK-TRIG-TRIANGLE-AREA-SAS · Introduce
I	Classify triangle data as SSS, SAS, ASA, AAS, or SSA. SK-TRIG-TRIANGLE-DATA · Introduce

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Polar Coordinates

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-POL-001 — I can plot and identify points in polar coordinates.

SUPPORTING SKILLS

I	Plot a point from radius and angle. SK-POL-PLOT · Introduce
A	Determine a reference angle and its quadrant. (first introduced in Math 32) SK-TRIG-ANGLE-REFERENCE · Apply

M32-POL-002 — I can represent a polar point using equivalent coordinate pairs.

SUPPORTING SKILLS

I	Generate equivalent polar coordinate pairs. SK-POL-EQUIVALENT · Introduce
A	Generate and recognize coterminal angles. (first introduced in Math 32) SK-TRIG-ANGLE-COTERMINAL · Apply

M32-POL-003 — I can convert points between polar and rectangular coordinates.

SUPPORTING SKILLS

I	Use coordinate relationships to convert a point between systems. SK-POL-CONVERT-POINT · Introduce
A	Use coordinates and radius to evaluate trigonometric functions. (first introduced in Math 32) SK-TRIG-CIRCLE-POINT · Apply
A	Evaluate an inverse trigonometric expression exactly or approximately. (first introduced in Math 32) SK-TRIG-INVERSE-EVALUATE · Apply

SKILL PROGRESSION

I Introduce R Reinforce D Deepen A Apply

Polar Functions

EXTENSION

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-PLX-001 I can graph and interpret basic polar functions.

SUPPORTING SKILLS

A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply
D	Plot a point from radius and angle. (first introduced in Math 32) SK-POL-PLOT · Deepen
A	Locate quarter-period key points of a periodic function. (first introduced in Math 32) SK-TRIG-GRAPH-KEYPOINTS · Apply

M32-PLX-002 I can convert equations between polar and rectangular forms.

SUPPORTING SKILLS

A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Substitute coordinate identities to convert an equation between systems. SK-POL-CONVERT-EQUATION · Introduce
A	Apply the Pythagorean trigonometric identities. (first introduced in Math 32) SK-TRIG-IDENTITY-PYTHAGOREAN · Apply

Math 32 In-Depth Curriculum

Precalculus: Trigonometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Vectors

EXTENSION

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-VEC-001 — **I can represent a vector geometrically and in component form.**

SUPPORTING SKILLS

A	Add congruence, parallel, perpendicular, and angle markings to a diagram. (first introduced in Math 22) SK-GEO-DIAGRAM-MARK · Apply
I	Write a vector in component form. SK-VEC-COMPONENTS · Introduce

M32-VEC-002 — **I can add, subtract, and multiply vectors by scalars.**

SUPPORTING SKILLS

A	Add, subtract, multiply, and divide signed integers. (inherited from middle school) SK-NUM-INTEGER-OPERATE · Apply
I	Perform component-wise vector operations. SK-VEC-OPERATE · Introduce

M32-VEC-003 — **I can determine vector magnitude, direction, and unit vector.**

SUPPORTING SKILLS

A	Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula. (first introduced in Math 22) SK-COORD-DISTANCE-PYTHAGOREAN · Apply
I	Determine a vector's direction angle with correct quadrant. SK-VEC-DIRECTION · Introduce
I	Calculate vector magnitude using the distance formula. SK-VEC-MAGNITUDE · Introduce
I	Normalize a nonzero vector to obtain a unit vector. SK-VEC-UNIT · Introduce

M32-VEC-004 — **I can resolve a vector into horizontal and vertical components.**

SUPPORTING SKILLS

A	Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. (first introduced in Math 22) SK-TRI-TRIG-RATIO · Apply
D	Write a vector in component form. (first introduced in Math 32) SK-VEC-COMPONENTS · Deepen
A	Determine a vector's direction angle with correct quadrant. (first introduced in Math 32) SK-VEC-DIRECTION · Apply

M32-VEC-005 — **I can perform basic operations with vectors in three dimensions.**

SUPPORTING SKILLS

I	Represent and operate on vectors with three components. SK-VEC-3D · Introduce
D	Calculate vector magnitude using the distance formula. (first introduced in Math 32) SK-VEC-MAGNITUDE · Deepen
D	Perform component-wise vector operations. (first introduced in Math 32) SK-VEC-OPERATE · Deepen

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Parametric Equations

EXTENSION

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-PAR-001 — I can graph a plane curve defined by parametric equations with its orientation.

SUPPORTING SKILLS	
A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply
I	Graph a parametric curve with its direction of motion. SK-PAR-GRAPH · Introduce
I	Generate ordered pairs from parametric equations. SK-PAR-TABLE · Introduce

M32-PAR-002 — I can eliminate a parameter to obtain a rectangular equation.

SUPPORTING SKILLS	
A	Isolate a specified variable in an equation or formula. (first introduced in Math 12) SK-ALG-VARIABLE-ISOLATE · Apply
I	Eliminate a parameter algebraically. SK-PAR-ELIMINATE · Introduce
A	Substitute an equivalent expression for a variable. (first introduced in Math 12) SK-SYS-SUBSTITUTE · Apply

M32-PAR-003 — I can create or interpret parametric equations for motion in the plane.

SUPPORTING SKILLS	
A	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Apply
I	Interpret or construct parametric equations for planar motion. SK-PAR-MODEL · Introduce
D	Generate ordered pairs from parametric equations. (first introduced in Math 32) SK-PAR-TABLE · Deepen
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

Math 39

Precalculus with Data Analysis

Math 39 Course at a Glance

Precalculus with Data Analysis

Linear Modeling and Residuals

REQUIRED

LEARNING OBJECTIVES

M39-LIN-001	I can calculate and interpret average rate of change with appropriate units.
M39-LIN-002	I can construct a linear function from two data points, a table, a graph, or a context.
M39-LIN-003	I can fit a linear regression model to data using technology and interpret its parameters.
M39-LIN-004	I can interpret a correlation coefficient as evidence about the direction and strength of a linear relationship.
M39-LIN-005	I can calculate residuals, construct a residual plot, and use its pattern to evaluate a model.
M39-LIN-006	I can use a linear model to make and interpret predictions, including solving for either variable.

Exponential Modeling

REQUIRED

LEARNING OBJECTIVES

M39-EXP-001	I can distinguish linear and exponential change using rates, multipliers, tables, graphs, or contexts.
M39-EXP-002	I can construct an exponential function from two data points, a table, or a context.
M39-EXP-003	I can interpret the initial value, growth or decay factor, and growth or decay rate of an exponential model.
M39-EXP-004	I can determine and interpret doubling time or half-life from an exponential model.
M39-EXP-005	I can solve exponential and logarithmic equations in modeling contexts.
M39-EXP-006	I can fit an exponential regression model using technology and evaluate its fit.
M39-EXP-007	I can compare linear and exponential models and justify which is more appropriate for a data set or context.

Power Functions and Function Comparison

REQUIRED

LEARNING OBJECTIVES

M39-POW-001	I can identify a power function and interpret the parameters in y = kx^p.
M39-POW-002	I can construct a power function from two data points or a proportional relationship.
M39-POW-003	I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.
M39-POW-004	I can recognize and model direct, inverse, and other power-variation relationships.
M39-POW-005	I can solve equations involving integer or rational powers.
M39-POW-006	I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.
M39-POW-007	I can fit a power regression model and interpret its parameters and correlation evidence.

Function Combinations and Polynomials

REQUIRED

LEARNING OBJECTIVES

M39-POL-001	I can add, subtract, multiply, and divide functions represented by tables, graphs, or formulas.
M39-POL-002	I can determine where a combination of functions is positive, zero, or undefined.
M39-POL-003	I can determine polynomial end behavior from degree and leading coefficient.
M39-POL-004	I can identify zeros and multiplicities and relate them to polynomial graph behavior.
M39-POL-005	I can estimate a polynomial's possible degree from its zeros, turning points, inflection points, and end behavior.
M39-POL-006	I can construct a polynomial function from specified zeros, multiplicities, and a point.
M39-POL-007	I can match polynomial formulas and graphs using intercepts, multiplicity, degree, and end behavior.
M39-POL-008	I can build and optimize a polynomial model in a geometric or economic context.

Math 39 Course at a Glance

Precalculus with Data Analysis

Describing Data

REQUIRED

LEARNING OBJECTIVES

M39-STA-001	I can distinguish categorical and quantitative data.
M39-STA-002	I can construct and interpret frequency and relative-frequency tables.
M39-STA-003	I can construct and interpret bar charts, pie charts, and histograms using appropriate technology.
M39-STA-004	I can identify misleading features in a data display.
M39-STA-005	I can describe a distribution's modality, symmetry, skewness, and outliers.
M39-STA-006	I can calculate and interpret mean, median, and mode and select an appropriate measure of center.
M39-STA-007	I can calculate and interpret range, standard deviation, quartiles, and interquartile range.
M39-STA-008	I can construct and interpret a five-number summary and boxplot.
M39-STA-009	I can use percentiles and z-scores to locate and compare values within distributions.
M39-STA-010	I can compare distributions using their shape, center, spread, and unusual features.

Probability

REQUIRED

LEARNING OBJECTIVES

M39-PRO-001	I can organize categorical data in a contingency table or Venn diagram.
M39-PRO-002	I can calculate and interpret empirical, marginal, joint, and conditional probabilities from a contingency table.
M39-PRO-003	I can identify an experiment, outcomes, events, and a sample space.
M39-PRO-004	I can calculate theoretical probabilities when outcomes are equally likely.
M39-PRO-005	I can use complements to calculate probabilities.
M39-PRO-006	I can calculate probabilities of compound 'and' events and distinguish independent from dependent events.
M39-PRO-007	I can calculate probabilities of compound 'or' events and distinguish overlapping from disjoint events.
M39-PRO-008	I can compare experimental and theoretical probability and explain the law of large numbers.

Expected Value

EXTENSION

LEARNING OBJECTIVES

M39-EV-001	I can calculate the expected value of a discrete probability situation.
M39-EV-002	I can interpret expected value to evaluate a game, risk, or long-run decision.

Linearization of Nonlinear Data

EXTENSION

LEARNING OBJECTIVES

M39-POW-008	I can linearize power or exponential data and translate between the linearized and original models.

SKILL PROGRESSION

I Introduce | R Reinforce | D Deepen | A Apply

Linear Modeling and Residuals

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M39-LIN-001 I can calculate and interpret average rate of change with appropriate units.

SUPPORTING SKILLS

D	Calculate average rate of change over an interval. (first introduced in Math 21) SK-FUNC-AROC · Deepen
D	Interpret average rate of change with contextual units. (first introduced in Math 21) SK-FUNC-AROC-INTERPRET · Deepen
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

M39-LIN-002 I can construct a linear function from two data points, a table, a graph, or a context.

SUPPORTING SKILLS

R	Convert among common forms of a linear equation. (first introduced in Math 12) SK-LIN-FORM-CONVERT · Reinforce
A	Calculate slope from two points. (first introduced in Math 12) SK-LIN-SLOPE-POINTS · Apply
A	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Apply

M39-LIN-003 I can fit a linear regression model to data using technology and interpret its parameters.

SUPPORTING SKILLS

D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
I	Use technology to determine a least-squares linear regression model. SK-REG-LINEAR-FIT · Introduce

M39-LIN-004 I can interpret a correlation coefficient as evidence about the direction and strength of a linear relationship.

SUPPORTING SKILLS

I	Interpret the sign and magnitude of a correlation coefficient in context. SK-REG-CORRELATION-INTERPRET · Introduce

M39-LIN-005 I can calculate residuals, construct a residual plot, and use its pattern to evaluate a model.

SUPPORTING SKILLS

I	Use residual patterns and contextual evidence to judge whether a model is appropriate. SK-REG-MODEL-ASSESS · Introduce
I	Calculate an observed value minus its model-predicted value. SK-REG-RESIDUAL-CALCULATE · Introduce
I	Construct and interpret a residual plot. SK-REG-RESIDUAL-PLOT · Introduce

M39-LIN-006 I can use a linear model to make and interpret predictions, including solving for either variable.

SUPPORTING SKILLS

A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply
D	Solve for an input associated with a specified output. (first introduced in Math 21) SK-FUNC-SOLVE-INPUT · Deepen
A	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Apply

SKILL PROGRESSION

I Introduce | R Reinforce | D Deepen | A Apply

Exponential Modeling

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M39-EXP-001 I can distinguish linear and exponential change using rates, multipliers, tables, graphs, or contexts.

SUPPORTING SKILLS

D	Use constant differences or ratios to distinguish linear and exponential patterns. (first introduced in Math 31) SK-EXP-DIFFERENCE-RATIO · Deepen
D	Compare functions shown in different representations. (first introduced in Math 21) SK-FUNC-COMPARE-REP · Deepen

M39-EXP-002 I can construct an exponential function from two data points, a table, or a context.

SUPPORTING SKILLS

D	Identify initial amount, rate, and time interval in context. (first introduced in Math 31) SK-EXP-MODEL-CONTEXT · Deepen
D	Determine an exponential model from tabular values. (first introduced in Math 31) SK-EXP-MODEL-TABLE · Deepen
D	Solve for an exponential model using two points. (first introduced in Math 31) SK-EXP-MODEL-TWO-POINTS · Deepen

M39-EXP-003 I can interpret the initial value, growth or decay factor, and growth or decay rate of an exponential model.

SUPPORTING SKILLS

D	Convert a percent increase or decrease to a growth or decay factor. (first introduced in Math 31) SK-EXP-PERCENT-FACTOR · Deepen
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen

M39-EXP-004 I can determine and interpret doubling time or half-life from an exponential model.

SUPPORTING SKILLS

A	Identify initial amount, rate, and time interval in context. (first introduced in Math 31) SK-EXP-MODEL-CONTEXT · Apply
D	Translate a doubling-time or half-life question into an exponential equation. (first introduced in Math 31) SK-LOG-DOUBLING-HALF · Deepen

M39-EXP-005 I can solve exponential and logarithmic equations in modeling contexts.

SUPPORTING SKILLS

A	Reject solutions that make a logarithm argument nonpositive. (first introduced in Math 31) SK-LOG-DOMAIN-CHECK · Apply
R	Isolate an exponential expression and apply a logarithm. (first introduced in Math 31) SK-LOG-SOLVE-EXP · Reinforce
R	Combine or isolate logarithms before converting to exponential form. (first introduced in Math 31) SK-LOG-SOLVE-LOG · Reinforce

M39-EXP-006 I can fit an exponential regression model using technology and evaluate its fit.

SUPPORTING SKILLS

A	Interpret the sign and magnitude of a correlation coefficient in context. (first introduced in Math 39) SK-REG-CORRELATION-INTERPRET · Apply
I	Use technology to determine an exponential regression model. SK-REG-EXPONENTIAL-FIT · Introduce
D	Use residual patterns and contextual evidence to judge whether a model is appropriate. (first introduced in Math 39) SK-REG-MODEL-ASSESS · Deepen

Math 39 In-Depth Curriculum

Precalculus with Data Analysis

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M39-EXP-007	I can compare linear and exponential models and justify which is more appropriate for a data set or context.
SUPPORTING SKILLS	
A	Compare functions shown in different representations. (first introduced in Math 21) SK-FUNC-COMPARE-REP · Apply
I	Select a function family whose behavior fits a table, graph, or context. SK-MODEL-FAMILY-SELECT · Introduce
D	Use residual patterns and contextual evidence to judge whether a model is appropriate. (first introduced in Math 39) SK-REG-MODEL-ASSESS · Deepen

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Power Functions and Function Comparison

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M39-POW-001 — I can identify a power function and interpret the parameters in y = kx^p.

SUPPORTING SKILLS

I	Interpret the coefficient and exponent in a power function. SK-POWER-PARAMETER-INTERPRET · Introduce
I	Represent and interpret a power-variation relationship. SK-VAR-POWER · Introduce

M39-POW-002 — I can construct a power function from two data points or a proportional relationship.

SUPPORTING SKILLS

A	Isolate a specified variable in an equation or formula. (first introduced in Math 12) SK-ALG-VARIABLE-ISOLATE · Apply
I	Determine a power function from two points or a proportional relationship. SK-POWER-CONSTRUCT · Introduce

M39-POW-003 — I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.

SUPPORTING SKILLS

I	Determine power-function behavior as the input approaches zero or positive or negative infinity. SK-POWER-END-BEHAVIOR · Introduce
I	Relate the coefficient and exponent of a power function to its graph shape. SK-POWER-SHAPE · Introduce
D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen

M39-POW-004 — I can recognize and model direct, inverse, and other power-variation relationships.

SUPPORTING SKILLS

I	Represent and interpret a direct-variation relationship. SK-VAR-DIRECT · Introduce
I	Represent and interpret an inverse-variation relationship. SK-VAR-INVERSE · Introduce
D	Represent and interpret a power-variation relationship. (first introduced in Math 39) SK-VAR-POWER · Deepen

M39-POW-005 — I can solve equations involving integer or rational powers.

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
A	Interpret numerator and denominator roles in a rational exponent. (first introduced in Math 21) SK-EXP-RATIONAL · Apply

M39-POW-006 — I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.

SUPPORTING SKILLS

D	Compare functions shown in different representations. (first introduced in Math 21) SK-FUNC-COMPARE-REP · Deepen
I	Compare the long-run growth or decay of functions from different families. SK-FUNC-DOMINANCE · Introduce

Math 39 In-Depth Curriculum

Precalculus with Data Analysis

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M39-POW-007	I can fit a power regression model and interpret its parameters and correlation evidence.
SUPPORTING SKILLS	
D	Interpret the coefficient and exponent in a power function. (first introduced in Math 39) SK-POWER-PARAMETER-INTERPRET · Deepen
A	Interpret the sign and magnitude of a correlation coefficient in context. (first introduced in Math 39) SK-REG-CORRELATION-INTERPRET · Apply
I	Use technology to determine a power regression model. SK-REG-POWER-FIT · Introduce

SKILL PROGRESSION

I Introduce | R Reinforce | D Deepen | A Apply

Function Combinations and Polynomials

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M39-POL-001 — **I can add, subtract, multiply, and divide functions represented by tables, graphs, or formulas.**

SUPPORTING SKILLS

A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply
I	Add, subtract, multiply, and divide function outputs. SK-FUNC-OPERATIONS · Introduce

M39-POL-002 — **I can determine where a combination of functions is positive, zero, or undefined.**

SUPPORTING SKILLS

I	Determine the domain of a sum, difference, product, quotient, or composition. SK-FUNC-DOMAIN-COMBINE · Introduce
D	Add, subtract, multiply, and divide function outputs. (first introduced in Math 39) SK-FUNC-OPERATIONS · Deepen

M39-POL-003 — **I can determine polynomial end behavior from degree and leading coefficient.**

SUPPORTING SKILLS

I	Infer polynomial end behavior from degree and leading coefficient. SK-POLY-END-BEHAVIOR · Introduce
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

M39-POL-004 — **I can identify zeros and multiplicities and relate them to polynomial graph behavior.**

SUPPORTING SKILLS

I	Relate zero multiplicity to local graph behavior. SK-POLY-MULTIPLICITY · Introduce
D	Determine real and complex zeros using an appropriate method. (first introduced in Math 21) SK-POLY-ZEROS · Deepen

M39-POL-005 — **I can estimate a polynomial's possible degree from its zeros, turning points, inflection points, and end behavior.**

SUPPORTING SKILLS

D	Infer polynomial end behavior from degree and leading coefficient. (first introduced in Math 39) SK-POLY-END-BEHAVIOR · Deepen
D	Relate zero multiplicity to local graph behavior. (first introduced in Math 39) SK-POLY-MULTIPLICITY · Deepen

M39-POL-006 — **I can construct a polynomial function from specified zeros, multiplicities, and a point.**

SUPPORTING SKILLS

D	Build a polynomial from zeros, multiplicities, and a point. (first introduced in Math 31) SK-POLY-CONSTRUCT · Deepen
A	Relate zero multiplicity to local graph behavior. (first introduced in Math 39) SK-POLY-MULTIPLICITY · Apply
A	Determine real and complex zeros using an appropriate method. (first introduced in Math 21) SK-POLY-ZEROS · Apply

Math 39 In-Depth Curriculum

Precalculus with Data Analysis

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M39-POL-007 — I can match polynomial formulas and graphs using intercepts, multiplicity, degree, and end behavior.

SUPPORTING SKILLS

D	Infer polynomial end behavior from degree and leading coefficient. (first introduced in Math 39) SK-POLY-END-BEHAVIOR · Deepen
A	Relate zero multiplicity to local graph behavior. (first introduced in Math 39) SK-POLY-MULTIPLICITY · Apply
A	Determine real and complex zeros using an appropriate method. (first introduced in Math 21) SK-POLY-ZEROS · Apply

M39-POL-008 — I can build and optimize a polynomial model in a geometric or economic context.

SUPPORTING SKILLS

A	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Apply
A	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Apply
D	Interpret vertex and zeros in an application. (first introduced in Math 21) SK-QUAD-MODEL-FEATURES · Deepen

Math 39 In-Depth Curriculum

Precalculus with Data Analysis

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Describing Data

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M39-STA-001 I can distinguish categorical and quantitative data.

SUPPORTING SKILLS

I	Classify data as categorical or quantitative. SK-STAT-DATA-TYPE · Introduce

M39-STA-002 I can construct and interpret frequency and relative-frequency tables.

SUPPORTING SKILLS

I	Construct and interpret a frequency table. SK-STAT-FREQUENCY-TABLE · Introduce
I	Calculate and interpret relative frequency. SK-STAT-RELATIVE-FREQUENCY · Introduce

M39-STA-003 I can construct and interpret bar charts, pie charts, and histograms using appropriate technology.

SUPPORTING SKILLS

I	Construct and interpret bar charts and pie charts. SK-STAT-CATEGORICAL-DISPLAY · Introduce
I	Construct and interpret a histogram for quantitative data. SK-STAT-HISTOGRAM · Introduce

M39-STA-004 I can identify misleading features in a data display.

SUPPORTING SKILLS

I	Identify perceptual distortion, misleading scales, and other misleading features in data displays. SK-STAT-GRAPH-CRITIQUE · Introduce

M39-STA-005 I can describe a distribution's modality, symmetry, skewness, and outliers.

SUPPORTING SKILLS

I	Describe a distribution by modality, symmetry, and skewness. SK-STAT-DISTRIBUTION-SHAPE · Introduce
I	Identify and interpret unusual observations or outliers. SK-STAT-OUTLIER-IDENTIFY · Introduce

M39-STA-006 I can calculate and interpret mean, median, and mode and select an appropriate measure of center.

SUPPORTING SKILLS

I	Select an appropriate measure of center based on distribution shape and context. SK-STAT-CENTER-SELECT · Introduce
I	Calculate and interpret the arithmetic mean. SK-STAT-MEAN · Introduce
I	Calculate and interpret the median. SK-STAT-MEDIAN · Introduce
I	Identify and interpret the mode or modes. SK-STAT-MODE · Introduce

M39-STA-007 I can calculate and interpret range, standard deviation, quartiles, and interquartile range.

SUPPORTING SKILLS

I	Determine quartiles and calculate and interpret the interquartile range. SK-STAT-QUARTILES-IQR · Introduce
I	Calculate and interpret the range. SK-STAT-RANGE · Introduce
I	Calculate and interpret standard deviation as a measure of spread. SK-STAT-STANDARD-DEVIATION · Introduce

Math 39 In-Depth Curriculum

Precalculus with Data Analysis

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M39-STA-008	I can construct and interpret a five-number summary and boxplot.
SUPPORTING SKILLS	
I	Construct and interpret a boxplot. SK-STAT-BOXPLOT · Introduce
I	Determine and interpret a five-number summary. SK-STAT-FIVE-NUMBER · Introduce

M39-STA-009	I can use percentiles and z-scores to locate and compare values within distributions.
SUPPORTING SKILLS	
I	Determine and interpret the percentile rank of a value. SK-STAT-PERCENTILE · Introduce
I	Calculate and interpret a z-score. SK-STAT-ZSCORE · Introduce

M39-STA-010	I can compare distributions using their shape, center, spread, and unusual features.
SUPPORTING SKILLS	
A	Select an appropriate measure of center based on distribution shape and context. (first introduced in Math 39) SK-STAT-CENTER-SELECT · Apply
I	Compare distributions using shape, center, spread, and unusual features. SK-STAT-DISTRIBUTION-COMPARE · Introduce
A	Describe a distribution by modality, symmetry, and skewness. (first introduced in Math 39) SK-STAT-DISTRIBUTION-SHAPE · Apply

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Probability

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M39-PRO-001 I can organize categorical data in a contingency table or Venn diagram.

SUPPORTING SKILLS

I	Construct and interpret a contingency table. SK-PROB-CONTINGENCY-TABLE · Introduce
I	Represent categorical events and their overlap with a Venn diagram. SK-PROB-VENN-DIAGRAM · Introduce

M39-PRO-002 I can calculate and interpret empirical, marginal, joint, and conditional probabilities from a contingency table.

SUPPORTING SKILLS

I	Calculate and interpret a conditional probability. SK-PROB-CONDITIONAL · Introduce
I	Calculate probability from observed relative frequency. SK-PROB-EMPIRICAL · Introduce
I	Calculate and interpret marginal and joint probabilities. SK-PROB-MARGINAL-JOINT · Introduce

M39-PRO-003 I can identify an experiment, outcomes, events, and a sample space.

SUPPORTING SKILLS

I	Identify outcomes, events, and the sample space of an experiment. SK-PROB-SAMPLE-SPACE · Introduce

M39-PRO-004 I can calculate theoretical probabilities when outcomes are equally likely.

SUPPORTING SKILLS

A	Identify outcomes, events, and the sample space of an experiment. (first introduced in Math 39) SK-PROB-SAMPLE-SPACE · Apply
I	Calculate theoretical probability for equally likely outcomes. SK-PROB-THEORETICAL · Introduce

M39-PRO-005 I can use complements to calculate probabilities.

SUPPORTING SKILLS

I	Use the complement rule to calculate probability. SK-PROB-COMPLEMENT · Introduce

M39-PRO-006 I can calculate probabilities of compound 'and' events and distinguish independent from dependent events.

SUPPORTING SKILLS

I	Calculate the probability of a compound and event. SK-PROB-AND · Introduce
D	Calculate and interpret a conditional probability. (first introduced in Math 39) SK-PROB-CONDITIONAL · Deepen
I	Determine whether events are independent or dependent. SK-PROB-INDEPENDENCE · Introduce

M39-PRO-007 I can calculate probabilities of compound 'or' events and distinguish overlapping from disjoint events.

SUPPORTING SKILLS

I	Determine whether events are overlapping or disjoint. SK-PROB-DISJOINT · Introduce
I	Calculate the probability of a compound or event. SK-PROB-OR · Introduce

Math 39 In-Depth Curriculum

Precalculus with Data Analysis

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M39-PRO-008	I can compare experimental and theoretical probability and explain the law of large numbers.
SUPPORTING SKILLS	
D	Calculate probability from observed relative frequency. (first introduced in Math 39) SK-PROB-EMPIRICAL · Deepen
I	Explain how experimental probability stabilizes with many trials. SK-PROB-LARGE-NUMBERS · Introduce
D	Calculate theoretical probability for equally likely outcomes. (first introduced in Math 39) SK-PROB-THEORETICAL · Deepen

Expected Value

EXTENSION

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M39-EV-001	**I can calculate the expected value of a discrete probability situation.**
SUPPORTING SKILLS	
I	Calculate the expected value of a discrete probability situation. SK-PROB-EXPECTED-VALUE · Introduce

M39-EV-002	**I can interpret expected value to evaluate a game, risk, or long-run decision.**
SUPPORTING SKILLS	
I	Interpret expected value as a long-run average for decision-making. SK-PROB-EXPECTED-INTERPRET · Introduce
D	Calculate the expected value of a discrete probability situation. (first introduced in Math 39) SK-PROB-EXPECTED-VALUE · Deepen

SKILL PROGRESSION

I	Introduce
R	Reinforce
D	Deepen
A	Apply

Linearization of Nonlinear Data

EXTENSION

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M39-POW-008	**I can linearize power or exponential data and translate between the linearized and original models.**
SUPPORTING SKILLS	
I	Transform exponential data so that a linear model can be fit to its logarithms. SK-REG-LOG-LINEARIZE · Introduce
I	Transform power data so that a linear model can be fit on logarithmic axes. SK-REG-LOGLOG-LINEARIZE · Introduce
I	Translate a linearized regression equation back to its original nonlinear form. SK-REG-UNLINEARIZE · Introduce

Math 49

Precalculus with Limits

MIDDLESEX SCHOOL | MATHEMATICS

Math 49 Course at a Glance

Precalculus with Limits

TEXTBOOK | OpenStax Precalculus 2e (Jay Abramson, 2021)

Review Content

REVIEW — SECTIONS 1.2, 1.5, 4.1, 4.2, 4.3, 4.4, 4.5, 4.7

LEARNING OBJECTIVES

M49-FND-001	I can determine the domain and range of a function from a formula, graph, or table.
M49-FND-002	I can describe and apply transformations of a parent function.
M49-FND-004	I can analyze exponential functions using their equations, graphs, and contexts.
M49-FND-005	I can analyze logarithmic functions using their equations, graphs, and contexts.
M49-FND-006	I can solve exponential and logarithmic equations using equivalent forms and logarithm properties.

Function Operations, Composition, and Inverses

REQUIRED — SECTIONS 1.4, 1.7

LEARNING OBJECTIVES

M49-FND-003	I can determine whether a function is even, odd, or neither.
M49-OPS-001	I can add, subtract, multiply, and divide functions and state the resulting domain.
M49-OPS-002	I can evaluate a composition from formulas, graphs, or tables.
M49-OPS-003	I can write a formula for the composition of two functions and determine its domain.
M49-OPS-004	I can decompose a function into a composition of simpler functions.
M49-OPS-005	I can determine whether a function has an inverse on a given domain.
M49-OPS-006	I can find and verify an inverse function algebraically.
M49-OPS-007	I can interpret the relationship between a function and its inverse from graphs or tables.

Power, Polynomial, and Rational Functions

REQUIRED — SECTIONS 3.3, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, 4.2, 4.4

LEARNING OBJECTIVES

M49-PWR-001	I can identify a power function and interpret the parameters in y = kx^p.
M49-PWR-002	I can construct a power function from two data points or a proportional relationship.
M49-PWR-003	I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.
M49-PWR-004	I can recognize and model direct, inverse, and other power-variation relationships.
M49-PWR-005	I can solve equations involving integer or rational powers.
M49-PWR-006	I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.
M49-PWR-008	I can analyze power functions with fractional exponents, including their domains, graph shapes, and behavior near zero.
M49-ALG-002	I can predict the end behavior of a polynomial from its degree and leading coefficient.
M49-ALG-003	I can determine polynomial zeros and their multiplicities.
M49-ALG-004	I can construct a polynomial function from specified zeros or graphical features.
M49-ALG-005	I can sketch and analyze a polynomial function using intercepts, multiplicity, and end behavior.
M49-ALG-006	I can identify holes and vertical, horizontal, or slant asymptotes of a rational function.
M49-ALG-007	I can sketch and analyze a rational function using its algebraic and asymptotic features.
M49-ALG-008	I can build or select a power, polynomial, or rational model for a context.

Limits and Function Comparison

REQUIRED — SECTIONS 12.1

LEARNING OBJECTIVES

M49-LIM-001	I can interpret limit notation as a statement about function behavior near an input.
M49-LIM-002	I can estimate a limit from a graph or table.
M49-LIM-003	I can evaluate a limit algebraically using direct substitution or simplification.
M49-LIM-005	I can interpret an infinite limit as vertical-asymptotic behavior.
M49-LIM-006	I can determine limits at infinity and relate them to long-run behavior.
M49-LIM-007	I can compare the long-run growth of polynomial, exponential, logarithmic, and rational functions.

Math 49 Course at a Glance

Precalculus with Limits

TEXTBOOK | OpenStax Precalculus 2e (Jay Abramson, 2021)

Sequences and Series

REQUIRED — SECTIONS 11.1, 11.2, 11.3, 11.4

LEARNING OBJECTIVES

M49-SEQ-001	I can evaluate and graph a sequence defined explicitly or recursively.
M49-SEQ-002	I can write explicit and recursive formulas for arithmetic sequences.
M49-SEQ-003	I can write explicit and recursive formulas for geometric sequences.
M49-SEQ-004	I can distinguish arithmetic, geometric, and other sequences from formulas, tables, or contexts.
M49-SEQ-005	I can create and interpret a sequence model for discrete change.
M49-SER-001	I can interpret and use summation notation.
M49-SER-002	I can calculate a finite arithmetic series.
M49-SER-003	I can calculate a finite geometric series.
M49-SER-004	I can determine whether an infinite geometric series converges and find its sum when it does.
M49-SER-005	I can create and interpret a series model for accumulated change.

Additional Limit Concepts

EXTENSION — SECTIONS 12.1

LEARNING OBJECTIVES

M49-LIM-004	I can determine one-sided limits and decide whether a two-sided limit exists.
M49-LIM-008	I can distinguish a function value from a limit and determine continuity at a point.

Parametric Equations

EXTENSION — SECTIONS 8.6, 8.7

LEARNING OBJECTIVES

M49-PAR-001	I can graph a plane curve defined by parametric equations with its orientation.
M49-PAR-002	I can eliminate a parameter to obtain a rectangular equation.
M49-PAR-003	I can create or interpret parametric equations for motion in the plane.

Conic Sections

EXTENSION — SECTIONS 10.1, 10.2, 10.3, 10.5

LEARNING OBJECTIVES

M49-CON-001	I can classify a circle, ellipse, or hyperbola from its equation or defining geometric relationship.
M49-CON-002	I can use completing the square to rewrite a general-form conic equation in standard form.
M49-CON-003	I can write, interpret, and graph implicit and parametric equations of a circle.
M49-CON-004	I can write, interpret, and graph implicit and parametric equations of an ellipse.
M49-CON-005	I can write, interpret, and graph implicit and parametric equations of a hyperbola.
M49-CON-006	I can determine the center, vertices, foci, axes, or asymptotes needed to describe a circle, ellipse, or hyperbola.
M49-CON-007	I can explain the geometric definitions of circles, ellipses, and hyperbolas using foci and distance relationships.
M49-CON-008	I can explain a parabola as the set of points equidistant from a focus and directrix.

Math 49 In-Depth Curriculum

Precalculus with Limits

TEXTBOOK | OpenStax Precalculus 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I	Introduce
R	Reinforce
D	Deepen
A	Apply

Review Content

REVIEW

UNIT TEXTBOOK COVERAGE

1.2 Domain and Range; 1.5 Transformation of Functions; 4.1 Exponential Functions; 4.2 Graphs of Exponential Functions; 4.3 Logarithmic Functions; 4.4 Graphs of Logarithmic Functions; 4.5 Logarithmic Properties; 4.7 Exponential and Logarithmic Models

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M49-FND-001 — I can determine the domain and range of a function from a formula, graph, or table.

TEXTBOOK SECTIONS 1.2 Domain and Range

SUPPORTING SKILLS

D	Determine domain restrictions from denominators, even roots, and logarithms. (first introduced in Math 31) SK-FUNC-DOMAIN-FORMULA · Deepen
R	Read domain and range from a graph. (first introduced in Math 12) SK-FUNC-DOMAIN-RANGE-GRAPH · Reinforce
R	Read domain and range from a table. (first introduced in Math 12) SK-FUNC-DOMAIN-RANGE-TABLE · Reinforce

M49-FND-002 — I can describe and apply transformations of a parent function.

TEXTBOOK SECTIONS 1.5 Transformation of Functions

SUPPORTING SKILLS

R	Interpret transformations applied to function inputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-HORIZONTAL · Reinforce
R	Distinguish stretches from compressions and horizontal effects from vertical effects. (first introduced in Math 31) SK-FUNC-TRANSFORM-SCALE · Reinforce
R	Interpret transformations applied to function outputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-VERTICAL · Reinforce

M49-FND-004 — I can analyze exponential functions using their equations, graphs, and contexts.

TEXTBOOK SECTIONS 4.1 Exponential Functions; 4.2 Graphs of Exponential Functions; 4.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

R	Identify horizontal asymptotic behavior of an exponential graph. (first introduced in Math 31) SK-EXP-ASYMPTOTE · Reinforce
R	Identify initial amount, rate, and time interval in context. (first introduced in Math 31) SK-EXP-MODEL-CONTEXT · Reinforce
A	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Apply

M49-FND-005 — I can analyze logarithmic functions using their equations, graphs, and contexts.

TEXTBOOK SECTIONS 4.3 Logarithmic Functions; 4.4 Graphs of Logarithmic Functions; 4.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

R	Identify vertical asymptotic behavior of a logarithmic graph. (first introduced in Math 31) SK-LOG-ASYMPTOTE · Reinforce
R	Interpret a logarithm as an exponent. (first introduced in Math 31) SK-LOG-DEFINITION · Reinforce
A	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Apply

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M49-FND-006

I can solve exponential and logarithmic equations using equivalent forms and logarithm properties.

TEXTBOOK SECTIONS 4.3 Logarithmic Functions; 4.5 Logarithmic Properties; 4.7 Exponential and Logarithmic Models

SUPPORTING SKILLS

R	Rewrite exponential expressions using a common base when possible. (first introduced in Math 31) SK-EXP-REWRITE-BASE · Reinforce
A	Reject solutions that make a logarithm argument nonpositive. (first introduced in Math 31) SK-LOG-DOMAIN-CHECK · Apply
A	Apply the logarithm power property in either direction. (first introduced in Math 31) SK-LOG-POWER · Apply
A	Apply the logarithm product property in either direction. (first introduced in Math 31) SK-LOG-PRODUCT · Apply
A	Apply the logarithm quotient property in either direction. (first introduced in Math 31) SK-LOG-QUOTIENT · Apply
R	Isolate an exponential expression and apply a logarithm. (first introduced in Math 31) SK-LOG-SOLVE-EXP · Reinforce
R	Combine or isolate logarithms before converting to exponential form. (first introduced in Math 31) SK-LOG-SOLVE-LOG · Reinforce

SKILL PROGRESSION

I	Introduce
R	Reinforce
D	Deepen
A	Apply

Function Operations, Composition, and Inverses

REQUIRED

UNIT TEXTBOOK COVERAGE
1.4 Composition of Functions; 1.7 Inverse Functions

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M49-FND-003 — I can determine whether a function is even, odd, or neither.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

D	Test a formula or graph for even or odd symmetry. (first introduced in Math 31) SK-FUNC-SYMMETRY · Deepen

M49-OPS-001 — I can add, subtract, multiply, and divide functions and state the resulting domain.

TEXTBOOK SECTIONS 1.4 Composition of Functions

SUPPORTING SKILLS

A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Determine the domain of a sum, difference, product, quotient, or composition. SK-FUNC-DOMAIN-COMBINE · Introduce
I	Add, subtract, multiply, and divide function outputs. SK-FUNC-OPERATIONS · Introduce

M49-OPS-002 — I can evaluate a composition from formulas, graphs, or tables.

TEXTBOOK SECTIONS 1.4 Composition of Functions

SUPPORTING SKILLS

D	Substitute one function formula into another. (first introduced in Math 31) SK-FUNC-COMPOSE-FORMULA · Deepen
D	Follow intermediate values to compose functions from tables. (first introduced in Math 31) SK-FUNC-COMPOSE-TABLE · Deepen
A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply

M49-OPS-003 — I can write a formula for the composition of two functions and determine its domain.

TEXTBOOK SECTIONS 1.4 Composition of Functions

SUPPORTING SKILLS

D	Substitute one function formula into another. (first introduced in Math 31) SK-FUNC-COMPOSE-FORMULA · Deepen
D	Determine the domain of a sum, difference, product, quotient, or composition. (first introduced in Math 39) SK-FUNC-DOMAIN-COMBINE · Deepen

M49-OPS-004 — I can decompose a function into a composition of simpler functions.

TEXTBOOK SECTIONS 1.4 Composition of Functions

SUPPORTING SKILLS

A	Substitute one function formula into another. (first introduced in Math 31) SK-FUNC-COMPOSE-FORMULA · Apply
I	Identify simpler inner and outer functions in a composition. SK-FUNC-DECOMPOSE · Introduce

M49-OPS-005 — I can determine whether a function has an inverse on a given domain.

TEXTBOOK SECTIONS 1.7 Inverse Functions

SUPPORTING SKILLS

D	Apply the horizontal line test for invertibility. (first introduced in Math 31) SK-FUNC-INVERSE-HLT · Deepen
R	Test whether a function is one-to-one on a domain. (first introduced in Math 31) SK-FUNC-ONE-TO-ONE · Reinforce

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M49-OPS-006 — I can find and verify an inverse function algebraically.

TEXTBOOK SECTIONS 1.7 Inverse Functions

SUPPORTING SKILLS

A	Substitute one function formula into another. (first introduced in Math 31) SK-FUNC-COMPOSE-FORMULA · Apply
D	Exchange variables and solve to obtain an inverse formula. (first introduced in Math 31) SK-FUNC-INVERSE-ALGEBRA · Deepen
D	Verify inverse functions using composition. (first introduced in Math 31) SK-FUNC-INVERSE-VERIFY · Deepen

M49-OPS-007 — I can interpret the relationship between a function and its inverse from graphs or tables.

TEXTBOOK SECTIONS 1.7 Inverse Functions

SUPPORTING SKILLS

D	Exchange domain and range between a function and its inverse. (first introduced in Math 31) SK-FUNC-INVERSE-DOMAIN-RANGE · Deepen
D	Exchange input and output values when interpreting an inverse. (first introduced in Math 31) SK-FUNC-INVERSE-SWAP · Deepen
D	Reverse input and output units when interpreting an inverse. (first introduced in Math 31) SK-FUNC-INVERSE-UNITS · Deepen

Power, Polynomial, and Rational Functions

REQUIRED

UNIT TEXTBOOK COVERAGE

3.3 Power Functions and Polynomial Functions; 3.4 Graphs of Polynomial Functions; 3.5 Dividing Polynomials; 3.6 Zeros of Polynomial Functions; 3.7 Rational Functions; 3.8 Inverses and Radical Functions; 3.9 Modeling Using Variation; 4.2 Graphs of Exponential Functions; 4.4 Graphs of Logarithmic Functions

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M49-PWR-001 **I can identify a power function and interpret the parameters in y = kx^p.**

TEXTBOOK SECTIONS 3.3 Power Functions and Polynomial Functions

SUPPORTING SKILLS

D	Interpret the coefficient and exponent in a power function. (first introduced in Math 39) SK-POWER-PARAMETER-INTERPRET · Deepen
D	Represent and interpret a power-variation relationship. (first introduced in Math 39) SK-VAR-POWER · Deepen

M49-PWR-002 **I can construct a power function from two data points or a proportional relationship.**

TEXTBOOK SECTIONS 3.9 Modeling Using Variation

SUPPORTING SKILLS

A	Isolate a specified variable in an equation or formula. (first introduced in Math 12) SK-ALG-VARIABLE-ISOLATE · Apply
D	Determine a power function from two points or a proportional relationship. (first introduced in Math 39) SK-POWER-CONSTRUCT · Deepen

M49-PWR-003 **I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.**

TEXTBOOK SECTIONS 3.3 Power Functions and Polynomial Functions

SUPPORTING SKILLS

D	Determine power-function behavior as the input approaches zero or positive or negative infinity. (first introduced in Math 39) SK-POWER-END-BEHAVIOR · Deepen
D	Relate the coefficient and exponent of a power function to its graph shape. (first introduced in Math 39) SK-POWER-SHAPE · Deepen
A	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Apply

M49-PWR-004 **I can recognize and model direct, inverse, and other power-variation relationships.**

TEXTBOOK SECTIONS 3.9 Modeling Using Variation

SUPPORTING SKILLS

D	Represent and interpret a direct-variation relationship. (first introduced in Math 39) SK-VAR-DIRECT · Deepen
D	Represent and interpret an inverse-variation relationship. (first introduced in Math 39) SK-VAR-INVERSE · Deepen
D	Represent and interpret a power-variation relationship. (first introduced in Math 39) SK-VAR-POWER · Deepen

M49-PWR-005 **I can solve equations involving integer or rational powers.**

TEXTBOOK SECTIONS 3.8 Inverses and Radical Functions

SUPPORTING SKILLS

A	Use inverse operations to maintain an equivalent equation. (first introduced in Math 12) SK-ALG-INVERSE-OPERATIONS · Apply
D	Interpret numerator and denominator roles in a rational exponent. (first introduced in Math 21) SK-EXP-RATIONAL · Deepen

Math 49 In-Depth Curriculum

Precalculus with Limits

TEXTBOOK | OpenStax Precalculus 2e (Jay Abramson, 2021)

SKILL PROGRESSION: I Introduce · R Reinforce · D Deepen · A Apply

M49-PWR-006
I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.
TEXTBOOK SECTIONS 4.2 Graphs of Exponential Functions; 4.4 Graphs of Logarithmic Functions

SUPPORTING SKILLS

A	Compare functions shown in different representations. (first introduced in Math 21) SK-FUNC-COMPARE-REP · Apply
D	Compare the long-run growth or decay of functions from different families. (first introduced in Math 39) SK-FUNC-DOMINANCE · Deepen

M49-PWR-008
I can analyze power functions with fractional exponents, including their domains, graph shapes, and behavior near zero.
TEXTBOOK SECTIONS 3.8 Inverses and Radical Functions

SUPPORTING SKILLS

A	Interpret numerator and denominator roles in a rational exponent. (first introduced in Math 21) SK-EXP-RATIONAL · Apply
A	Require an even-root radicand to be nonnegative when determining domain. (first introduced in Math 21) SK-FUNC-DOMAIN-EVEN-ROOT · Apply
I	Analyze the domain, graph shape, and behavior near zero of a power function with a fractional exponent. SK-POWER-FRACTIONAL-ANALYZE · Introduce

M49-ALG-002
I can predict the end behavior of a polynomial from its degree and leading coefficient.
TEXTBOOK SECTIONS 3.3 Power Functions and Polynomial Functions; 3.4 Graphs of Polynomial Functions

SUPPORTING SKILLS

I	Infer polynomial end behavior from degree and leading coefficient. SK-POLY-END-BEHAVIOR · Introduce
A	Identify terms, factors, coefficients, variables, and degree in a polynomial expression. (first introduced in Math 12) SK-POLY-VOCABULARY · Apply

M49-ALG-003
I can determine polynomial zeros and their multiplicities.
TEXTBOOK SECTIONS 3.4 Graphs of Polynomial Functions; 3.5 Dividing Polynomials; 3.6 Zeros of Polynomial Functions

SUPPORTING SKILLS

I	Relate zero multiplicity to local graph behavior. SK-POLY-MULTIPLICITY · Introduce
D	Determine real and complex zeros using an appropriate method. (first introduced in Math 21) SK-POLY-ZEROS · Deepen

M49-ALG-004
I can construct a polynomial function from specified zeros or graphical features.
TEXTBOOK SECTIONS 3.4 Graphs of Polynomial Functions; 3.5 Dividing Polynomials; 3.6 Zeros of Polynomial Functions

SUPPORTING SKILLS

D	Build a polynomial from zeros, multiplicities, and a point. (first introduced in Math 31) SK-POLY-CONSTRUCT · Deepen
A	Relate zero multiplicity to local graph behavior. (first introduced in Math 39) SK-POLY-MULTIPLICITY · Apply
A	Determine real and complex zeros using an appropriate method. (first introduced in Math 21) SK-POLY-ZEROS · Apply

M49-ALG-005
I can sketch and analyze a polynomial function using intercepts, multiplicity, and end behavior.
TEXTBOOK SECTIONS 3.4 Graphs of Polynomial Functions; 3.6 Zeros of Polynomial Functions

SUPPORTING SKILLS

D	Infer polynomial end behavior from degree and leading coefficient. (first introduced in Math 39) SK-POLY-END-BEHAVIOR · Deepen
D	Relate zero multiplicity to local graph behavior. (first introduced in Math 39) SK-POLY-MULTIPLICITY · Deepen
A	Determine real and complex zeros using an appropriate method. (first introduced in Math 21) SK-POLY-ZEROS · Apply

Math 49 In-Depth Curriculum

Precalculus with Limits

TEXTBOOK | OpenStax Precalculus 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M49-ALG-006 I can identify holes and vertical, horizontal, or slant asymptotes of a rational function.

TEXTBOOK SECTIONS 3.7 Rational Functions

SUPPORTING SKILLS

A	Cancel common factors rather than individual terms. (first introduced in Math 21) SK-RAT-CANCEL-FACTORS · Apply
I	Determine horizontal or slant asymptotic behavior. SK-RAT-END-BEHAVIOR · Introduce
A	Factor numerators and denominators completely. (first introduced in Math 21) SK-RAT-FACTOR · Apply
I	Identify a removable discontinuity and its coordinates. SK-RAT-HOLE · Introduce
I	Determine vertical asymptotes from noncanceling denominator zeros. SK-RAT-VERTICAL-ASYMPTOTE · Introduce

M49-ALG-007 I can sketch and analyze a rational function using its algebraic and asymptotic features.

TEXTBOOK SECTIONS 3.7 Rational Functions

SUPPORTING SKILLS

D	Determine horizontal or slant asymptotic behavior. (first introduced in Math 49) SK-RAT-END-BEHAVIOR · Deepen
D	Identify a removable discontinuity and its coordinates. (first introduced in Math 49) SK-RAT-HOLE · Deepen
A	Determine excluded values from an original denominator. (first introduced in Math 21) SK-RAT-RESTRICTIONS · Apply
I	Determine the sign of a rational function on intervals. SK-RAT-SIGN-INTERVAL · Introduce
D	Determine vertical asymptotes from noncanceling denominator zeros. (first introduced in Math 49) SK-RAT-VERTICAL-ASYMPTOTE · Deepen

M49-ALG-008 I can build or select a power, polynomial, or rational model for a context.

TEXTBOOK SECTIONS 3.9 Modeling Using Variation

SUPPORTING SKILLS

I	Select a function family whose behavior fits a table, graph, or context. SK-MODEL-FAMILY-SELECT · Introduce
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Apply

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Limits and Function Comparison

REQUIRED

UNIT TEXTBOOK COVERAGE

12.1 Finding Limits: Numerical and Graphical Approaches

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M49-LIM-001 — I can interpret limit notation as a statement about function behavior near an input.

TEXTBOOK SECTIONS 12.1 Finding Limits: Numerical and Graphical Approaches

SUPPORTING SKILLS

I	Read and write limit notation with correct approach direction. SK-LIM-NOTATION · Introduce

M49-LIM-002 — I can estimate a limit from a graph or table.

TEXTBOOK SECTIONS 12.1 Finding Limits: Numerical and Graphical Approaches

SUPPORTING SKILLS

I	Estimate one-sided and two-sided limits from a graph. SK-LIM-GRAPH · Introduce
I	Estimate a limit from input-output values approaching from both sides. SK-LIM-TABLE · Introduce

M49-LIM-003 — I can evaluate a limit algebraically using direct substitution or simplification.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

I	Resolve an indeterminate algebraic form by factoring or rationalizing. SK-LIM-SIMPLIFY · Introduce
I	Test a limit using direct substitution. SK-LIM-SUBSTITUTE · Introduce
A	Factor numerators and denominators completely. (first introduced in Math 21) SK-RAT-FACTOR · Apply

M49-LIM-005 — I can interpret an infinite limit as vertical-asymptotic behavior.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

I	Interpret unbounded output near a finite input as an infinite limit. SK-LIM-INFINITE · Introduce
D	Determine vertical asymptotes from noncanceling denominator zeros. (first introduced in Math 49) SK-RAT-VERTICAL-ASYMPTOTE · Deepen

M49-LIM-006 — I can determine limits at infinity and relate them to long-run behavior.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

I	Compare dominant terms to determine behavior at infinity. SK-LIM-INFINITY · Introduce
D	Determine horizontal or slant asymptotic behavior. (first introduced in Math 49) SK-RAT-END-BEHAVIOR · Deepen

M49-LIM-007 — I can compare the long-run growth of polynomial, exponential, logarithmic, and rational functions.

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

A	Compare functions shown in different representations. (first introduced in Math 21) SK-FUNC-COMPARE-REP · Apply
I	Apply the long-run growth hierarchy among logarithmic, polynomial, and exponential functions. SK-LIM-GROWTH-ORDER · Introduce

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Sequences and Series

REQUIRED

UNIT TEXTBOOK COVERAGE
11.1 Sequences and Their Notations; 11.2 Arithmetic Sequences; 11.3 Geometric Sequences; 11.4 Series and Their Notations

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M49-SEQ-001 I can evaluate and graph a sequence defined explicitly or recursively.

TEXTBOOK SECTIONS 11.1 Sequences and Their Notations

	SUPPORTING SKILLS
I	Generate an explicit formula from a sequence pattern. SK-SEQ-EXPLICIT · Introduce
I	Graph a sequence as discrete ordered pairs. SK-SEQ-GRAPH · Introduce
I	Interpret subscripts and evaluate terms of a sequence. SK-SEQ-NOTATION · Introduce
I	Generate a recursive formula with an initial condition. SK-SEQ-RECURSIVE · Introduce

M49-SEQ-002 I can write explicit and recursive formulas for arithmetic sequences.

TEXTBOOK SECTIONS 11.2 Arithmetic Sequences

	SUPPORTING SKILLS
I	Recognize constant difference in an arithmetic sequence. SK-SEQ-DIFFERENCE · Introduce
D	Generate an explicit formula from a sequence pattern. (first introduced in Math 49) SK-SEQ-EXPLICIT · Deepen
D	Generate a recursive formula with an initial condition. (first introduced in Math 49) SK-SEQ-RECURSIVE · Deepen

M49-SEQ-003 I can write explicit and recursive formulas for geometric sequences.

TEXTBOOK SECTIONS 11.3 Geometric Sequences

	SUPPORTING SKILLS
D	Generate an explicit formula from a sequence pattern. (first introduced in Math 49) SK-SEQ-EXPLICIT · Deepen
I	Recognize constant ratio in a geometric sequence. SK-SEQ-RATIO · Introduce
D	Generate a recursive formula with an initial condition. (first introduced in Math 49) SK-SEQ-RECURSIVE · Deepen

M49-SEQ-004 I can distinguish arithmetic, geometric, and other sequences from formulas, tables, or contexts.

TEXTBOOK SECTIONS 11.1 Sequences and Their Notations; 11.2 Arithmetic Sequences; 11.3 Geometric Sequences

	SUPPORTING SKILLS
A	Compare functions shown in different representations. (first introduced in Math 21) SK-FUNC-COMPARE-REP · Apply
D	Recognize constant difference in an arithmetic sequence. (first introduced in Math 49) SK-SEQ-DIFFERENCE · Deepen
D	Recognize constant ratio in a geometric sequence. (first introduced in Math 49) SK-SEQ-RATIO · Deepen

M49-SEQ-005 I can create and interpret a sequence model for discrete change.

TEXTBOOK SECTIONS 11.2 Arithmetic Sequences; 11.3 Geometric Sequences

	SUPPORTING SKILLS
A	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Apply
A	Generate an explicit formula from a sequence pattern. (first introduced in Math 49) SK-SEQ-EXPLICIT · Apply
I	Identify initial value and pattern of change in a discrete context. SK-SEQ-MODEL · Introduce
A	Generate a recursive formula with an initial condition. (first introduced in Math 49) SK-SEQ-RECURSIVE · Apply

Math 49 In-Depth Curriculum

Precalculus with Limits

TEXTBOOK | OpenStax Precalculus 2e (Jay Abramson, 2021)

SKILL PROGRESSION: I Introduce · R Reinforce · D Deepen · A Apply

M49-SER-001 — I can interpret and use summation notation.

TEXTBOOK SECTIONS 11.4 Series and Their Notations

SUPPORTING SKILLS	
A	Interpret subscripts and evaluate terms of a sequence. (first introduced in Math 49) SK-SEQ-NOTATION · Apply
I	Expand and evaluate a finite sum written in sigma notation. SK-SER-SIGMA · Introduce

M49-SER-002 — I can calculate a finite arithmetic series.

TEXTBOOK SECTIONS 11.4 Series and Their Notations

SUPPORTING SKILLS	
A	Recognize constant difference in an arithmetic sequence. (first introduced in Math 49) SK-SEQ-DIFFERENCE · Apply
I	Apply an arithmetic-series sum formula. SK-SER-ARITHMETIC · Introduce

M49-SER-003 — I can calculate a finite geometric series.

TEXTBOOK SECTIONS 11.4 Series and Their Notations

SUPPORTING SKILLS	
A	Recognize constant ratio in a geometric sequence. (first introduced in Math 49) SK-SEQ-RATIO · Apply
I	Apply a finite geometric-series sum formula. SK-SER-GEOMETRIC-FINITE · Introduce

M49-SER-004 — I can determine whether an infinite geometric series converges and find its sum when it does.

TEXTBOOK SECTIONS 11.4 Series and Their Notations

SUPPORTING SKILLS	
A	Recognize constant ratio in a geometric sequence. (first introduced in Math 49) SK-SEQ-RATIO · Apply
I	Test convergence and sum an infinite geometric series. SK-SER-GEOMETRIC-INFINITE · Introduce

M49-SER-005 — I can create and interpret a series model for accumulated change.

TEXTBOOK SECTIONS 11.4 Series and Their Notations

SUPPORTING SKILLS	
A	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Apply
A	Apply an arithmetic-series sum formula. (first introduced in Math 49) SK-SER-ARITHMETIC · Apply
A	Apply a finite geometric-series sum formula. (first introduced in Math 49) SK-SER-GEOMETRIC-FINITE · Apply
A	Test convergence and sum an infinite geometric series. (first introduced in Math 49) SK-SER-GEOMETRIC-INFINITE · Apply
I	Identify the initial term, change factor, and number of terms in an application. SK-SER-MODEL · Introduce

Additional Limit Concepts

EXTENSION

UNIT TEXTBOOK COVERAGE

12.1 Finding Limits: Numerical and Graphical Approaches

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M49-LIM-004 — I can determine one-sided limits and decide whether a two-sided limit exists.

TEXTBOOK SECTIONS 12.1 Finding Limits: Numerical and Graphical Approaches

SUPPORTING SKILLS

D	Estimate one-sided and two-sided limits from a graph. (first introduced in Math 49) SK-LIM-GRAPH · Deepen
D	Read and write limit notation with correct approach direction. (first introduced in Math 49) SK-LIM-NOTATION · Deepen
A	Estimate a limit from input-output values approaching from both sides. (first introduced in Math 49) SK-LIM-TABLE · Apply

M49-LIM-008 — I can distinguish a function value from a limit and determine continuity at a point.

TEXTBOOK SECTIONS 12.1 Finding Limits: Numerical and Graphical Approaches

SUPPORTING SKILLS

I	Check the three conditions for continuity at a point. SK-LIM-CONTINUITY · Introduce
A	Estimate one-sided and two-sided limits from a graph. (first introduced in Math 49) SK-LIM-GRAPH · Apply
D	Test a limit using direct substitution. (first introduced in Math 49) SK-LIM-SUBSTITUTE · Deepen

Parametric Equations

EXTENSION

UNIT TEXTBOOK COVERAGE

8.6 Parametric Equations; 8.7 Parametric Equations: Graphs

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M49-PAR-001 — I can graph a plane curve defined by parametric equations with its orientation.

TEXTBOOK SECTIONS 8.7 Parametric Equations: Graphs

SUPPORTING SKILLS

A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply
D	Graph a parametric curve with its direction of motion. (first introduced in Math 32) SK-PAR-GRAPH · Deepen
D	Generate ordered pairs from parametric equations. (first introduced in Math 32) SK-PAR-TABLE · Deepen

M49-PAR-002 — I can eliminate a parameter to obtain a rectangular equation.

TEXTBOOK SECTIONS 8.6 Parametric Equations

SUPPORTING SKILLS

A	Isolate a specified variable in an equation or formula. (first introduced in Math 12) SK-ALG-VARIABLE-ISOLATE · Apply
D	Eliminate a parameter algebraically. (first introduced in Math 32) SK-PAR-ELIMINATE · Deepen
A	Substitute an equivalent expression for a variable. (first introduced in Math 12) SK-SYS-SUBSTITUTE · Apply

M49-PAR-003 — I can create or interpret parametric equations for motion in the plane.

TEXTBOOK SECTIONS 8.6 Parametric Equations; 8.7 Parametric Equations: Graphs

SUPPORTING SKILLS

D	Interpret or construct parametric equations for planar motion. (first introduced in Math 32) SK-PAR-MODEL · Deepen
A	Generate ordered pairs from parametric equations. (first introduced in Math 32) SK-PAR-TABLE · Apply
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Conic Sections

EXTENSION

UNIT TEXTBOOK COVERAGE

10.1 The Ellipse; 10.2 The Hyperbola; 10.3 The Parabola; 10.5 Conic Sections in Polar Coordinates

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M49-CON-001 — **I can classify a circle, ellipse, or hyperbola from its equation or defining geometric relationship.**

TEXTBOOK SECTIONS 10.1 The Ellipse; 10.2 The Hyperbola; 10.5 Conic Sections in Polar Coordinates

SUPPORTING SKILLS

I	Classify a circle, ellipse, hyperbola, or parabola from an equation or geometric definition. SK-CONIC-CLASSIFY · Introduce
I	Use fixed-distance and focal-distance relationships to explain circles, ellipses, and hyperbolas. SK-CONIC-DISTANCE-DEFINITION · Introduce

M49-CON-002 — **I can use completing the square to rewrite a general-form conic equation in standard form.**

TEXTBOOK SECTIONS 10.1 The Ellipse; 10.2 The Hyperbola; 10.3 The Parabola

SUPPORTING SKILLS

A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Complete squares to convert a general conic equation to standard form. SK-CONIC-GENERAL-STANDARD · Introduce
D	Create a perfect-square trinomial by completing the square. (first introduced in Math 21) SK-QUAD-COMPLETE-SQUARE · Deepen

M49-CON-003 — **I can write, interpret, and graph implicit and parametric equations of a circle.**

TEXTBOOK SECTIONS No mapped textbook section

SUPPORTING SKILLS

I	Connect implicit, standard, and parametric equations of a circle. SK-CONIC-CIRCLE-EQUATIONS · Introduce
I	Determine and use a circle's center and radius. SK-CONIC-CIRCLE-FEATURES · Introduce
A	Graph a parametric curve with its direction of motion. (first introduced in Math 32) SK-PAR-GRAPH · Apply
A	Generate ordered pairs from parametric equations. (first introduced in Math 32) SK-PAR-TABLE · Apply

M49-CON-004 — **I can write, interpret, and graph implicit and parametric equations of an ellipse.**

TEXTBOOK SECTIONS 10.1 The Ellipse

SUPPORTING SKILLS

I	Connect implicit, standard, and parametric equations of an ellipse. SK-CONIC-ELLIPSE-EQUATIONS · Introduce
I	Determine and use an ellipse's center, axes, vertices, and foci. SK-CONIC-ELLIPSE-FEATURES · Introduce
A	Graph a parametric curve with its direction of motion. (first introduced in Math 32) SK-PAR-GRAPH · Apply
A	Generate ordered pairs from parametric equations. (first introduced in Math 32) SK-PAR-TABLE · Apply

Math 49 In-Depth Curriculum

Precalculus with Limits

TEXTBOOK | OpenStax Precalculus 2e (Jay Abramson, 2021)

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M49-CON-005

I can write, interpret, and graph implicit and parametric equations of a hyperbola.

TEXTBOOK SECTIONS 10.2 The Hyperbola

SUPPORTING SKILLS

I	Connect implicit, standard, and parametric equations of a hyperbola. SK-CONIC-HYPERBOLA-EQUATIONS · Introduce
I	Determine and use a hyperbola's center, axes, vertices, foci, and asymptotes. SK-CONIC-HYPERBOLA-FEATURES · Introduce
A	Graph a parametric curve with its direction of motion. (first introduced in Math 32) SK-PAR-GRAPH · Apply
A	Generate ordered pairs from parametric equations. (first introduced in Math 32) SK-PAR-TABLE · Apply

M49-CON-006

I can determine the center, vertices, foci, axes, or asymptotes needed to describe a circle, ellipse, or hyperbola.

TEXTBOOK SECTIONS 10.1 The Ellipse; 10.2 The Hyperbola; 10.5 Conic Sections in Polar Coordinates

SUPPORTING SKILLS

D	Determine and use a circle's center and radius. (first introduced in Math 49) SK-CONIC-CIRCLE-FEATURES · Deepen
A	Classify a circle, ellipse, hyperbola, or parabola from an equation or geometric definition. (first introduced in Math 49) SK-CONIC-CLASSIFY · Apply
D	Determine and use an ellipse's center, axes, vertices, and foci. (first introduced in Math 49) SK-CONIC-ELLIPSE-FEATURES · Deepen
D	Determine and use a hyperbola's center, axes, vertices, foci, and asymptotes. (first introduced in Math 49) SK-CONIC-HYPERBOLA-FEATURES · Deepen

M49-CON-007

I can explain the geometric definitions of circles, ellipses, and hyperbolas using foci and distance relationships.

TEXTBOOK SECTIONS 10.1 The Ellipse; 10.2 The Hyperbola; 10.5 Conic Sections in Polar Coordinates

SUPPORTING SKILLS

D	Use fixed-distance and focal-distance relationships to explain circles, ellipses, and hyperbolas. (first introduced in Math 49) SK-CONIC-DISTANCE-DEFINITION · Deepen
A	Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula. (first introduced in Math 22) SK-COORD-DISTANCE-PYTHAGOREAN · Apply

M49-CON-008

I can explain a parabola as the set of points equidistant from a focus and directrix.

TEXTBOOK SECTIONS 10.3 The Parabola; 10.5 Conic Sections in Polar Coordinates

SUPPORTING SKILLS

I	Explain a parabola using its focus-directrix distance relationship. SK-CONIC-PARABOLA-DEFINITION · Introduce
A	Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula. (first introduced in Math 22) SK-COORD-DISTANCE-PYTHAGOREAN · Apply